



SECJ2203: Software Engineering

System Documentation (SD)

Koperasi KADA Automated System

Version 5.0

12 February 2025

Faculty of Computing

Prepared by: Synapse

Revision Page

a. Overview

The System Documentation (SD) of the current version of Koperasi KADA's Automated System has the system introduction comprising the scope, purpose, objective and feature to be included in the system. SD provides user personas, user stories and system features. Product backlogs, sprint deliverables, user story with support of sequence and activity diagram. Also covered are non-functional requirements

b. Target Audience

The target audience are applicants, admin and Ahli Lembaga Koperasi of KADA.

c. Project Team Members

Member Name	Role	Task	Status
Farra Nurzahin	Project Manager	Project planning	Complete
		Project management	Incomplete
		Create system architecture	Complete
Dayang Farah Farzana	Technical Writer	Gather system and user requirements	Complete
		Create SRS document	Complete
		System documentation	Complete
Nawarrah Auni	System Developer	Develop system	Incomplete
		System Testing	Incomplete
		System Implementation	Incomplete
Safiya Nursyahadah	Designer	System Design	Incomplete
		Design system functional diagram	Complete
		Design draft system	Complete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Farra Nurzahin binti Zaharil Anuar	Completed Topic 1(Introduction), Scope, Section 2.1.2 Section 2.2.1, Section 2.3, Section 2.4.2, Section 2.5	30/11/2024
	Dayang Farah Farzana binti Abang Idham	Completed Overview, Section 1.3, Section 2.1.3, Section 2.2	30/11/2024

		(System Features), Section 2.3, Section 2.4.3	
	Nawwarah Auni binti Nazrudin	Completed Purpose, Specific Requirements, Section 2.1, Section 2.2.2, Section 2.3, Section 2.4, Section 2.4.1, Section 2.6.1, Section 2.6.4	29/11/2024
	Safiya Nursyahadah binti Masnoor	Completed Target Audience, Section 1.5 (overview), Section 2.1.1, Section 2.2.3 (state diagram), Section 2.3, Section 2.6,	30/11/2024
2.0	Farra Nurzahin binti Zaharil Anuar	Completed Section 4.1, Section 4.2, Section 4.3, Section 4.4 Section 5.1, Section 5.1.1, Section 5.1.7, Section 5.1.8, Section 5.1.9, Section 6.1	19/12/2024
	Dayang Farah Farzana binti Abang Idham	Completed Section 4, Section 4.2, Section 4.3, Section 4.4 Section 5, Section 5.1.1, Section 5.1.2, Section 5.1.4, Section 5.1.6, Section 5.1.12, Section 5.1.13, Section 6.1, Section 7	19/12/2024
	Nawwarah Auni binti Nazrudin	Completed Section 3.1, Section 4.2, Section 4.3, Section 4.4 Section 4.5, Section 5.1.2, Section 5.1.10, Section 5.1. 11, Section 5.1.14, Section 6.1, Section 7	19/12/2024
	Safiya Nursyahadah binti Masnoor	Completed Section 3, Section 4.2, Section 4.3, Section 4.4, Section 5.1.3, Section 5.1.5, Section 5.1.15, Section 6, Section 6.1	19/12/2024
3.0	Dayang Farah Farzana binti Abang Idham	Correction of architectural views	31/12/2024
4.0	Farra Nurzahin binti Zaharil Anuar	Correction of ERD and Data Dictionary	14/1/2025
5.0	Farra Nurzahin binti Zaharil Anuar	Correction of 2.1.1, 2.2.1, 2.3, 2.4.1, 4.1, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, Completed UC010	12/2/2025
	Dayang Farah Farzana binti Abang Idham	Correction of 2.1.3, 2.2.1, 2.3, Section 4, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7,	12/2/2025

		Completed UC011	
	Nawwarah Auni binti Nazrudin	Correction of 2.1.2, 2.2.1, 2.2.2, 2.3, 2.4.2, 3.1, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, Completed UC009	12/2/2025
	Safiya Nursyahadah binti Masnoor	Correction of 2.1.2, 2.2.3, 2.4.3, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7	12/2/2025

Table of Contents

1	Introduction		7
	1.1	Purpose	7
	1.2	Scope	7
	1.3	Definitions, Acronyms and Abbreviations	8
	1.4	References	9
	1.5	Overview	9
2	Specific Requirements		10
	2.1	User characteristics	10
	2.2	System Features	13
	2.3	Use Case Details	20
	2.3.1	UC001: Use Case <Login to Apply Membership>	21
		Use Case Specification of UC001	19
		Activity Diagram of UC001	22
		System Sequence Diagram of UC001	23
	2.3.2	UC002: Use Case <Review Applicant's Form>	24
		Use Case Specification of UC002	25
		Activity Diagram of UC002	26
		System Sequence Diagram of UC002	26
	2.3.3	UC003: Use Case <Approve Membership>	27
		Use Case Specification of UC003	27
		Activity Diagram of UC003	28
		System Sequence Diagram of UC003	29
	2.3.4	UC004: Use Case <Apply Loan>	30

			Use Case Specification of UC004	31
			Activity Diagram of UC004	31
			System Sequence Diagram of UC004	32
		2.3.5	UC005: Use Case <Calculate Loan>	33
			Use Case Specification of UC005	33
			Activity Diagram of UC005	34
			System Sequence Diagram of UC005	35
		2.3.6	UC006: Use Case <Approve Loan Application>	36
			Use Case Specification of UC006	36
			Activity Diagram of UC006	37
			System Sequence Diagram of UC006	38
		2.3.7	UC007: Use Case <Prepare Statement and Report>	39
			Use Case Specification of UC007	39
			Activity Diagram of UC007	40
			System Sequence Diagram of UC007	41
		2.3.8	UC008: Use Case <Notify via Email>	42
			Use Case Specification of UC008	42
			Activity Diagram of UC008	44
			System Sequence Diagram of UC008	45
		2.3.9	UC009: Use Case <Manage Transaction>	46
			Use Case Specification of UC009	46
			Activity Diagram of UC009	47
			System Sequence Diagram of UC009	48
		2.3.10	UC010: Use Case <Terminate Membership>	49

		Use Case Specification of UC010	49
		Activity Diagram of UC010	49
		System Sequence Diagram of UC010	50
	2.3.11	UC011: Use Case <Approve Terminate Application>	51
		Use Case Specification of UC011	51
		Activity Diagram of UC011	52
		System Sequence Diagram of UC011	53
	2.4	Performance and Other Requirements	54
	2.5	Design Constraints	55

3	Architectural Rationale		57
	3.1	Architectural Style and Rationale	57
4	Architectural Views		58
	4.1	Use Case View	59
	4.2	Implementation View	60
	4.3	Logical View	70
	4.4	Process View	76
	4.5	Physical/ Deployment View	88
5	Data Design		89
	5.1	Data Dictionary	90
		5.1.1 Entity: <Admin>	91
		5.1.2 Entity: <AdminApplicationFeedback>	92
		5.1.3 Entity: <ALK>	92
		5.1.4 Entity: <AlkApplicationFeedback>	92
		5.1.5 Entity: <Applicant>	93

		5.1.6 Entity: <BankInfo>	94
		5.1.7 Entity: <FamilyInfo>	94
		5.1.8 Entity: <Guarantor>	95
		5.1.9 Entity: <HistoryModiMem>	95
		5.1.10 Entity: <HistoryModiLoan>	95
		5.1.11 Entity: <Loan>	96
		5.1.12 Entity: <LoanRepayment>	96
		5.1.13 Entity: <LoanType>	97
		5.1.14 Entity: <Managing>	97
		5.1.15 Entity: <Membership>	97
		5.1.16 Entity: <MembershipEnd>	97
		5.1.17 Entity: <Saving>	98
		5.1.18 Entity: <SavingType>	98
		5.1.19 Entity: <Status>	99
		5.1.20 Entity: <User>	99
		5.1.21 Entity: <UserType>	99
6		User Interface Design	100
	6.1	Screen Images	101
7		Traceability	109

1. Introduction

This part of System Documentation (SD) provides detailed information about the system's development and implementation processes. This document consists of Software Requirements Specification (SRS) to outline the functional and nonfunctional requirements of the system. This to make sure what kind of software in the Koperasi KADA's system must be achieved. The comprehension of Software Design Description (SDD) is also in this document to provide essential detailed design information and architecture for the system's development. Lastly, Software Test Documentation (STD) is included in this document to describe the test plans to validate and verify the software's functionality and performance.

1.1 Purpose

This SD describes the requirements for Koperasi KADA's Automated System. It explains how the system will handle membership and loan management. It also describes the system's features, how users will interact with it, and any rules or limits the system must follow. To help understand the system better, this document uses user stories that describe scenarios to show different users, like staff and members, will interact with the system to complete their tasks.

This document is meant for developers, system administrators, users and stakeholder, Koperasi KADA who need to understand how the system works and how to use it effectively.

1.2 Scope

The software product is the Koperasi KADA automated website system which is going to be built to make managing memberships and processing loans easier for cooperative members. It contains the following main software products including membership management system module, loan management system module, report generator module and notification system module.

Firstly, the membership management system module makes sure the processing of the submission, review and approval of membership applications are secured and also provides the applicants with a real-time update. Secondly, the loan management module eases the submission and approval of loan applications that includes the loan calculator as well as real-time update. This module only prioritizes loans within the Koperasi KADA and does not support other external loans. Next, the report generator module covers the preparation of detailed reports and statements regarding loans, membership and other

activities for the administrators. Lastly, the notification system module is built for sending an automated notification via applicant's email to keep members updated about their application statuses. It only covers the notification from email and not SMS.

This software is built to make managing memberships and processing loans easier for Koperasi KADA. This product ensures the improvement of operational efficiency by automating the manual processes as well as reduces errors and saves time for administrators and ensures data security for Koperasi KADA by having a built-in encryption and access control. The objective of this document is to:

- I. To design a web-based system for KADA's koperasi system.
- II. To develop and test the system to ensure the requirements are met.

1.3 Definitions, Acronyms and Abbreviation

This section provides the definitions, acronyms and abbreviations used in the System Documentation (SD)

- ALK : Ahli Lembaga Koperasi - Board members that responsible for decision-making
- Admin : Staff that responsible for managing and reviewing applications
- Applicant : A user that apply membership or loan through the system
- SD : System Documentation - Detailed document

Term	Definition
ALK	Ahli Lembaga Koperasi - Board members that responsible for decision-making
Admin	Staff that responsible for managing and reviewing applications
Applicant	A user that apply membership or loan through the system
SD	System Documentation - Detailed document

1.4 References

[1] A. Dennis, B. H. Wixom & D. Tegarden (2015). *System Analysis and Design, An Object-Oriented Approach with UML*, 5th Edition. [E-book]

1.5 Overview

The System Documentation (SD) for Koperasi KADA's Automated System provides a detailed description of the system's purpose, structure, and features. It includes a system introduction that defines its scope, purpose, objectives, and features. This document uses diagrams like use case, class diagram, state machine diagram to visualize the system's functionality and interactions. Additionally, it covers non-functional requirements and design constraints to ensure the system's quality, performance, and security standard.

The SD is organized into sections for better understanding. It begins with an introduction to the system's goals and requirements, followed by detailed descriptions of personas, system features, and workflows. Specific requirements are explained using user stories supported by diagrams for each use case. The document concludes with sections on performance expectations, design constraints and administrative details like version control and the project team's distributions. This structured approach ensures that the SD is user-friendly and informative for developers, stakeholders and end users.

2. Specific Requirements

This section explains the detailed software requirements necessary for building and testing the system. It provides enough information to guide designers in creating a system that meets these needs and helps testers ensure the system works as intended. Every requirement is described in a way that can be observed or experienced by users, operators, or connected systems. It includes details about all inputs, the action it performs and the outputs.

2.1 User characteristics

The personas for this system includes applicants, administrators, and members of the Koperasi KADA board (ALK). Each user has specific needs and responsibilities, which are described through their individual stories to provide clear understanding of the user requirements.

2.1.1 Persona 1

User Need

- Applicants need a way to apply for a membership, login into the system, apply for a loan and calculate loan as well as apply for a termination of membership.

User Stories (US001)

- As an applicant, I want to apply for a membership and login into the Koperasi Kada Automated System so that I can apply for a loan.
- As a member, I want to fill out a loan application form so that I can provide my details for loan approval.
- As a member, I want to calculate the monthly loan repayment and any applicable dividends so that I can make an informed decision.
- As a member, I want to terminate my membership application so that I can discontinue my association with Koperasi KADA.

When Applicant opens the Koperasi KADA Automated System's website, the interface shows the user's login details. After the Applicant login successfully, the applicant will be taken directly to the main page menu which includes activities such as membership application, loan application, and calculation of loan features.

If the applicant clicks on 'Apply for Membership', the applicants are taken to the membership form page from the menu option. The applicant has to complete details in the first part of the form and then press on the 'next' button to go to the next page. On the following page, applicants fill in other parts of the form and click the 'submit' button at the end of the page. Submission shows their membership status where applicants are informed if their application has been reviewed or not.

However, to apply for a loan, an applicant has to login as a member first. On the home page after login applicants will be directed to the loan application page where they can input the amount and type of loan, and the loan duration to get estimated monthly repayments using the loan calculator.

Additionally, in order to proceed further, the applicant has to accept the agreement by clicking on the agreement option. After that, the applicant has to give the details of a guarantor and make sure that guarantor agrees to the terms and conditions. Once an applicant completes the form and submits the application, the system will show the loan application status.

If a member wishes to terminate their membership, they can do so by selecting 'Borang Berhenti' under the 'Permohonan Berhenti' option from the menu. Once the member submitted their application, the system will show the application status.

2.1.2 Persona 2

User Need

Admin need a way to review membership, loan and terminate applications, manage transactions, generate an online statement and report and send automatic notifications to applicants

User Stories (US002)

- As an admin, I want to review submitted membership applications so that I can verify the applicant's details.
- As an admin, I want to review and validate loan applications so that I can ensure it aligns with the Koperasi KADA's loan criteria.
- As an admin, I want to review submitted membership termination so that I can verify the main reason for termination.
- As an admin, I want to manage transactions of the members of Koperasi KADA's so that financial records remain accurate and up-to-date for reporting.

- As an admin, I want to generate online statements and reports so that I can provide correct reports' records to the stakeholders.
- As an admin, I want to send automated email notifications to applicants so that they are updated about the status of their applications.
- As an admin, I want to send financial statement to applicants so that they are updated about their financial balance.

Admin has a task on managing the system effectively including reviewing submitted membership, membership termination and loan applications, generating statements and reports as well as sending notifications via email. Admin logs in the system as admin and redirects to the admin page.

If admin chooses 'Membership List' or 'Loan Application List' or 'Membership Termination', they will be redirected to the list of the applicant's name page. Admin click on the not reviewed button to review applicant's details and make sure all details provided are correct and complete.

In both cases, they can click on the 'Complete' if the application is complete and correct and 'Incomplete' if the application is incomplete and not accurate.

If the admin approves a member through 'Approve Member by ALK,' they will be redirected to the 'Status Submission' page to send the application status to the applicant. Approved applicants can then access the 'Financial Statement' page to generate their statement.

The admin can also generate the statement by clicking on 'Generate Statement'. After that, a pop-up of an generated statement will be displayed on the admin screen to confirm the statement. Automated notifications are then sent to the approved applicant through their registered email by clicking on the 'Send' button.

2.1.3 Persona 3

User Need

- ALK needs a way to approve membership and approve loan applications

User Stories

- As an ALK, I want to approve applicants' membership so that I can ensure that only eligible individuals can become membership and earn membership benefits

- As an ALK, I want to approve loan application so that I can let only the eligible applicants to receive loan in line with company's policies
- As an ALK, I want to approve membership termination application so that I can deactivate members' membership

ALK is responsible for membership and loan approval and the decision made is fair inline with company policies. ALK logs in to the system and then is forwarded to a personalized page for ALK.

If ALK decides to review membership applications, then the system will display a pending list for membership applications. The same applies to when ALK chose to review loan applications.

ALK then selects any applications that appear on the pending list of membership and loan approval. ALK evaluates the applications and decides for both membership and loan approval or rejection based on the company's policies. For membership, if the status of the reviewed application is 'Incomplete', then ALK will reject immediately.

In the case of loan approval, ALK is allowed to use the 'Compare' button to compare loan applications with loan terms. The system compares and displays whether or not applicants pass the loan terms. ALK also can directly approve or reject the application without using the button.

ALK is also responsible for terminating membership application approval. Similar to membership and loan application, the termination request is reviewed by Admin before forwarding it to ALK for final decision. ALK then decide to approve or reject the application.

After making decisions for membership, loan and membership termination application, the system will update the applicant's status, notify the applicants and the status is passed to admin for further report.

2.2 System Features

The Koperasi KADA Automated System is a software solution designed to enhance operation efficiency. Access to the system is possible on standard desktop and mobile devices, operating on widely used browsers and operating systems. It puts at the disposal of the staff to manage the membership, loan and termination application more effectively. In

changing from manual procedures to an automated system, the new system guarantees in increasing the efficiency of operation, security of data and more user-friendly in usage for all relevant parties. The system features a real-time database, integrated for easy tracking and management applications, with ease of access both for applicants and KADA's staff. The system features are as follows ;

1. Membership Management

Description :

- The system supports membership registration, review membership, membership approval and record membership data.

Requirements :

- Allow secure submission and review of membership applications
- Allow ALK to approve or reject membership applications
- Allow applicants to monitor the status of their application in real-time
- Record the data

2. Loan Management

Description :

- The system supports loan application, loan review and loan approval as well as record loan data.

Requirements :

- Allow members to submit loan application, admin to review and ALK to approve loan application
- Allow applicants to monitor the status of their application in real-time
- Record the data

3. Terminate Membership Management

Description :

- The system supports a termination membership application, review and approval as well as record data.

Requirements :

- Allow members to submit terminate membership applications, admin to review and ALK to approve the application.
- Allow applicants to monitor the status of their application in real-time
- Record the data

4. Report Generator

Description :

- Generate detailed report for members

Requirements :

- Allow administrators to generate detailed financial statements and reports for Koperasi KADA members focusing on savings balance and loan balance

5. Notification

Description :

- Automated email notification related to loan and membership status

Requirements :

- Sends automated email notification to applicants regarding membership and loan status of their applications
- Sends generated financial statements through email notification

6. Financial Transaction

Description :

- A system for recording, updating, and reconciling financial transactions

Requirements :

- Allow administrators to records all saving transactions in the koperasi
- Allow administrators to keep track of all loan repayments

7. Status Modification

Description

- A function to change membership and loan status

Requirements :

- Allow ALK to change membership status from Diterima to Ditolak and vice versa
- Allow ALK to change loan status from Diterima to Ditolak and vice versa

2.2.1 Use Case

Figure 2.1 below is a use case that illustrates the structure of the Koperasi KADA Automated System. This use case highlights key modules such as membership application module, loan application module, review module, approval module and statement report production module. Each module covers a particular use case such as LogIn to Apply Membership (UC01) in the membership application module, Apply for Loan (UC04) and

Calculate Loan and Dividend (UC05) in loan application module, Review Applicant's Form (UC02) in review module, Approve Membership (UC03) and Approve Loan Application (UC06) in approval module, Prepare Statement and report (UC07) and Notify via Email (UC08) in statement report generation module as well as Manage Transactions (UC09) in financial tracking module. The detailed description of each module and functions is tabulated in Table 2.1.

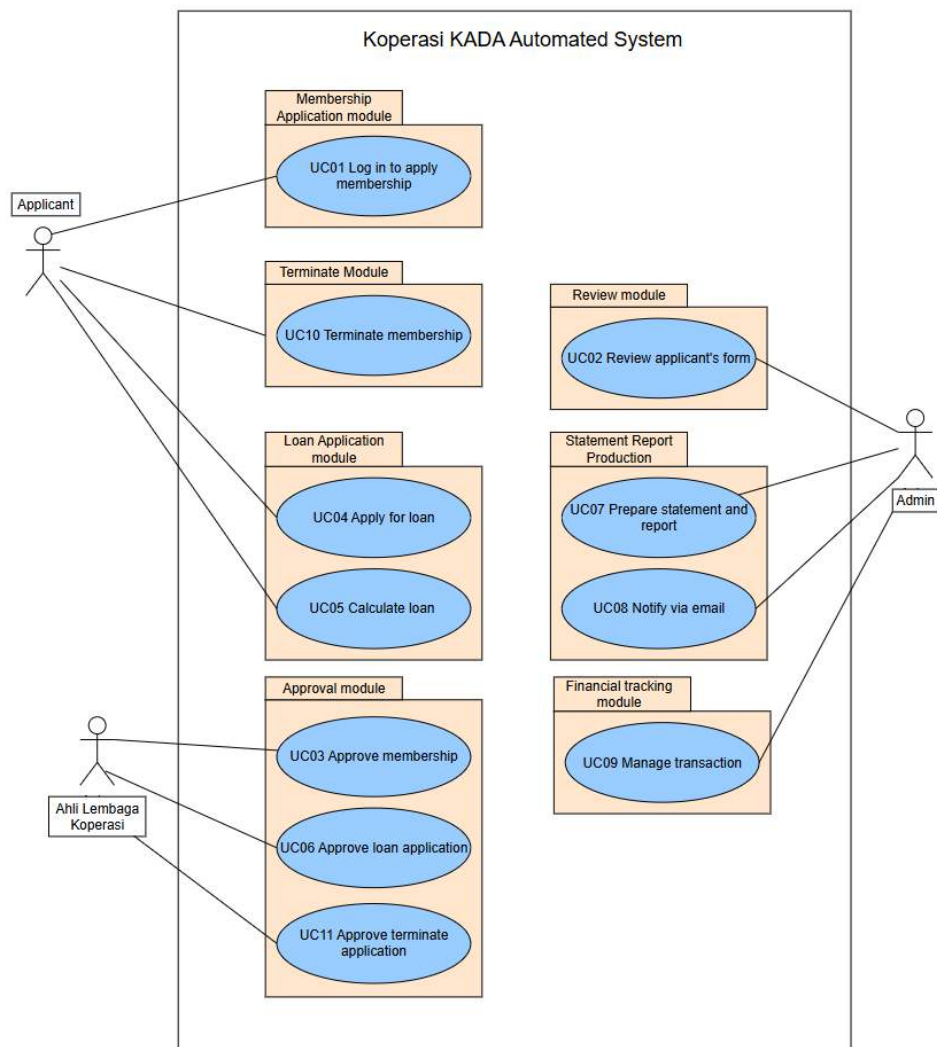


Figure 2.1: Use Case Diagram for Koperasi KADA's Automated System

Table 2.1: Description of Module and Functions for Koperasi KADA's Automated System

Use Case	Function	Description
UC01	Login to apply membership	This use case enables applicants to login, complete and submit a membership form within the system.

UC02	Review applicant's form	This use case ensures the admin to thoroughly check and validate each applicant's form before final submission.
UC03	Approve Membership	This use case enables Ahli Lembaga Koperasi (ALK) to approve or reject membership applications submitted by applicants.
UC04	Apply Loan	This use case enables members to submit loan applications online.
UC05	Calculate Loan and Dividend	This use case automates the calculation of monthly repayment and applicable dividends based on applicant's input.
UC06	Approve Loan Application	This use case enables the Ahli Lembaga Koperasi (ALK) staff to review and make a decision on loan applications submitted by Koperasi members.
UC07	Prepare Statement and Report	This use case allows the system to generate a detailed statement report for loan applicants based on their approved loan applications.
UC08	Notify via Email	This use case enables the system to send notifications to applicants regarding their loan statement and other important updates.
UC09	Manage Transaction	This use case enables the system to handle and record all savings and loan repayment transactions whether on a monthly or irregular basis efficiently.
UC10	Terminate Membership	This use case enables the applicant to request membership termination.
UC11	Approve Terminate Application	This use case enables the ALK to approve or reject membership termination applications for members that wish to end their membership.

2.2.2 Domain Model

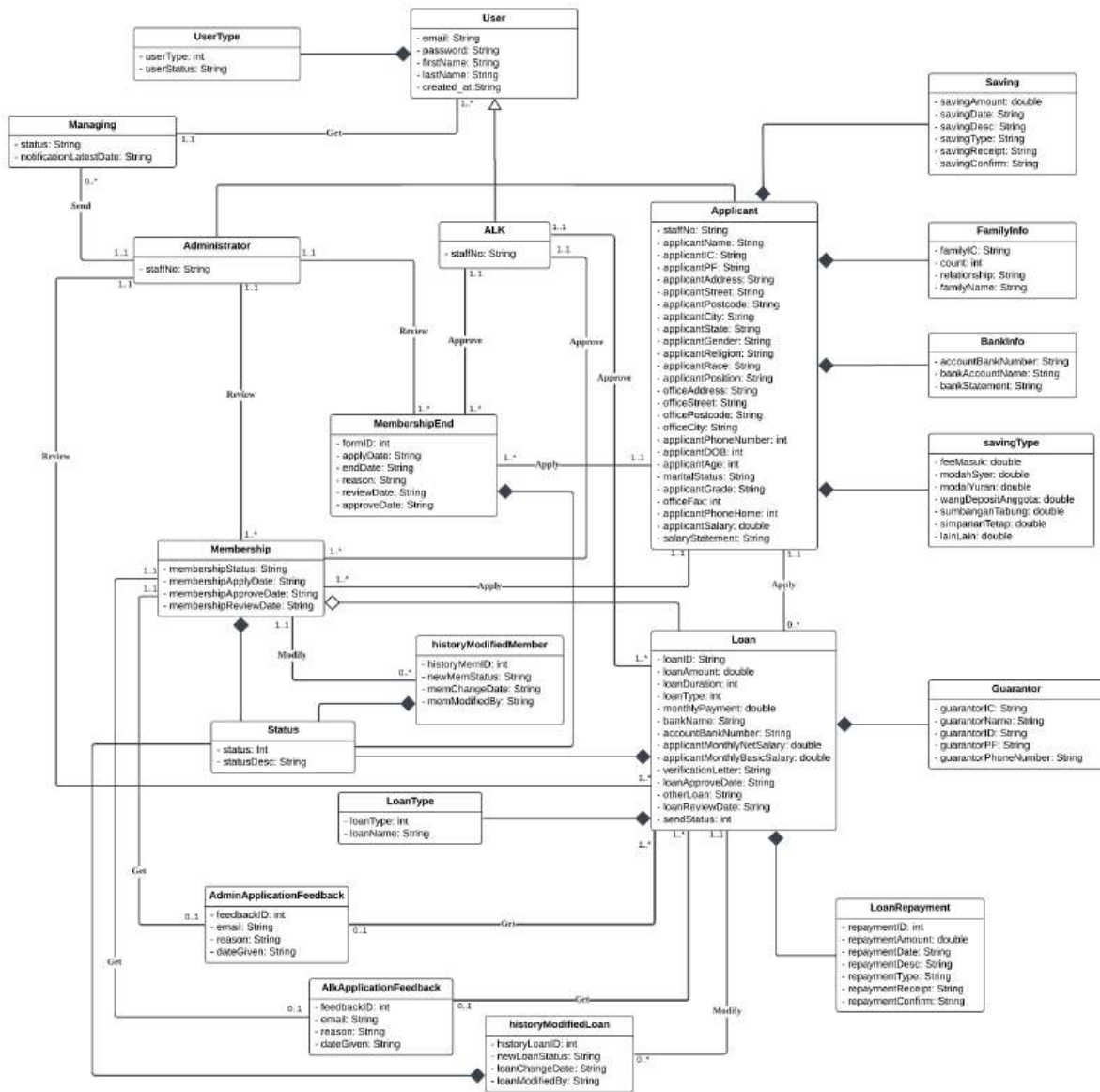


Figure 2.2: Domain Model for Koperasi KADA's Automated System

2.2.3 State Diagram

Figure 2.3 shows the Loan State Machine Diagram. It is the flow of loan application from the time an application is made to the time the final decision is made. It starts when an applicant completes the loan application and becomes the 'Loan Under Review' state where the ALK needs to approve the application. If the loan is rejected the applicant can reapply the application. If the loan has been approved, the loan status will change to 'Loan Active'. When the loan is fully paid, the process goes to 'Loan Completed', which closes the loan.

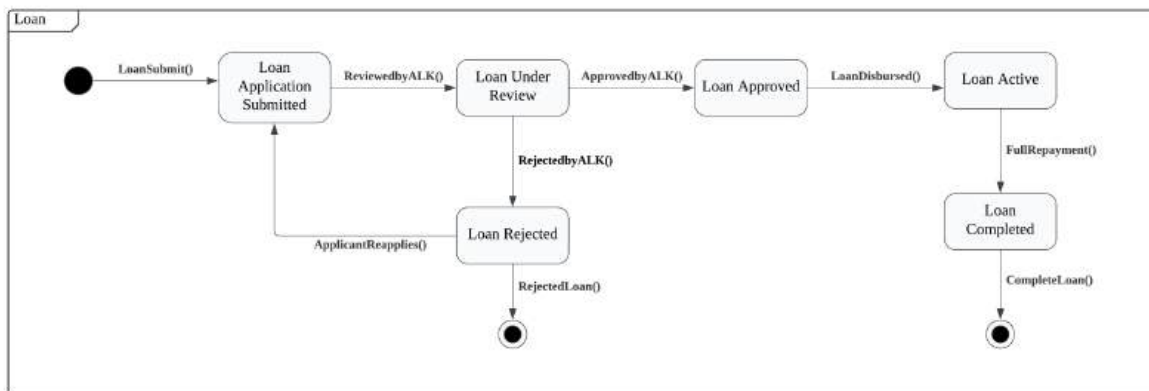


Figure 2.3: State Machine Diagram of Loan for Koperasi KADA's Automated System

Figure 2.4 shows the Membership State Machine Diagram's sequence of an application for membership. The process begins when a user submits the application, and it goes to 'Membership Under Review', the ALK either approves it or denies it. If approved, the membership becomes 'Membership Active'. From here, applicants can allow the membership to lapse and be 'Membership Expired'. The rejected applications enable applicants to reapply and go through the process all over again.

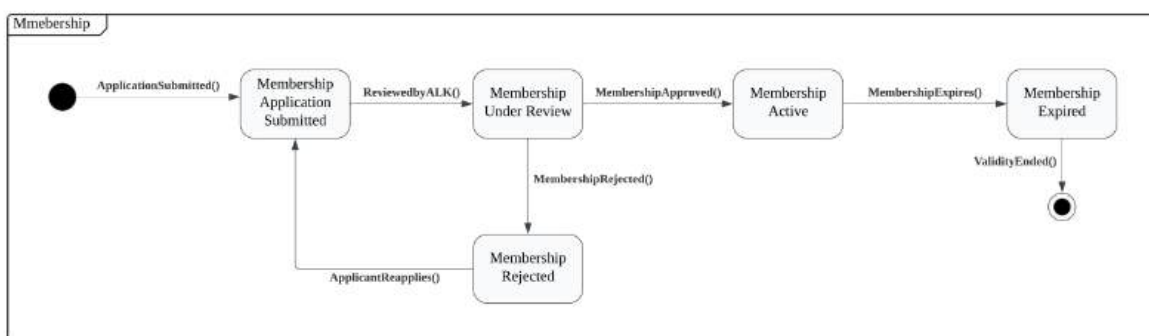


Figure 2.4: State Machine Diagram of Membership for Koperasi KADA's Automated System

Figure 2.5 shows the Report Generation State Machine Diagram. This is the process of creating and distributing the report. It starts with 'Report Initiated' each time the Admin

initiates the creation of a report. After being generated, the report can go to the recipient and be in the state of 'Report Sent' or be stored in the system at the state of 'Report Stored'. Notifications are sent with the report, and status changed to 'Notification Created' and when the recipient reads the notification, it ends at 'Notification Read'

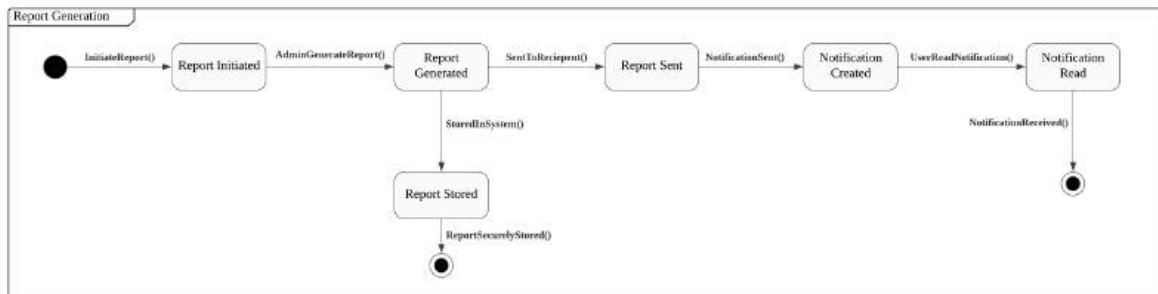


Figure 2.5: State Machine Diagram of Report Generation for Koperasi KADA's Automated System

Figure 2.6 shows the Membership Termination State Machine Diagram. The process begins when a member submits a termination request, entering the 'Membership Termination Application Submitted' state. The request is then reviewed, leading to either 'Membership Termination Approved' or 'Membership Termination Rejected'. If rejected, the applicant can reapply, restart the application process. Once approved, the membership ends and moves to 'Membership Termination Completed', concluding when validity expires.

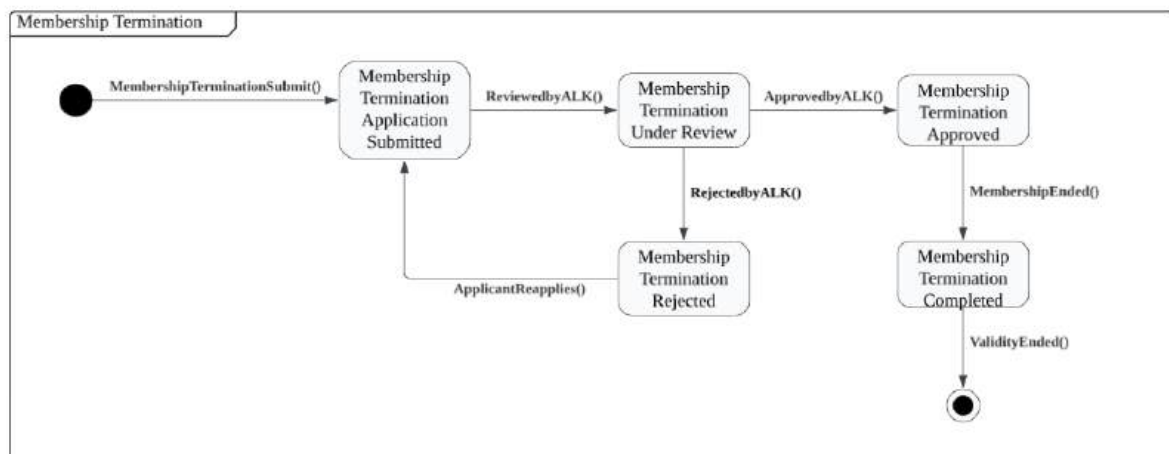


Figure 2.6: State Machine Diagram of Membership Termination for Koperasi KADA's Automated System

2.3 Use Case Details

This section explains how the system works in different situations. It includes a Use Case Description, which tells the purpose of the system, who uses it, and the steps to complete a task. A Sequence Diagram shows how different parts of the system interact with

each other step by step, like a story with messages being passed around. An Activity Diagram shows the flow of actions, including decisions and steps that can happen at the same time. These diagrams and descriptions make it easier to understand how the system works and what it can do.

2.3.1 UC001: Use Case Login to Apply Membership

Table 2.2 user story description allows applicants to access the membership application system securely. After entering valid login credentials, users are redirected to a dashboard where they can apply for membership by filling out a detailed application form. The system verifies the submitted information and ensures all required fields are completed. If the information is valid, the application is successfully submitted. If any issues are detected, the system provides feedback to guide users in correcting their errors.

Table 2.2: Use Case Description for Login to Apply Membership

Use Case: Login to Apply Membership
ID: UC001
User Story Description As an applicant, I want to log into the membership application system, So that I can apply for membership.
Flow of events: 1. The Applicant navigates to the login page of the system. 2. The system prompts the Applicant to enter their username and password. 3. The Applicant enters their login details and submits the form. 4. The system verifies the login information. 4.1 If the login details are correct, the system grants access to the Applicant. 4.2 If the login details are incorrect, the system displays an error message. 5. Once logged in, the Applicant can access the membership application form.
Alternative flow <i>n</i>: n.1 : If the applicant forgets their login details, they can click on the “Forgot Password” option. n.2 : The system guides the Applicant through a password recovery process by sending a recovery link to their registered email.
Acceptance Criteria Postcondition : 1. The Applicant is successfully logged into the system and gains access to the membership application form. Precondition : 1. The Applicant must be registered in the system with a valid username and password

2. The system must be accessible for login attempts. Other conditions : 1. The system must ensure that all login details are encrypted for security purposes. 2. The login page must support both desktop and mobile devices.
Exception flow: 1. If the Applicant enters invalid login details three times, the system locks the account temporarily and prompts Applicant to use the "Forgot Password" option. 2. If the system is unavailable due to maintenance, the Applicant can see a maintenance notice and cannot login until the system is recovered.

Figure 2.7 shows the workflow for the Login to Apply Membership. The process starts with the applicant logging in and the system checks their authorization. If unauthorized, an error message is displayed; otherwise, they are redirected to the dashboard. The applicant then fills out the membership application form and submits it. The system checks if the submission was successful. If not, it displays an error message; otherwise, it shows a confirmation message to indicate successful submission.

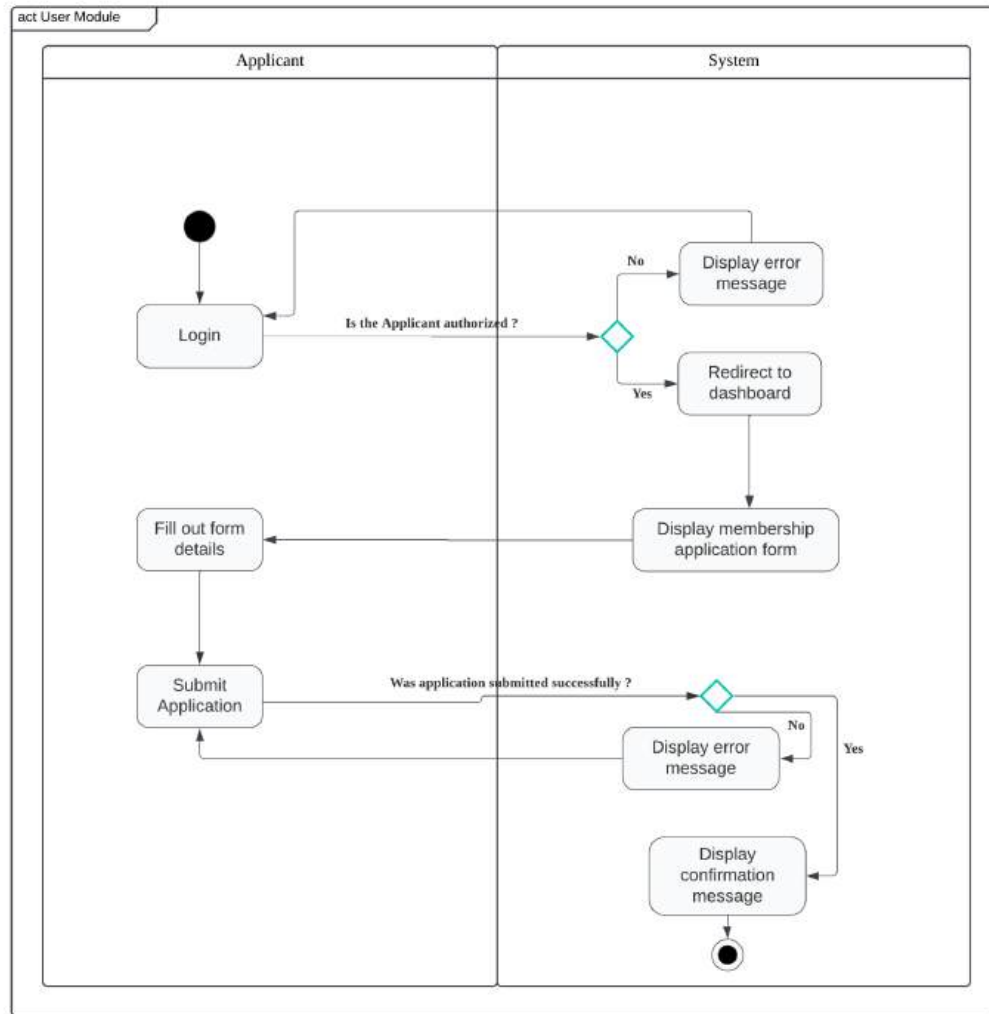


Figure 2.7 Activity Diagram for Login to Apply Membership

Figure 2.8 illustrates the applicant's login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

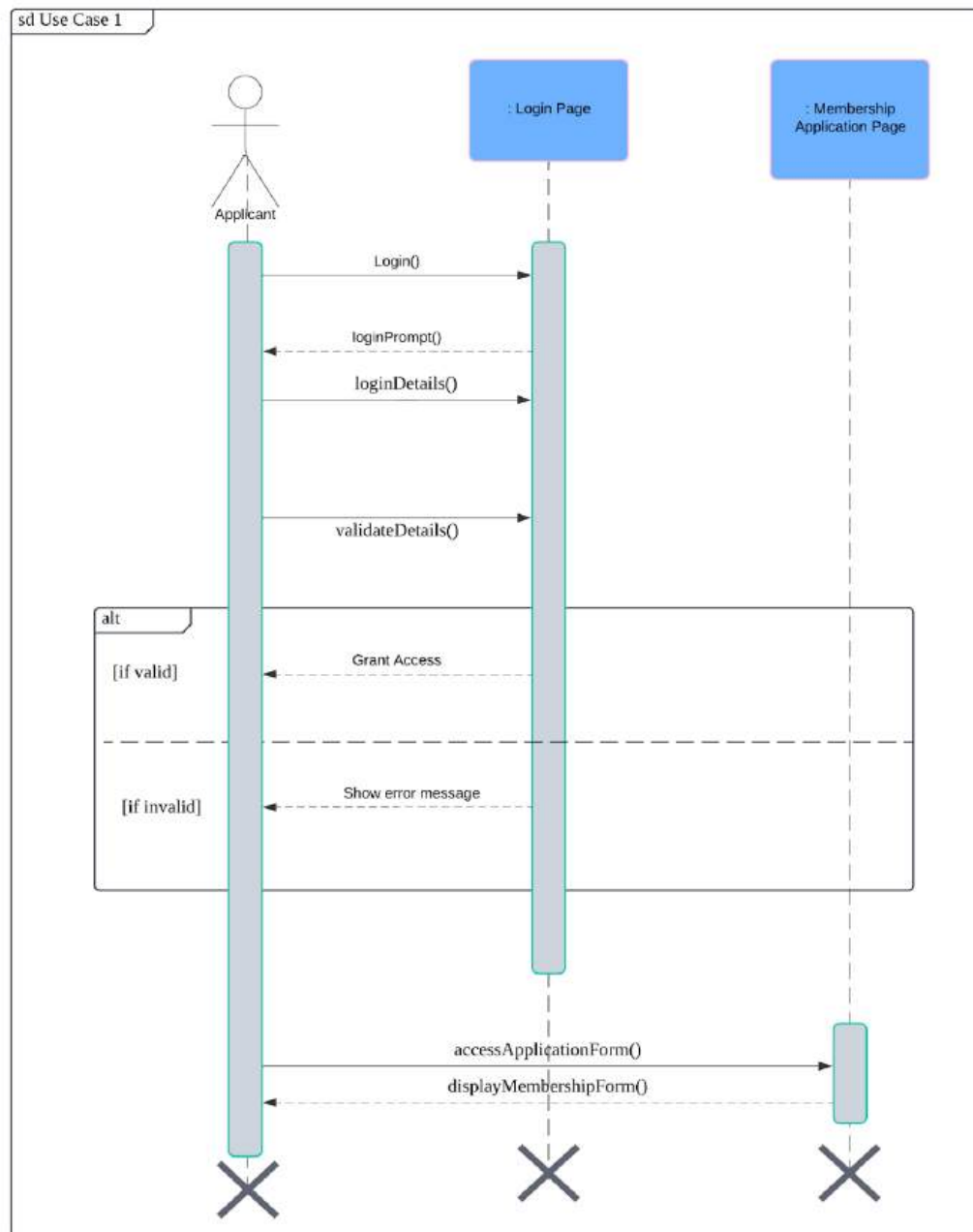


Figure 2.8: Sequence Diagram for Login to Apply Membership

2.3.2 UC002: User Case Review Applicant's Form

Table 2.3 user story description allows administrators to review multiple membership, loan, or membership termination applications in batches. The process begins with the administrator selecting a batch of applications for review. Based on the review, the admin can approve or reject the application based on the provided information. For rejected applications, the system allows the admin to provide specific reasons for rejection. Approved applications are sent to the next stage of processing.

Table 2.3: Use Case Description for Review Applicant's Form

Use Case: Review Applicant's Form
ID: UC002
User Story Description As an admin, I want to review the membership, loan and membership termination application forms submitted by applicants, So that I can verify applicant details and determine whether the forms are complete or incomplete for further processing.
Flow of events: 1. The Admin logs into the system. 2. The Admin navigates to the section containing submitted membership, loan or membership termination 3. The system displays a list of all pending applications. 4. The Admin selects an application form to review. 5. The system displays the details of the selected application. 6. The Admin reviews the information provided by the Applicant.
Alternative flow n: n.1 : If the Admin encounters an incomplete or invalid application form : 1. The Admin flags the application for follow-up and adds a note specifying the missing or incorrect information. n.2 : If the Admin needs to pause the review and continue later : 1. The Admin saves the progress of the review without finalizing the status. n.3 : If the Admin accidentally selects the wrong application : 1. The Admin uses the "Back" button to return to the list of applications.
Acceptance Criteria Postcondition : 1. The admin successfully reviews the application form and determines the validity of applicant's details. Precondition : 1. The Admin must have valid login details to access the system. 2. Applicants must have submitted their membership applications forms for review. Other conditions : 1. The system must allow the Admin to view all details of the selected application form. 2. The system must ensure that only pending applications are displayed for review.
Exception flow: 1. If the system is down, the Admin cannot access the application forms and is shown a maintenance message. 2. If an application form is incomplete or contains errors, the Admin flags it for further action.

Figure 2.9 illustrates the Review Applicant's Form for admin. The process begins with the admin logging in. If login details are incorrect, the system displays an error message;

otherwise, it grants access to the admin page. The admin chooses between the Membership List, Loan Application List, or Membership Termination List and reviews an applicant's form. If the form is complete, it is marked as reviewed; if not, it is marked as incomplete. The system then displays the current status, concluding the workflow.

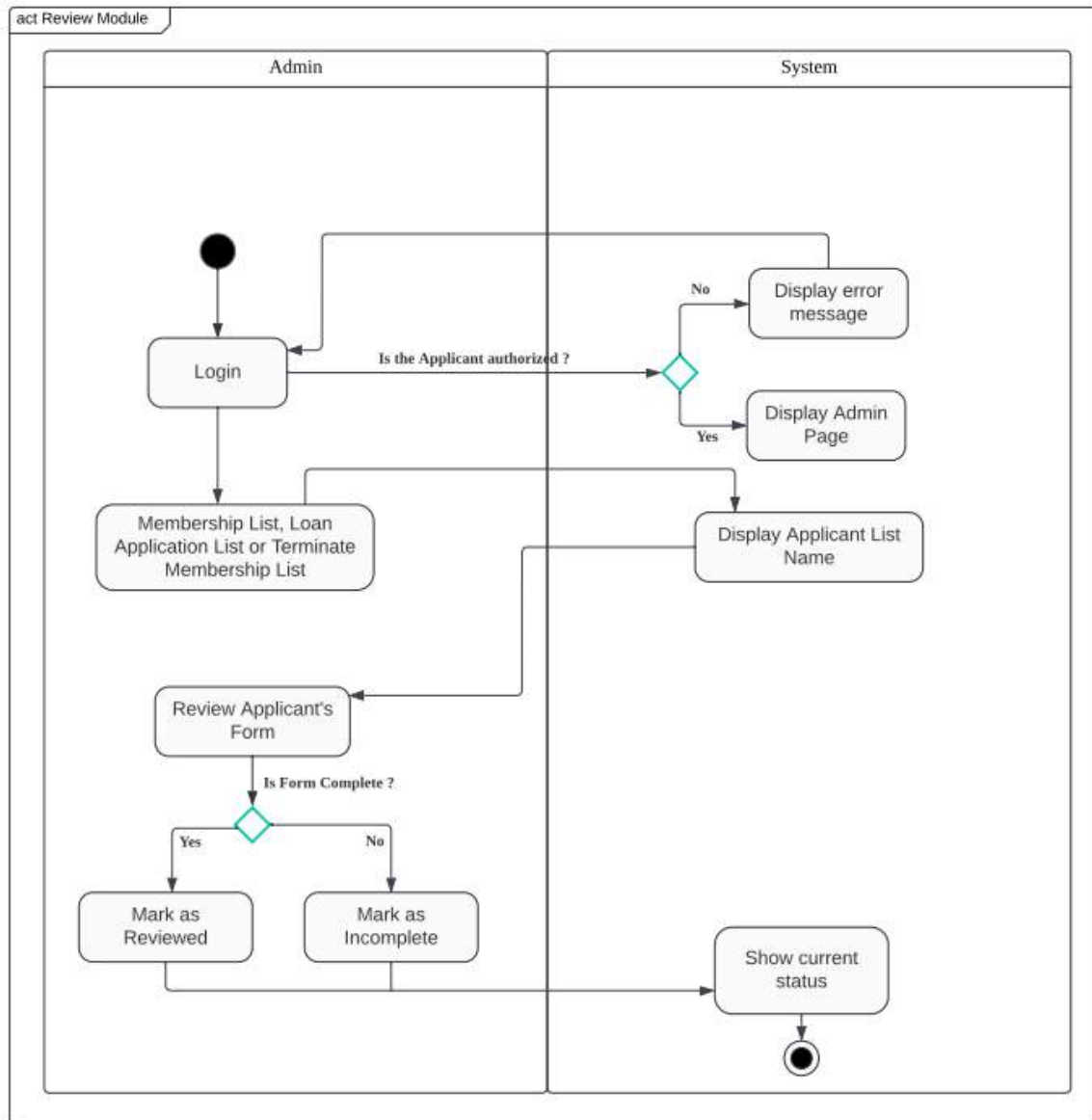


Figure 2.9: Activity Diagram for Review Applicant's Form

Figure 2.10 shows the process of an admin reviewing membership, loan and termination membership applications. The admin logs in, navigates to the application list, and views batch application details. The system loops through each application in the batch, marking it as either "reviewed" or "incomplete". A batch summary is displayed after all forms are processed.

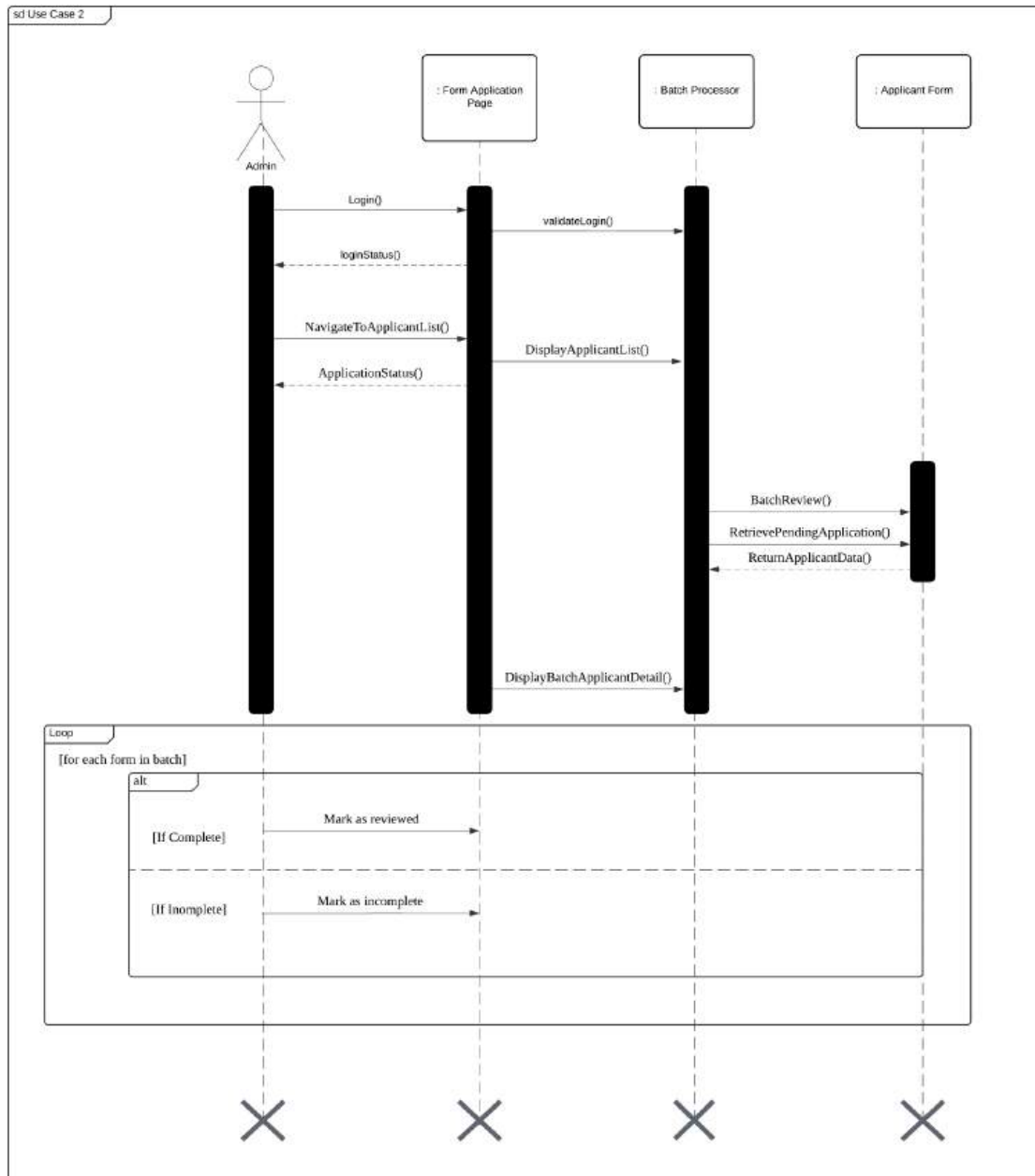


Figure 2.10: Sequence Diagram for Review Applicant's Form

2.3.3 UC003: Use Case Approve Membership

Table 2.4 describes the process of ALK in approving membership applications. ALK accesses the pending applications, assesses them and makes a decision on their approval or rejection, basing their judgement on the applicants' eligibility. If applications are incomplete, they are rejected. If further discussion is required, they are flagged for clarification. When a decision is reached, the system will update the applicant's status regarding membership and record this. The preconditions are that the admin must view the

applications beforehand and ALK would have logged on. System shall ensure applicant documents access is secure.

Table 2.4: Use Case Description for Approve Membership

Use Case: Approve Membership
ID: UC003
User Story Description As an Ahli Lembaga Koperasi (ALK), I want to approve applicants' membership , So that I can ensure that only eligible individuals can become membership and earn membership benefits
Flow of events: 1. ALK accesses the pending list of membership applications 2. ALK chooses a membership applications to evaluate 3. The ALK can choose batch processing for approval simultaneous 4. ALK makes decision to approve or reject the membership application 5. ALK chooses next membership application
Alternative flow n: n.1 : If the application status forwarded by admin is "Incomplete", system will reject immediately n.2 : If ALK requires additional discussion, the application is flagged for further clarification in meeting
Acceptance Criteria Postcondition : 1. The applicant's membership status is updated. 2. The system records the applicant's membership status. Precondition : 1. Admin has reviewed the membership applications. 2. The ALK must be logged into the system. Other conditions : 1. The system must allow ALK to view all documents uploaded by applicants
Exception flow: 1. If the system is unavailable, both admin and ALK cannot process applications and error message will be displayed

Figure 2.11 is an activity diagram for ALK's process flow for handling membership applications. It begins with ALK logging into the system. Upon successful login, access is granted, while unsuccessful attempts prompt an error message. ALK then selects the membership sections, views the pending membership applications and evaluates each

application. Depending on the evaluation result, the application is either approved or rejected and the system updates the membership status accordingly.

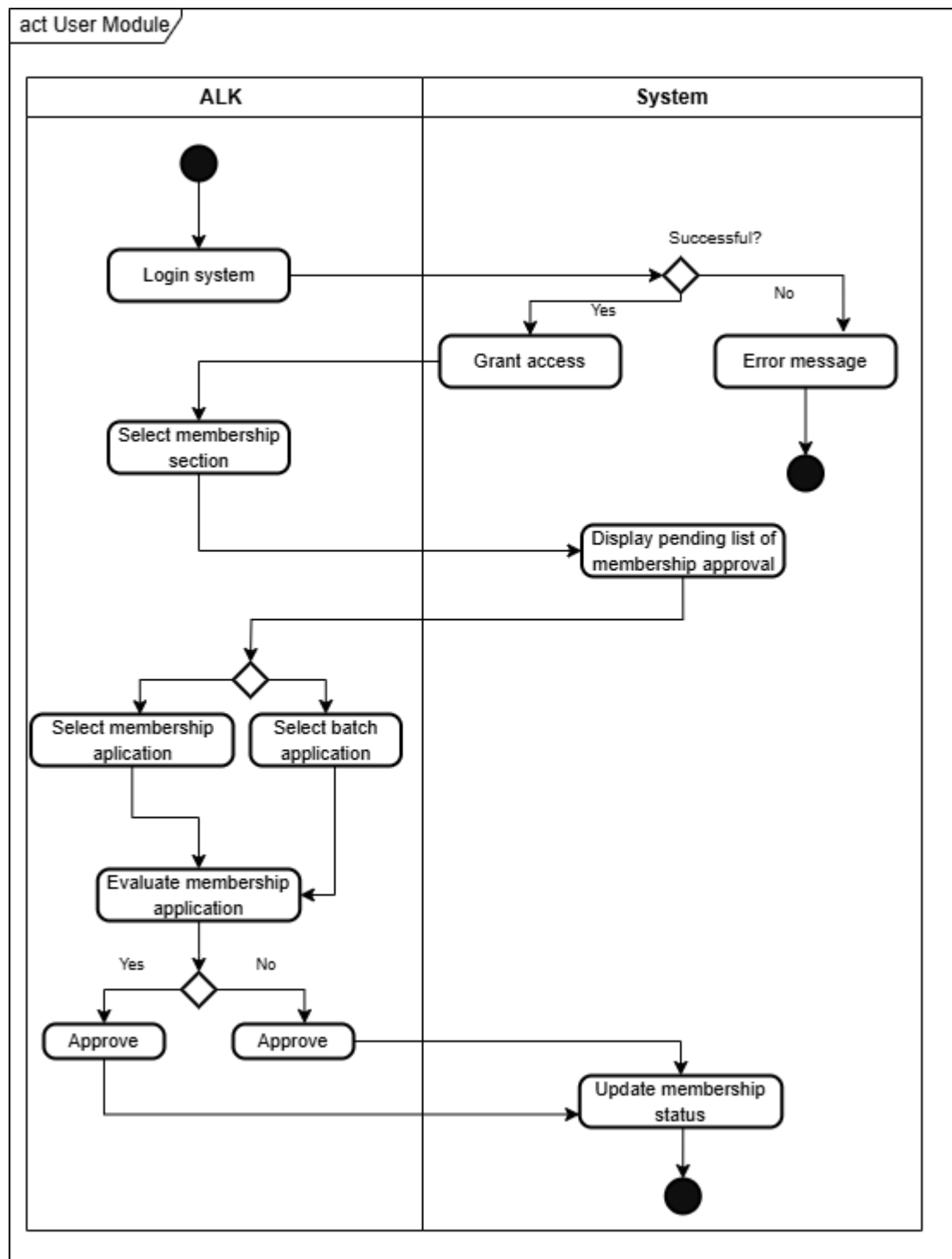


Figure 2.11: Activity Diagram for Approve Membership

Figure 2.12 shows a sequence diagram for use case approved membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any membership application from the pending list, evaluates and decides whether to approve or

not. ALK also can use batch processing to select more than one applications for approval simultaneously. Finally, the system updates the loan status

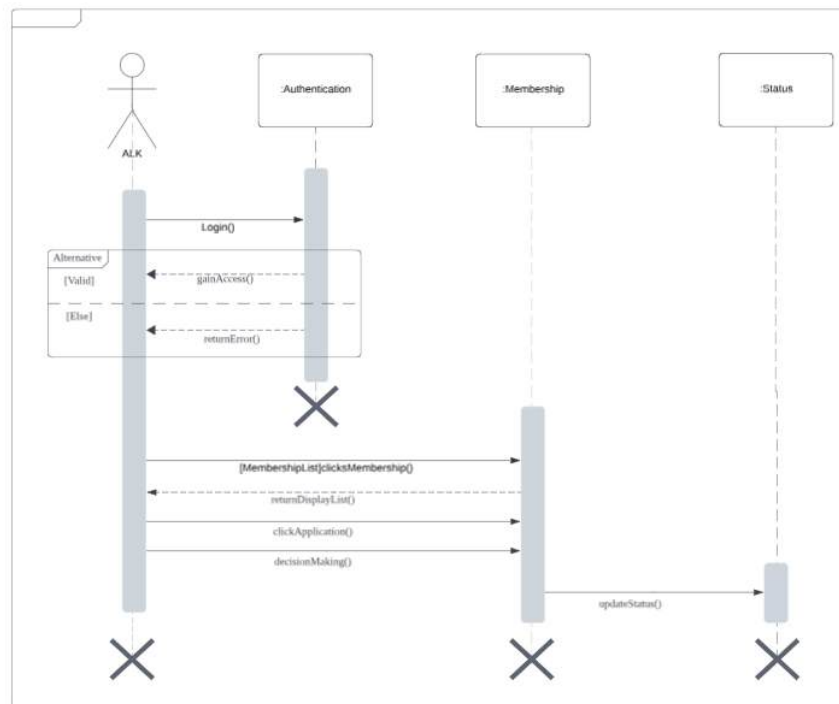


Figure 2.12: Sequence Diagram for Approve Membership

2.3.4 UC004: Use Case Apply Loan

Table 2.5 user story description emphasises on enabling members of Koperasi KADA to submit loan applications seamlessly through an online platform. The process starts with the applicant filling out the compulsory details in the loan application form. The system validates the application to ensure all required information is correct and complete once it is submitted. If the validation is successful, the application is stored in the loan database for further processing. Applicants will receive a confirmation message after the submission and can track their application status in real-time. This user story description plans for the data accuracy while aims to provide an outlined and user-friendly exposure for loan applications.

Table 2.5: Use Case Description for Apply Loan

Use Case: Apply Loan
ID: UC004
User Story Description

As an member of Koperasi KADA, I want to apply for loan through the Koperasi KADA Automated System, So that I can request financial support for my needs.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the “Apply Loan” option after logged in. 3. The system displays the loan application form. 4. The member completes the loan application form. 1. The member submits the loan application form to the system.
Alternative flow <i>n</i>: n.1 : If any required fields are empty or invalid, the system will ask the member to re-enter the correct or valid information before submit.
Acceptance Criteria Postcondition : 1. The loan application is successfully submitted and recorded in the system. 2. The member receives a confirmation notification. Precondition : 1. The member must be logged into the system. 2. The member must have an approved membership status. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system detects a problem with the member’s account, an error message is displayed and the members cannot proceed with the application.

Figure 2.13 is an activity diagram that streamlines the required process of members of Koperasi KADA to follow when submitting the loan application. The process starts with the member logging into the system where their credentials are validated. The member then selects the “Apply Loan” button once logged in and the system will display the application form. Next, the member fills in the compulsory details. If the details are complete and valid, the system shows the loan application status and notifies the member upon successful submission. If there are errors, users are required to correct and resubmit the form while the system displays an error message. The activity ends when the loan application is submitted successfully.

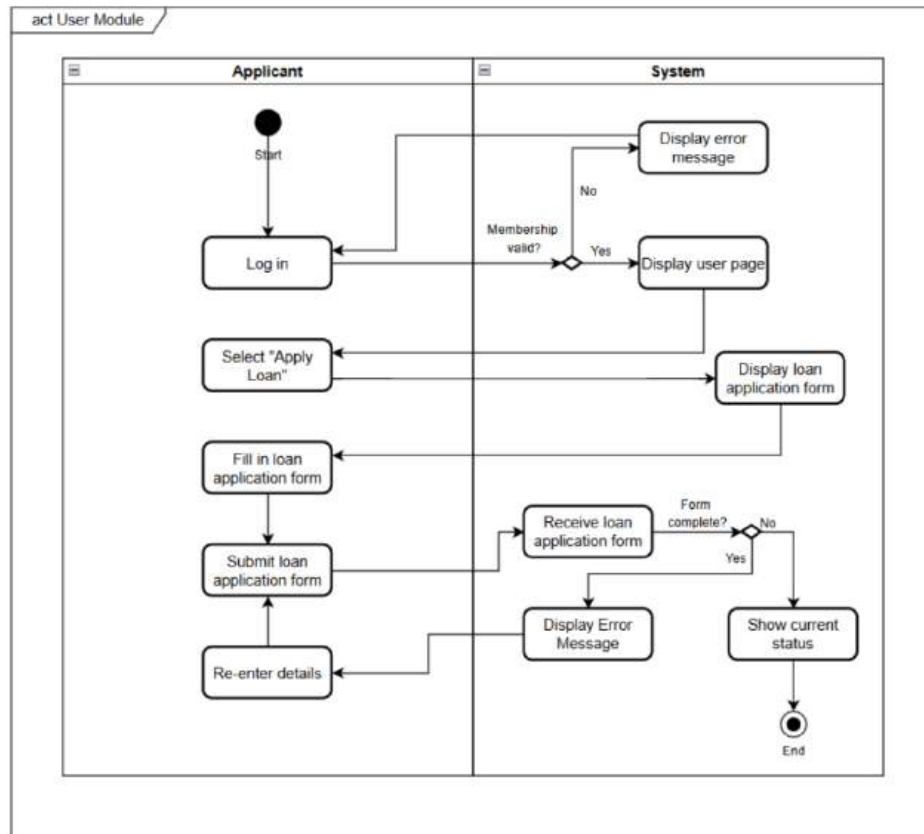


Figure 2.13: Sequence Diagram for Apply Loan

Figure 2.14 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system's database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

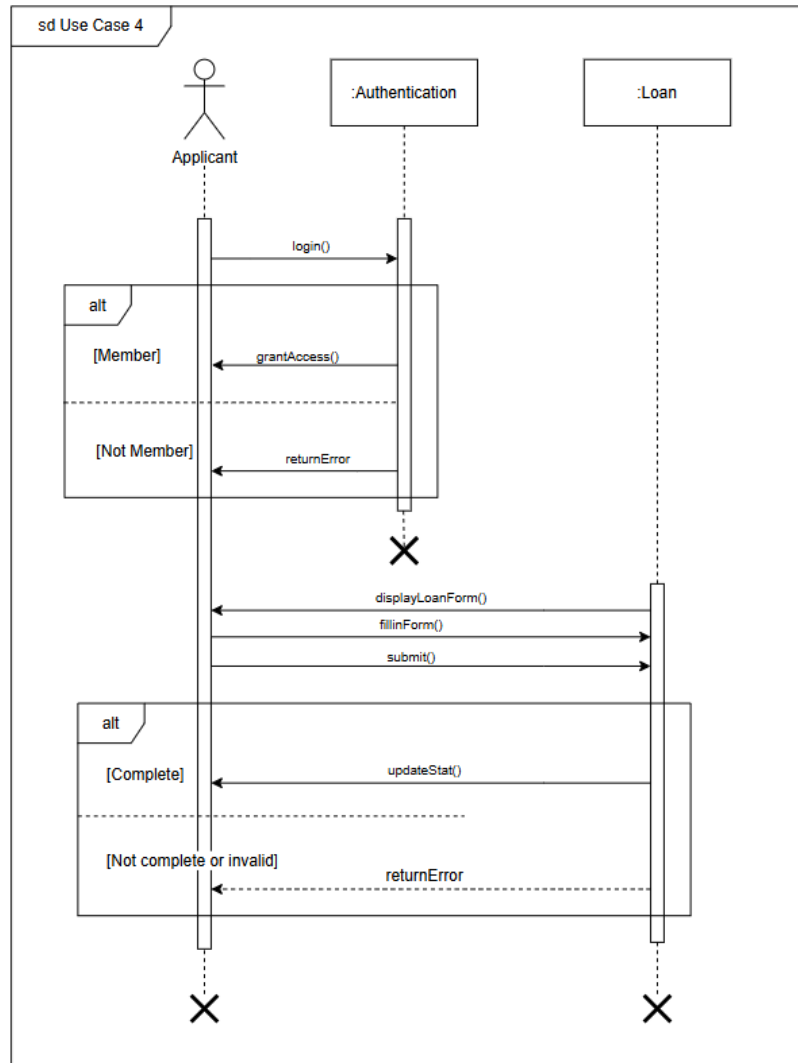


Figure 2.14: Sequence Diagram for Apply Loan

2.3.5 UC005: Use Case Calculate Loan

Table 2.6 focuses on the members of Koperasi KADA who want to calculate their loan repayment and dividend returns to know about their financial commitments and benefits. The table outlines the process that includes the member logging into the system, selects the “Loan Calculator” button features, enters the loan details like amount, duration and interest rate. Then, click the “Calculate” button and the system validates the inputs. If valid, the system displays the repayment results. If invalid, it shows an error message.

Table 2.6: Use Case Description for Calculate Loan

Use Case: Calculate Loan
ID: UC005

User Story Description As an member of Koperasi KADA, I want to calculate my loan repayment and dividend returns, So that I know about my financial commitments and also benefits.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the “Loan Calculator” option after logged in. 3. The system displays the loan calculator page. 4. The member enters the loan desired amount, duration. 5. The member chooses the interest rate from the list of chosen loan types. 6. The member clicks on the “Calculate” button.
Alternative flow <i>n</i>: n.1 : If the member enters invalid amount, an error message will be displayed.
Acceptance Criteria Postcondition : 1. The system calculates and displays the amount of loan to the member. Precondition : 1. The member must be logged into the system. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system is unable to process calculations, an error message will be displayed.

Figure 2.15 is an activity diagram for calculating loans to determine their loan repayment and dividend returns that require the steps a member should follow. The process begins with the member logging into the system, where their details are then validated. The member selects the “Loan Calculator” button and the system displays the loan calculator form. The members click the “Calculate” after entering the necessary loan details like amount, duration and interest rate. The system validates the input and processes the calculation. If the inputs are valid, the system displays the calculated repayment. If invalid, an error message is displayed that requires the member to reenter the details. The activity ends when the member gets the calculated results.

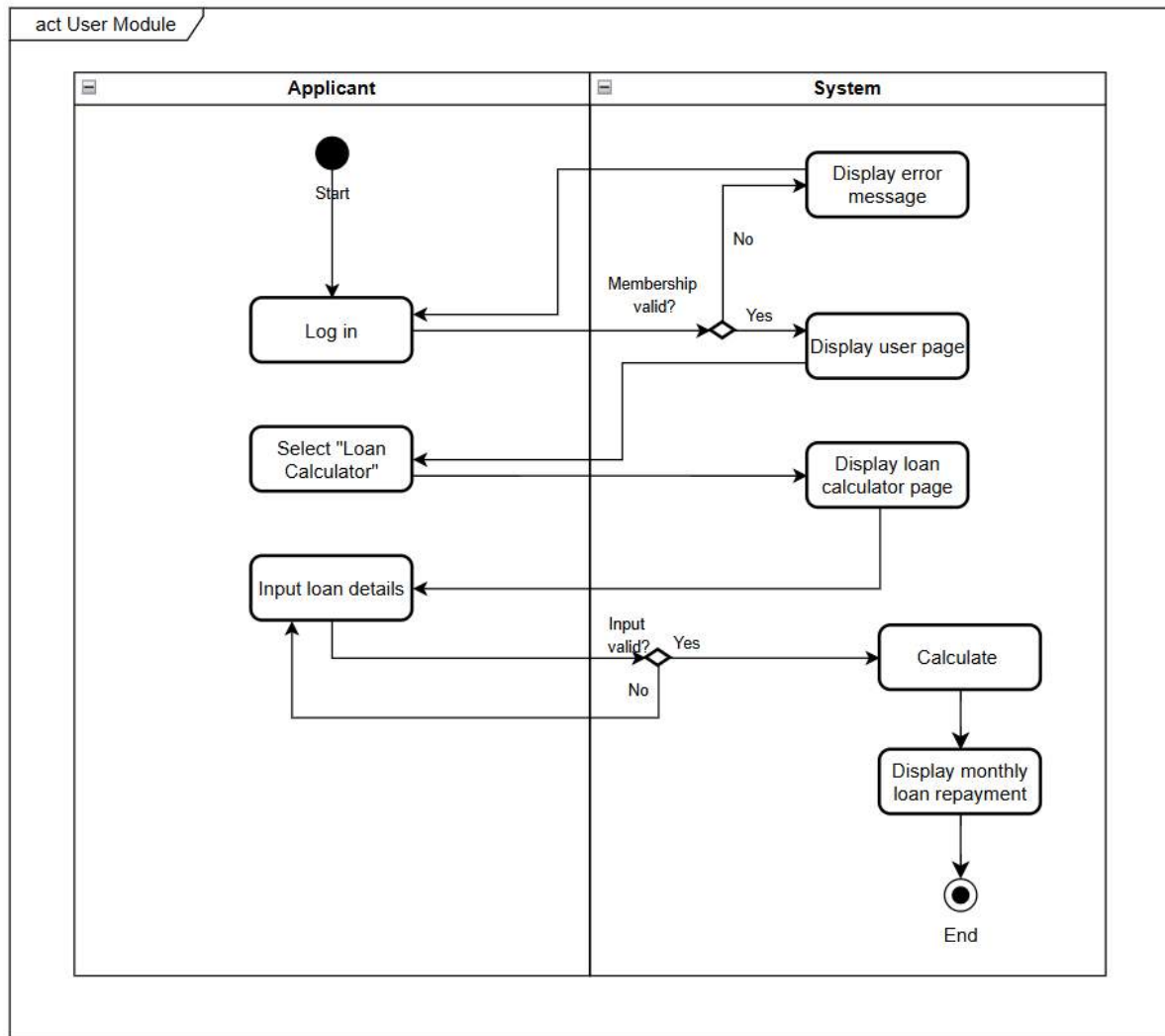


Figure 2.15: Sequence Diagram for Calculate Loan

Figure 2.16 is a sequence diagram for calculating loans that highlights the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects the "Loan Calculator" button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

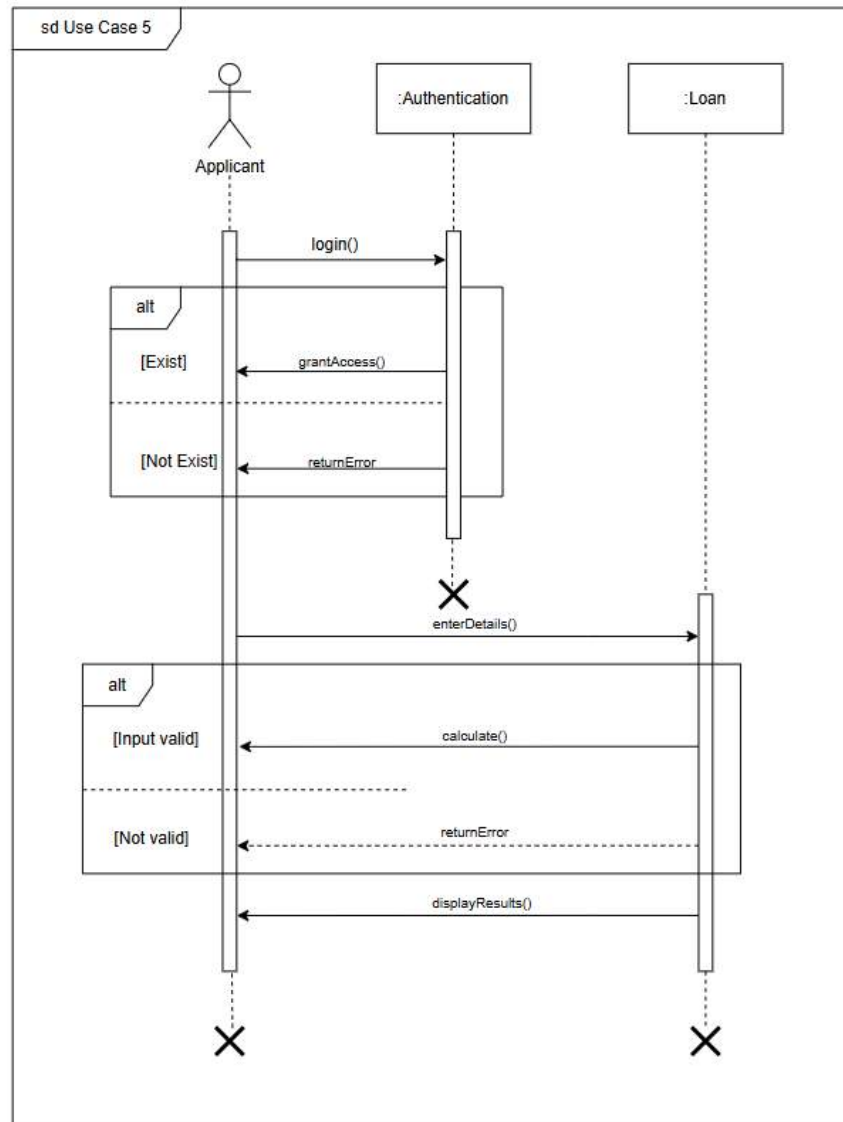


Figure 2.16: Sequence Diagram for Calculate Loan

2.3.6 UC006: Use Case Approve Loan Application

Table 2.7 shows a use case description of the procedure of loan application approval by ALK. The system is viewed for pending applications by ALK, evaluates applications based on the company's policies, after which a decision to either approve, reject or use the compare feature before finalizing. Then, ALK can proceed to the next application or log out. Alternative flows include rejecting incomplete applications reviewed by admin or postponing decisions. Postconditions ensure the loan status is updated, recorded and applicants notified. Preconditions required admin's prior review and ALK's login. Exception flows address system downtime or missing data, returning applications to admin for clarification.

Table 2.7: Use Case Description for Approve Loan Application

Use Case ID: UC006
User Story Description: As an Ahli Lembaga Koperasi I want to approve loan application So that i can let only the eligible applicants to receive loan in line with company's policies
Flow of Events: 1. The ALK staff member accesses the list of pending loan applications. 2. The ALK selects a specific loan application to review the applicant's details 3. The ALK can choose batch processing for approval simultaneous 4. The ALK staff member evaluates the application based on established criteria. 5. The ALK makes a decision to either directly approve or reject or use the compare button to compare the applicant's form with loan terms before approving or rejecting the loan form. 6. The ALK chooses the next application or log out of the system.
Alternative Flow n: n.1 : If the application is marked "Incomplete" by Admin, ALK will reject immediately n.2 : If ALK need more time to review particular application, ALK will press press back button and do decision later
Acceptance Criteria: Postcondition: 1. Loan application status is updated 2. The system record the loan application status 3. Applicants are notified of the status Precondition: 1. The Admin has reviewed the loan applications 2. ALK must have logged into the system Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel
Exception Flow: 1. If the system is down, ALK cannot evaluate the applications, causing delays 2. If ALK cannot make a decision due to missing data, the application is sent back to the Admin for further discussion

Figure 2.17 illustrates the activity diagram for approving a loan application. The process begins with a user logging into the system. If the login is successful, the user is granted access to proceed, otherwise, an error message is displayed. The user selects the loan section, which displays a pending list of loan applications. After selecting a specific loan application, the user can choose to compare it with others using the compare button, which triggers the comparison process and displays the results. ALK also can utilize batch processing which is selecting more than one applications to approve or reject

simultaneously. Next, the application is evaluated, leading to either approval or rejection. Finally, the loan status is updated to reflect the decision.

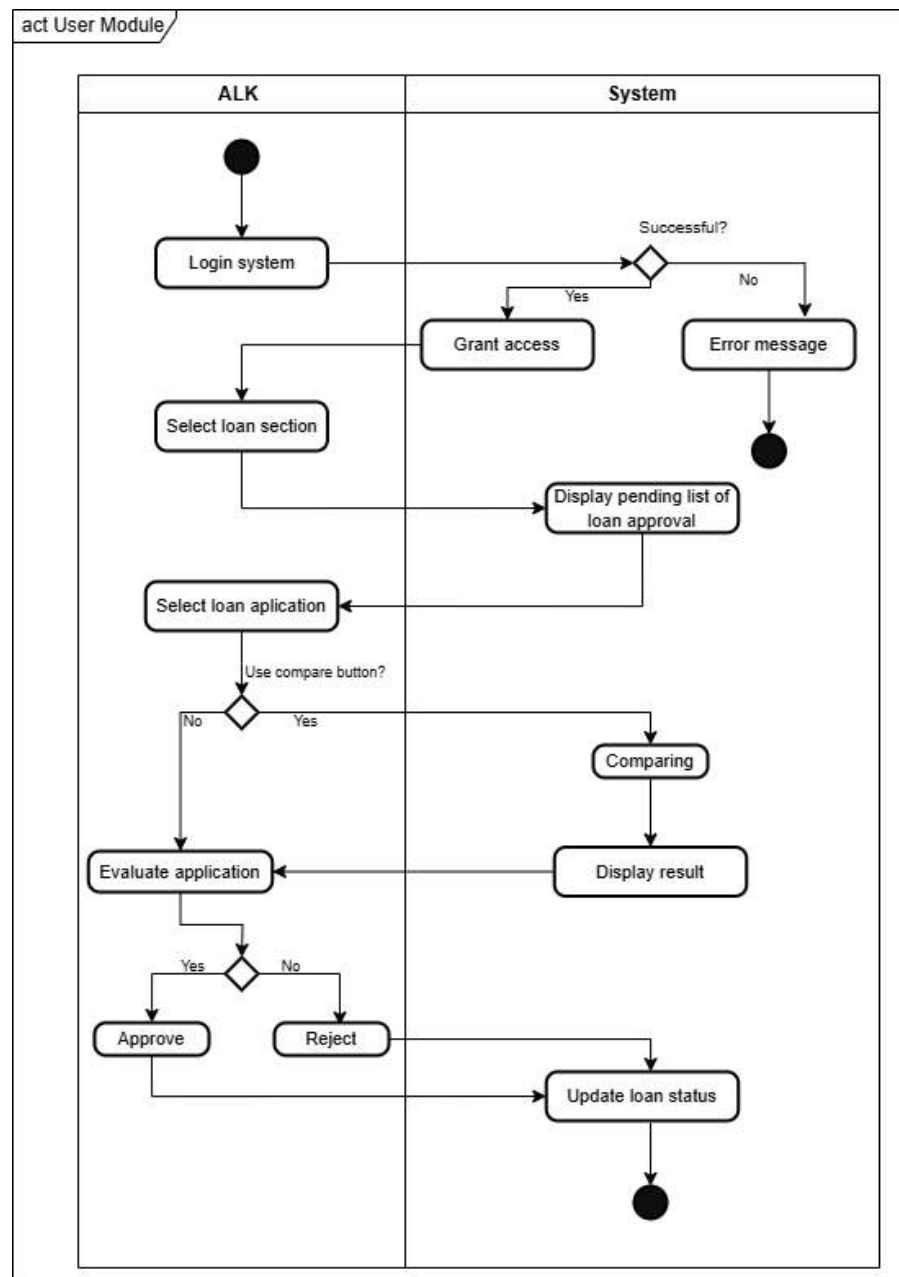


Figure 2.17: Sequence Diagram for Approve Loan Application

Figure 2.18 shows a sequence diagram for use case approved loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list or utilizes batch processing, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision. Finally, the system updates the loan status.

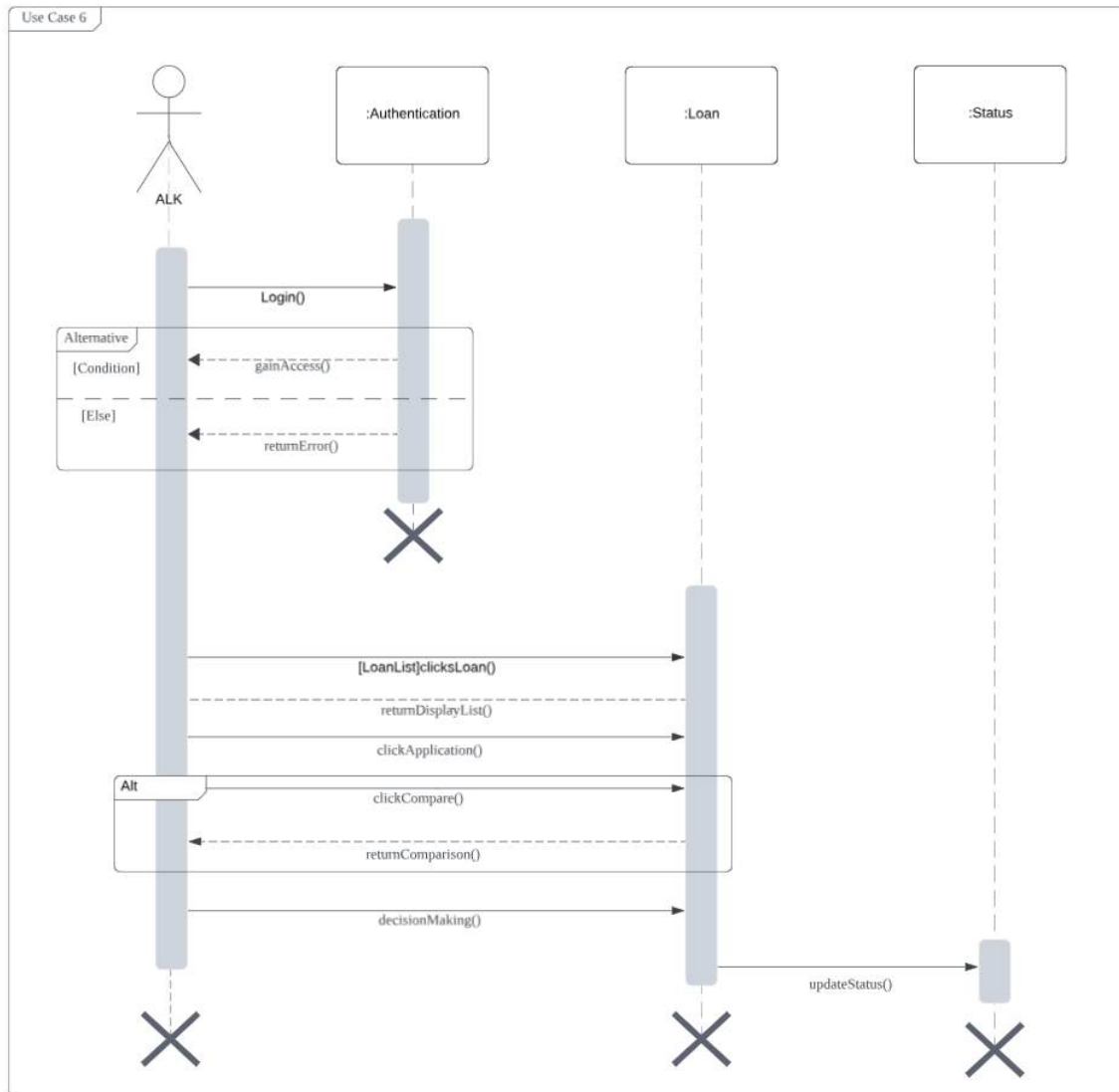


Figure 2.18: Sequence Diagram for Approve Loan Application

2.3.7 UC007: Use Case Prepare Statement and Report

Table 2.8 provides a detailed description of the use case for the Prepare Statement and Report feature, which describes how an admin generates statements and reports to provide applicants with details about their application status, loan history, and membership information. The admin can choose between a financial statement or a corporation report. The system retrieves relevant data from the database and processes the report based on the selected member. The generated report is then saved for future reference. If the database is unavailable or a system error occurs, the admin will be notified and given options to retry or troubleshoot the issue.

Table 2.8: Use Case Description for Prepare Statement and Report

Use Case: Prepare Statement and Report
ID: UC007
User Story Description As an admin, I want prepare statements and reports for applicants, So that I can provide them with detailed information about their application status, loan history, and membership details. Flow of events: 1. Select whether the report is for financial statement or corporation report 2. If the report is for generated statement: 2.1. Admin retrieves data for approved member and loan. 2.2. The system fetches the relevant applicant data from the database. 2.3. Admin clicks the "Generate Statement" button to initiate the statement preparation process. 2.4. The system prepares the statement report based on the retrieved data. 2.5. The system displays successful report generation. 3. If the report is for corporation: 3.1. The system fetches the relevant information about membership applications, loan applications and terminate membership applications. 3.2. The system displays the information based on monthly and yearly views. Acceptance Criteria Postcondition : 1. The statement or report is successfully generated and saved in the system for future reference. Precondition : 1. Admin must be logged into the system with the appropriate permissions. 2. Required data for the statement and report must be accessible within the system. Other conditions : 1. The system must have sufficient access control to ensure only authorized admins can generate reports. 2. The admin must be able to view and select the specific applicant for whom the statement or report is to be generated. Exception flow: 1. If the database is temporarily unavailable, the system will display a message indicating the issue and allow the admin to retry the process later. 2. If the system fails to generate the statement or report due to system errors, the admin will be notified with a detailed error message and instructed on the next steps.

Figure 2.19 shows the activity diagram for how an admin uses the system to create statements for approved loan applicants. The process starts with the admin logging into the

system. If the admin is authorized, they can choose to view "Laporan Koperasi" for corporate reports or "Penyata Kewangan" for financial statements. The system then shows the needed data, like a list of approved members. To generate a statement, the admin clicks "Hasilkan", and the system prepares the report. If the process is successful, the statement is displayed. If there is a problem, the system shows an error message so the admin can try again.

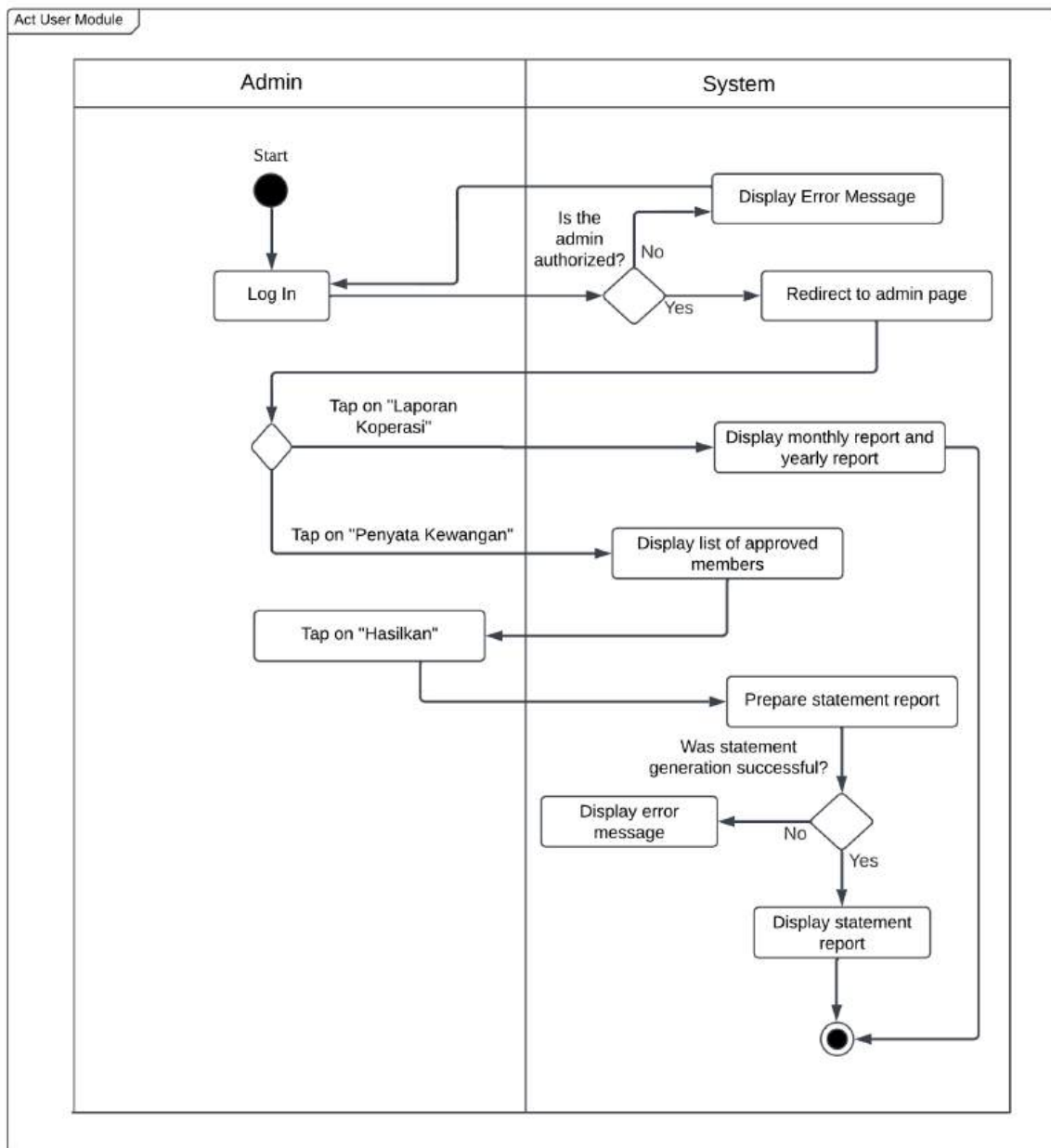


Figure 2.19: Sequence Diagram for Prepare Statement and Report

Figure 2.20 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems.

The admin begins by logging in, where successful authentication grants access to the admin page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

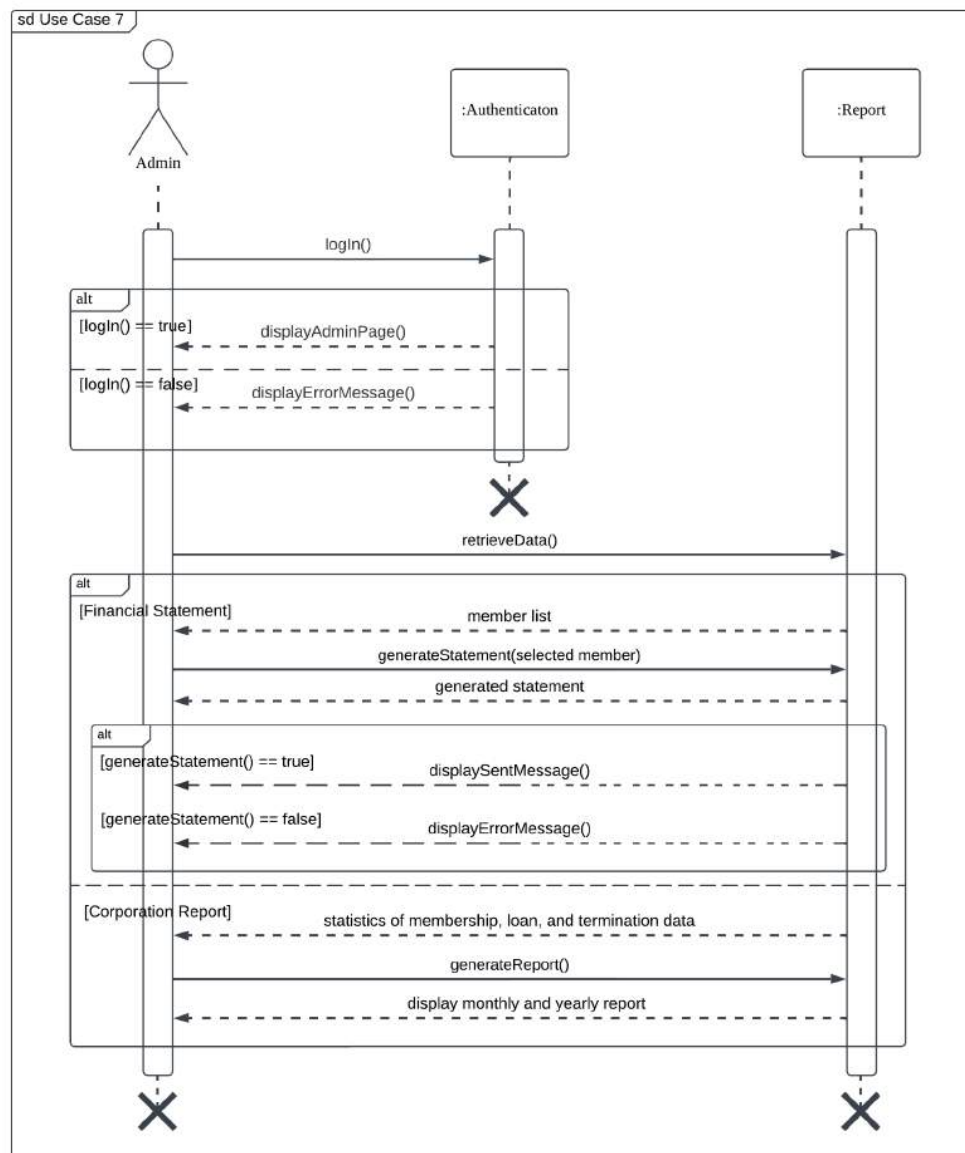


Figure 2.20: Sequence Diagram for Prepare Statement and Report

2.3.8 UC008: Use Case Notify via Email

Table 2.9 describes the Notify via Email use case, where the admin reviews prepared statements and sends notification emails to applicants. The system allows the admin to

select whether the email notification is for a generated statement or an updated application status. Once selected, the system generates and sends the appropriate email, logging the status for tracking purposes. The system also supports batch processing for multiple applicants and handles exceptions such as failed email generation or invalid email addresses by notifying the admin for corrective actions.

Table 2.9: Use Case Description for Notify via Email

Use Case: Notify via Email
ID: UC008
User Story Description As an admin, I want to review prepared statements and send notification emails to applicant, So that I can ensure the applicants are informed about their loan details.
Flow of events: 1. Select whether the email notification is for generated statement or states notification. 2. If the notification is for generated statement: 2.1. The system detects that a loan statement and report have been prepared. 2.2. The Admin reviews the prepared statement and clicks the "Send" button. 2.3. The system generates a notification email with statement details. 2.4. The system sends the email to the selected members. 2.5. The system logs the email status in the admin interface. 3. If the transaction is irregular: 3.1. The Admin tap on "Send" at desired applicants. 3.2. The system generates an email notification based on the updated status. 3.3. The system sends the email to the selected applicants.
Acceptance Criteria Postcondition : 1. Recipients are notified via email, with access to the statement or report as needed. Precondition : 1. The system must have a list of recipients with valid email addresses. 2. The statement or report has been prepared in the system. Other conditions : 1. The system must allow the Admin to view all details of the selected application form before sending the notification email.
Alternative Flow 1: 1. If the Admin wants to batch processing for application status or generated statement. 2. Admin selected and processes multiple applicants at once. 3. The system should then send email notifications to each applicant.
Exception flow:

1. If the system fails to generate a notification email, it will notify the admin of the failure and allow the admin to retry sending the email.
2. If the email address is invalid, the system will prompt the admin to correct the address before retrying the process.

Figure 2.21 shows an activity diagram for the steps an how an admin sends email notifications about financial statements and the status of applicant's membership, loan or terminate applications. The process starts when the admin logs into the system. If authorized, they can open "Penyata Kewangan" to view list of koperasi's members or select "Penghantaran Status" to view a list of approved and rejected applicants. After selecting a member, the system displays the statement report. The admin then clicks "Hantar" to send the email. If the email is sent successfully, a confirmation message appears. If there is an issue, the system shows an error message so the admin can take action.

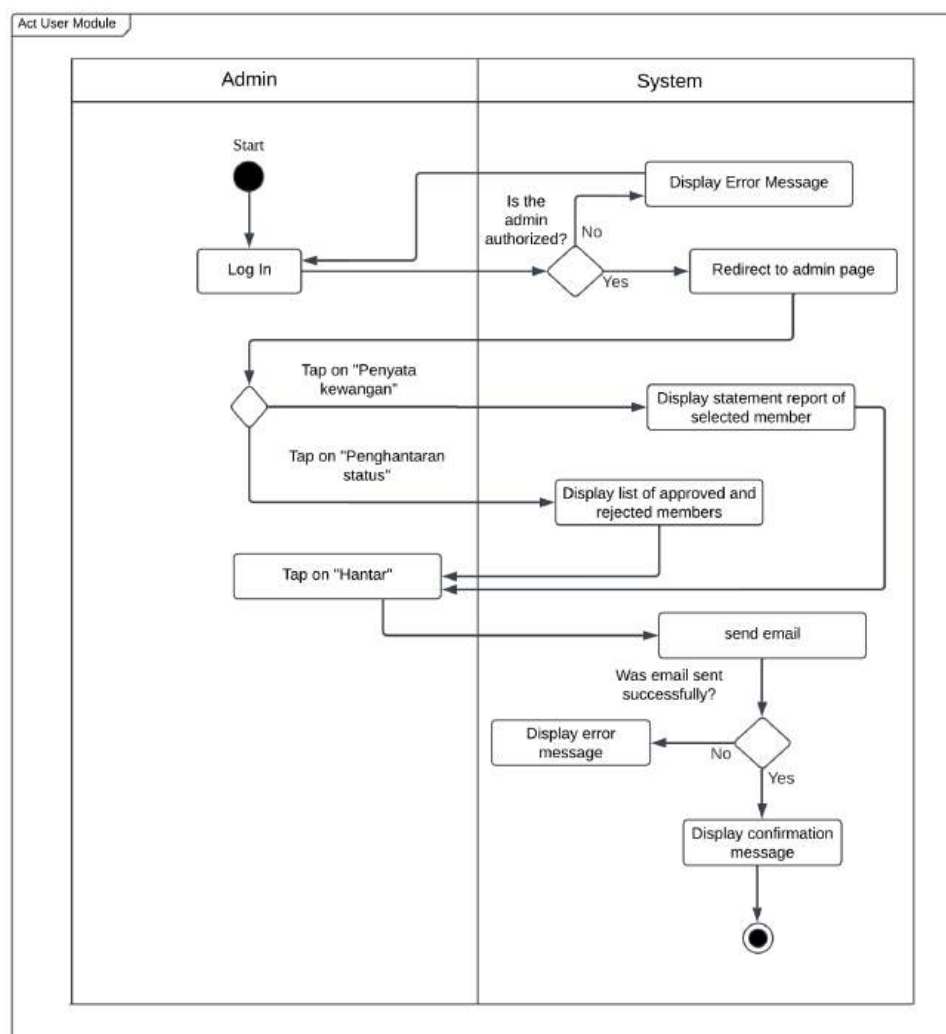


Figure 2.21: Sequence Diagram for Notify via Email

Figure 2.22 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

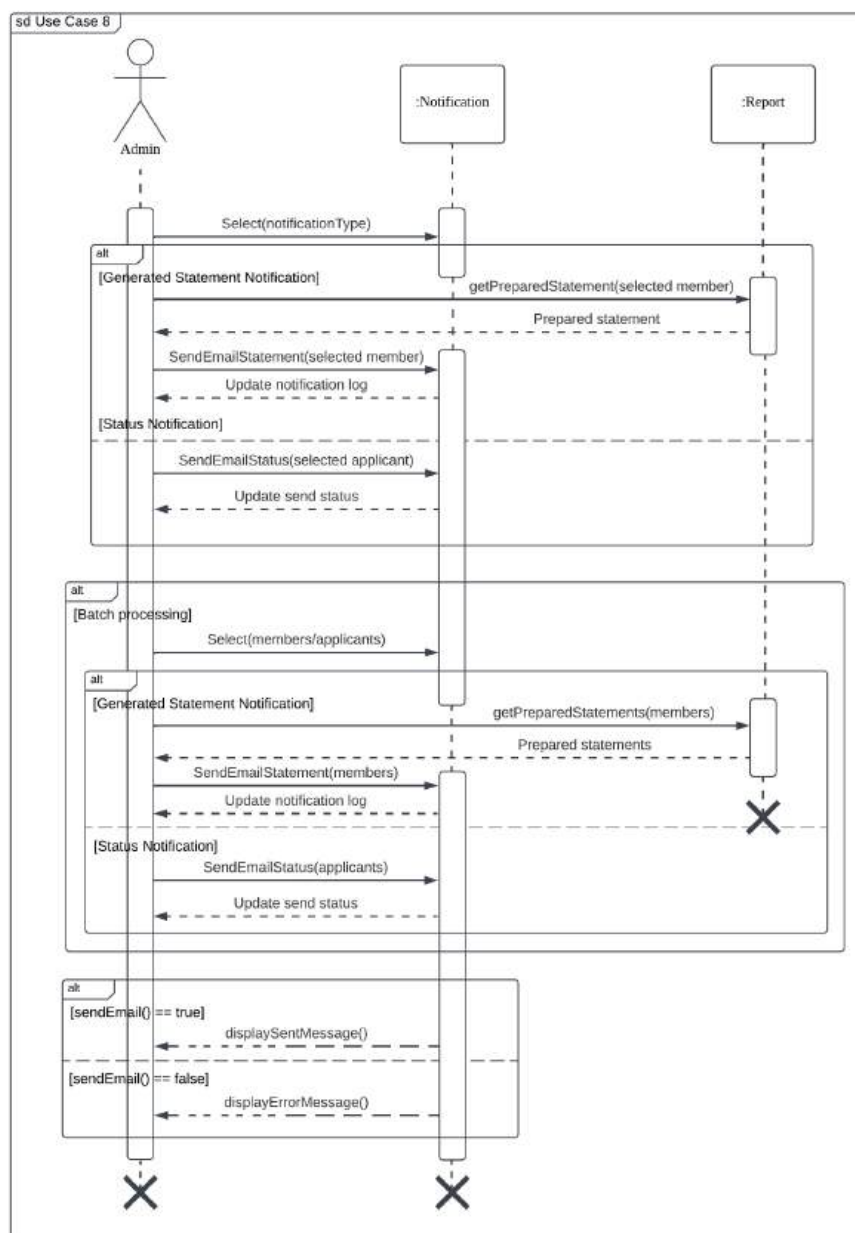


Figure 2.22: Sequence Diagram for Notify via Email

2.3.9 UC009: Use Case Manage Transaction

Table 2.10 describes the Manage Transaction use case. This use case explains how an admin handles and records savings and loan repayment transactions to keep financial records correct and updated. The admin can choose between monthly or irregular transactions. For monthly transactions, the admin selects a member and processes the update. For irregular transactions, the admin fills out a form and submits it. The system also allows batch processing for multiple monthly transactions. If there are missing details, incorrect member information, or system errors, the system will show a warning and prevent the transaction until fixed.

Table 2.10: Use Case Description for Manage Transactions

Use Case: Manage Transactions
ID: UC009
User Story Description As an admin, I want to manage and record all savings and loan repayment transactions, whether on a monthly or irregular basis, So that I can ensure the financial records are accurate and up to date.
Flow of events: 1. Select whether the transaction is monthly or irregular. 2. If the transaction is monthly: 2.1. Choose the member involved in the transaction and click "Keluarkan". 2.2. The system updates financial transactions. 3. If the transaction is irregular: 3.1. Select Savings or Loan Repayment. 3.2. Fill in the transaction form. 3.3. Submit the form for processing.
Acceptance Criteria Postcondition : 1. The transaction is successfully recorded in the system. Precondition : 1. The admin is logged into the system. 2. The membership application status is approved in the system. Other conditions : 1. Only authorized administrators can process transactions.
Alternative Flow 1: 1. If the Admin wants to batch processing for monthly transactions 2. Admin selected and processes multiple monthly transactions at once. 3. The transactions are recorded and updated.
Exception flow: 1. If required fields are missing in the transaction form, the system prompts an error. 2. If the selected member does not exist, the system prevents transaction processing.

3. If there is a system error, the transaction is logged for review.

Figure 2.23 illustrates the activity diagram for an admin to manage financial transactions. The process begins when the admin logs in. If authorized, they can access the "Transaksi" section and choose between "Bulanan" (monthly) or "Tidak Tetap" (non-fixed) transactions. If selecting "Tidak Tetap," the system displays a transaction form for the admin to fill in. After completing the form, they tap "Hantar Transaksi" to proceed. A receipt form appears, which must also be completed. Once the admin taps "OK," the system updates the transaction. If the update is successful, a confirmation message is displayed. If there is an issue, an error message appears, requiring further action.

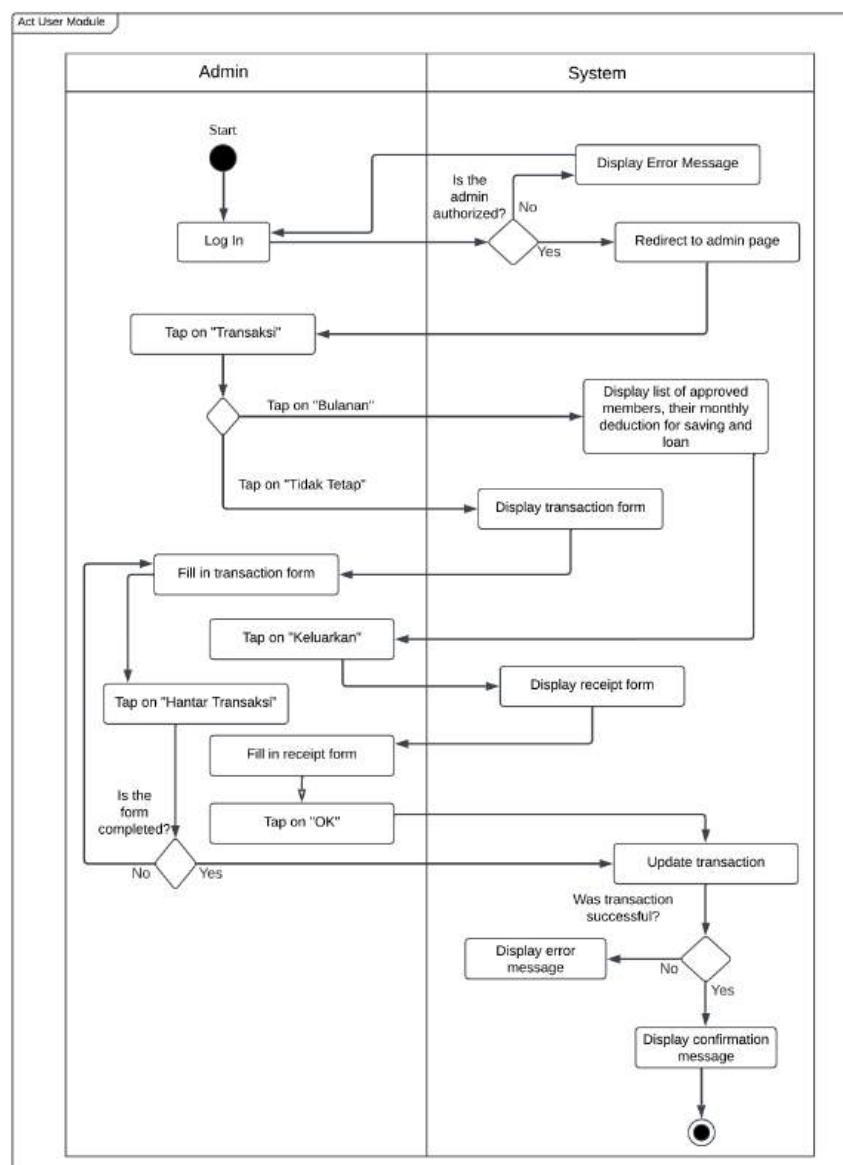


Figure 2.23: Activity Diagram for Manage Transactions

Figure 2.24 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

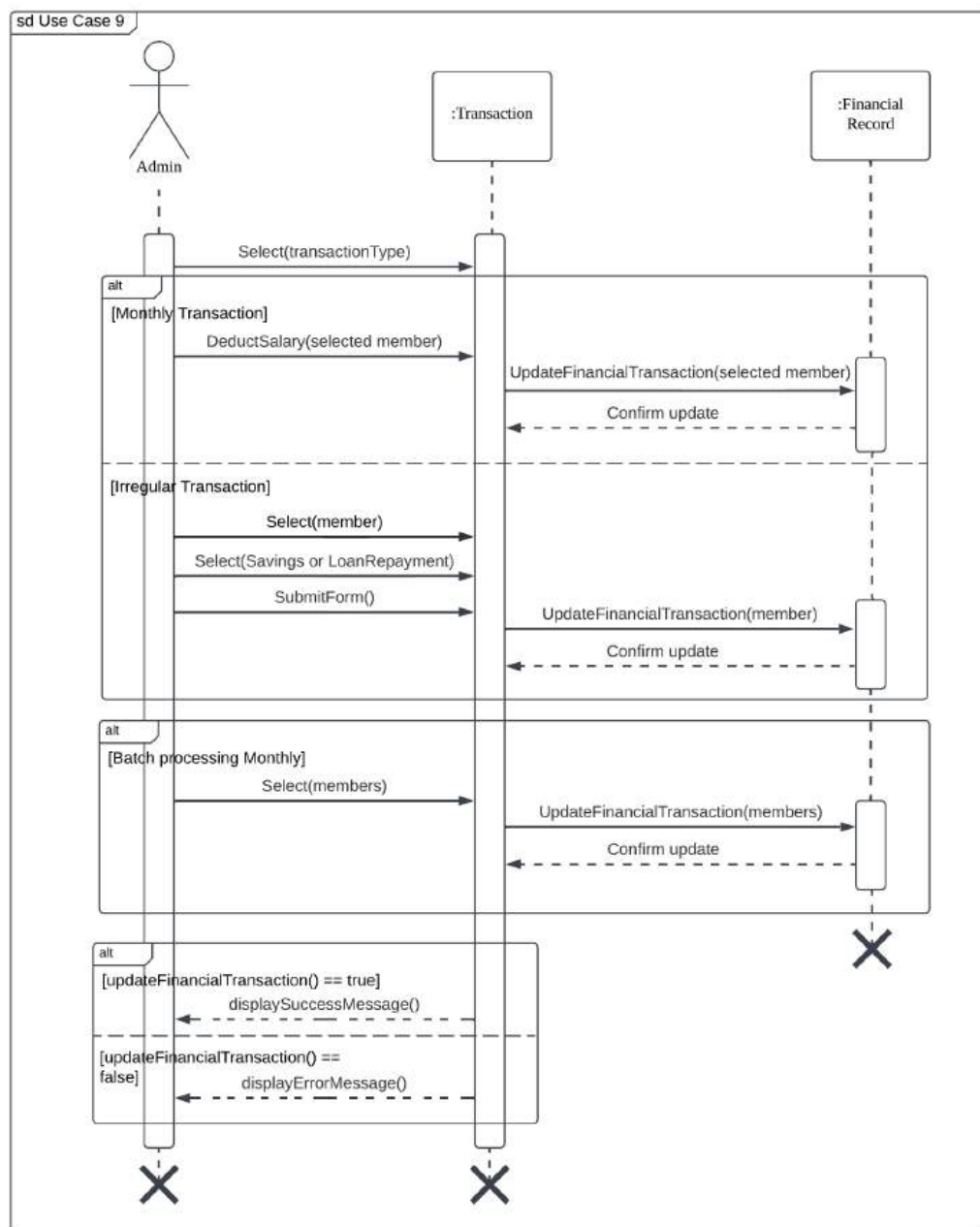


Figure 2.24: Sequence Diagram for Manage Transactions

2.3.10 UC010: Use Case Terminate Membership

Table 2.11 is a use case description that enables an Applicant to request termination of the Koperasi KADA membership. The Applicant submits the required details through the system, which validates the request and then forwards it to the Admin for review. Once they submit the form, the status of the application can be displayed on the status page.

Table 2.11: Use Case Description for Terminate Membership

Use Case ID: UC010
User Story Description: As an Applicant I want to terminate my membership So that i can discontinue my association with the cooperative
Flow of Events: 1. The Applicant logs into the system. 2. The Applicant navigates to the membership termination session. 3. The Applicant submits a termination request by providing the required details. 4. The system validates the termination request for completeness. 5. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked “Incomplete” by Admin, the system notifies the Applicant to resubmit with all required details.
Acceptance Criteria: Postcondition: 1. The membership termination request is processed. 2. The Applicant is notified of the outcome. Precondition: 1. The Applicant has an active membership. 2. The Applicant is logged into the system. Other condition: 1. All applicants’ data must be encrypted and only accessible to authorized personnel.
Exception Flow: 1. If the system is down, the Applicant cannot submit the termination request, causing delays.

Figure 2.25 is an activity diagram for terminate membership that outlines the interaction between the Applicant and the System. The Applicant logs into the system, navigates to the termination section and submits a termination request. The System then validates the request for completeness. If the request is incomplete, the Applicant is notified to resubmit. If complete, the request is forwarded to the Admin for review.

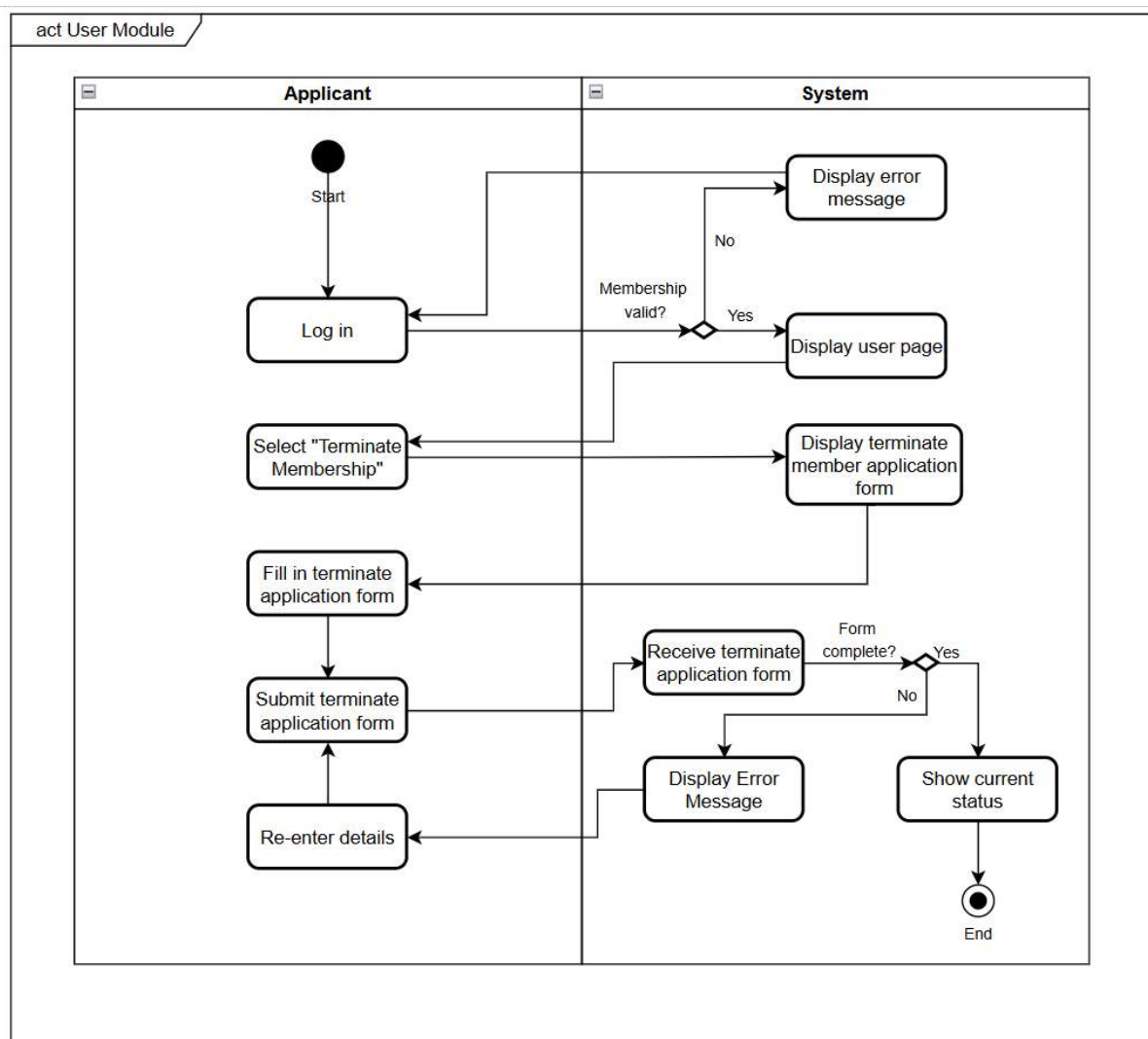


Figure 2.25: Activity Diagram for Terminate Membership

Figure 2.26 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

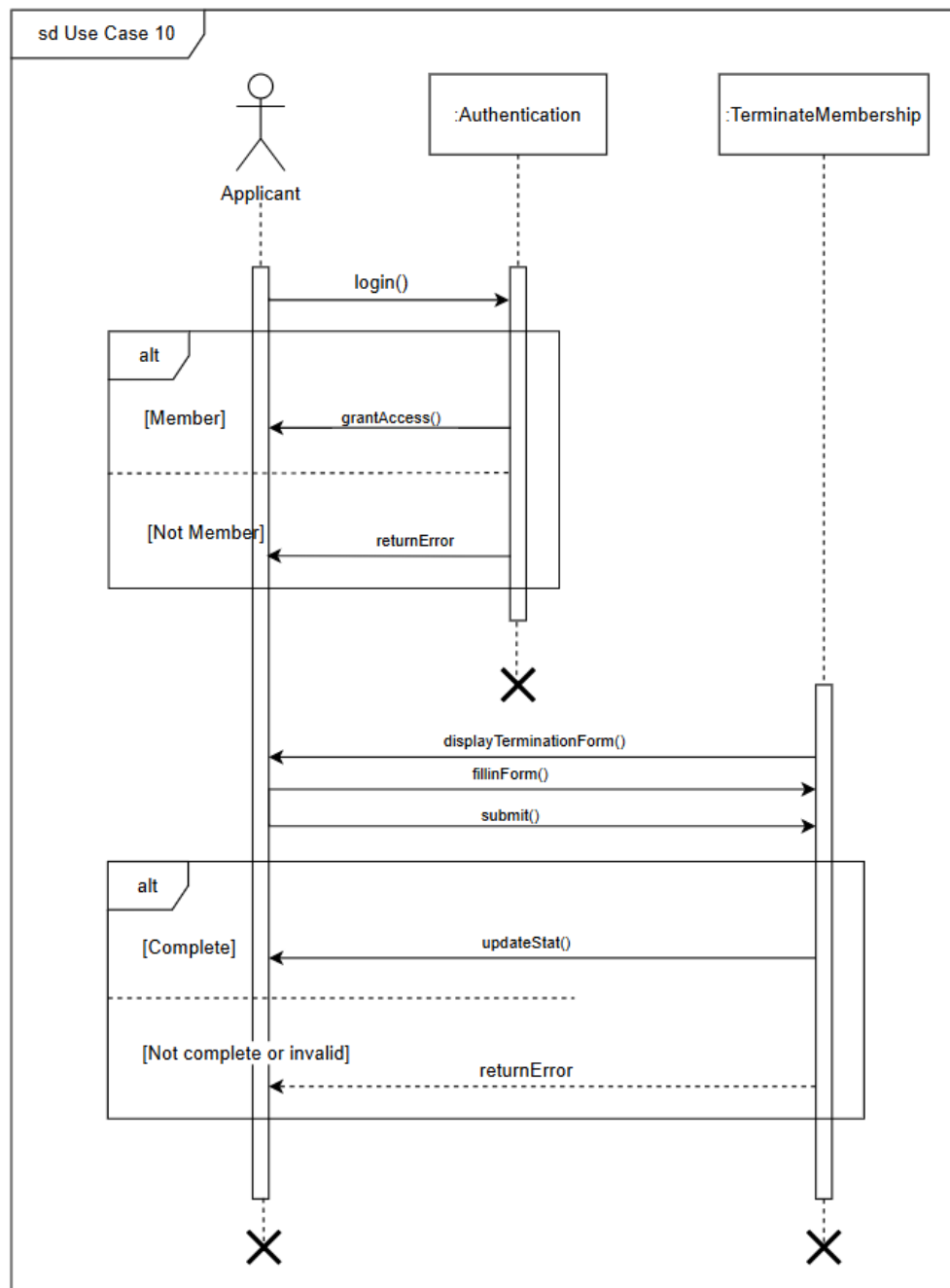


Figure 2.26: Sequence Diagram for Terminate Membership

2.3.11 UC011: Use Case Approve Terminate Application

Table 2.12 is a use case description that enables an ALK to approve termination membership application. The application must be reviewed by Admin beforehand. Then, ALK will make decisions based on circumstances to approve or reject applications. If the application is rejected, then the membership continues as usual otherwise the membership ends immediately.

Table 2.12: Use Case Description for Approve Terminate Application

Use Case ID: UC011
User Story Description: As an Alk I want to approve terminate application So that i can discontinue member's membership
Flow of Events: 1. The ALK logs into the system. 2. The ALK navigates to the terminate application page. 3. The ALK performs decision-making to approve or reject application. 4. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked "Incomplete" by Admin, the application will not be forwarded to ALK
Acceptance Criteria: Postcondition: 1. Status is updated 2. The Applicant is notified of the outcome. Precondition: 1. Application reviewed by Admin Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel.

Figure 2.27 is an activity diagram for approve terminate application that outlines the interaction between the ALK and the System. The ALK logs into the system, navigates to the termination section and clicks the termination application. The ALK then performs decision-making which is to approve or reject the application. Once the decision-making has been made, the latest status will be updated based on the ALK's decision.

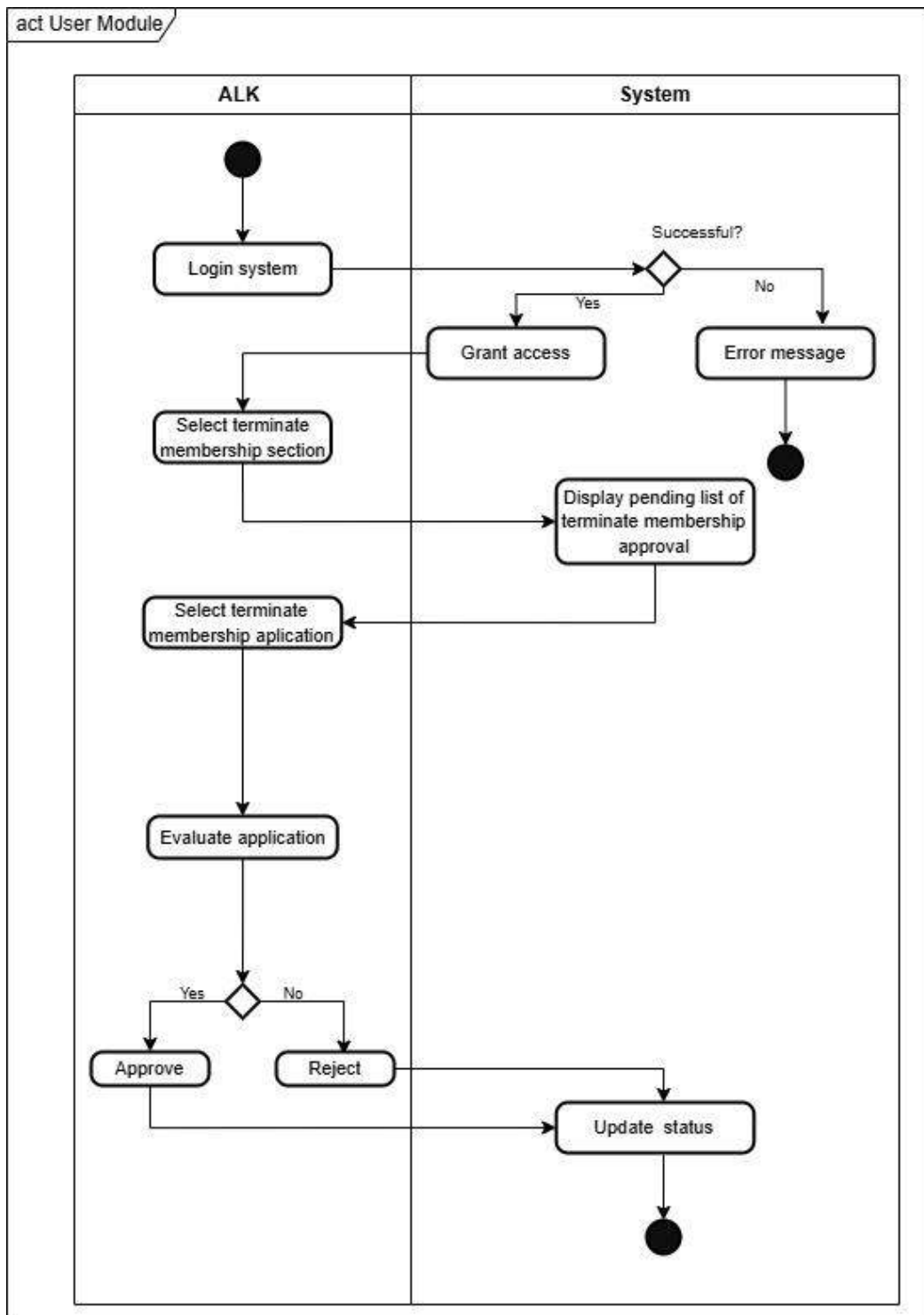


Figure 2.27: Activity Diagram for Approve Terminate Application

Figure 2.28 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin.

The ALK then performs decision-making whether to approve or reject the application. Then the latest status will be updated to keep the applicant informed of real-time status.

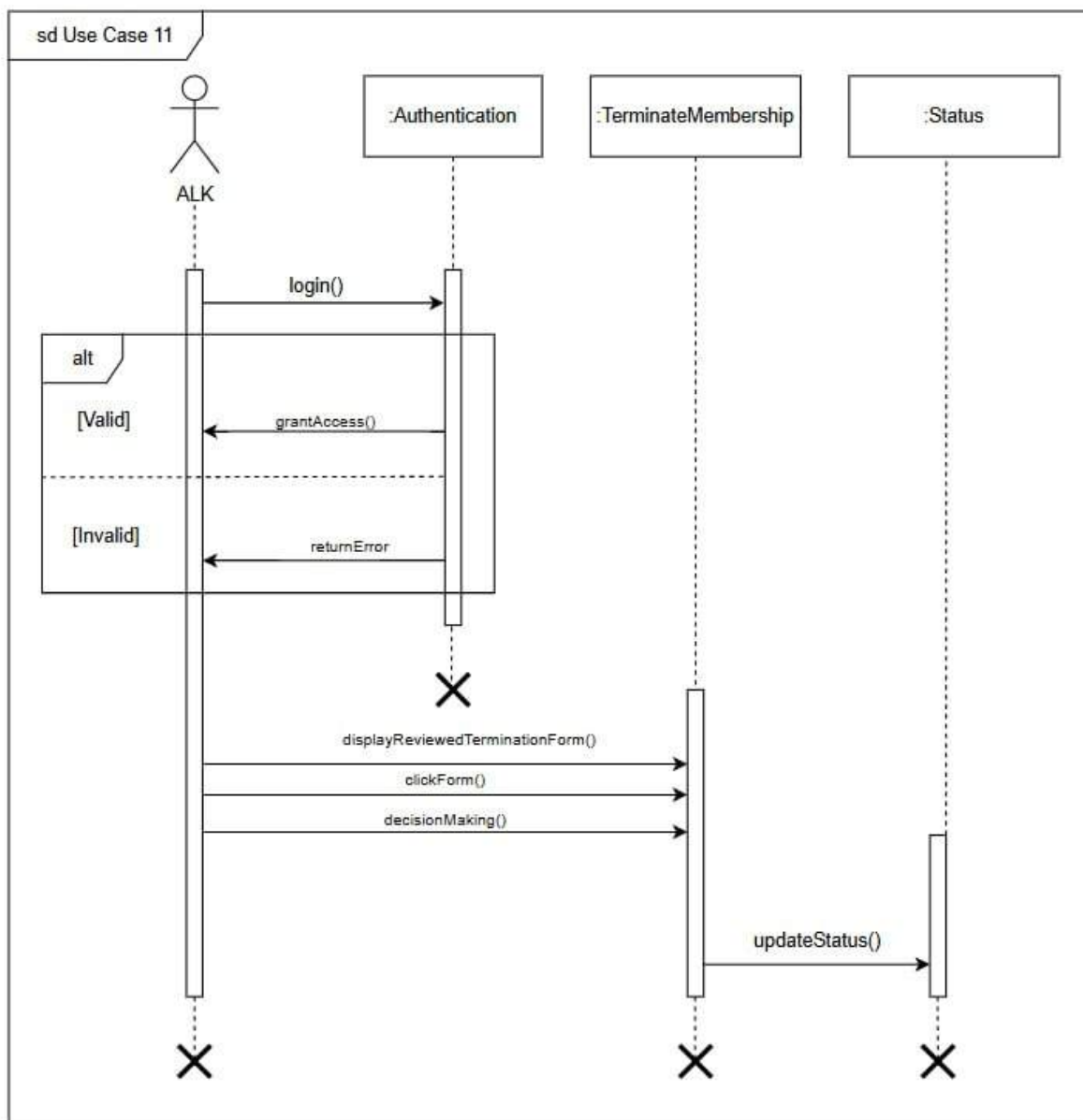


Figure 2.28: Sequence Diagram for Approve Terminate Application

2.4 Performance and Other Requirements

Non-functional requirements for this project system define how the system should work and focus on that rather than show its specific functionalities. It is essential to have non-functional requirements to make sure the system fulfills quality standards and can work in various environments. These requirements are classified into Software System Attributes, Performance and Other Requirements according to the ISO/IEC/IEEE 29148 standard.

Performance requirements specify how quickly the system can handle data and react to user requests. These are requirements that belong to the system's performance under different conditions such as:

- Response time: The response time for any user request should not exceed 2 seconds under normal operating conditions.
- Throughput: The system should handle a significant number of concurrent users without performance degradation to ensure timely processing of membership and loan applications.
- Capacity: The system must store a large number of member records and loan applications while maintaining optimal performance.
- Availability: The system should remain accessible and functional even under interruptions or heavy loads.

Software System Attributes define the overall quality and behavior of the software. These attributes include:

- Usability: The system's interface should enable new users to complete tasks without requiring a lot of training.
- Reliability: The system must secure data consistency and correctness, with a monthly downtime tolerance of no more than 0.5%.
- Maintainability: The system should enable updates such as bug fixes by laying objectives to finish major updates in a week and minor updates in 1 to 2 days.
- Portability: The system's software needs to work with a variety of hardware and software operating systems, including different Windows and macOS versions.
- Safety: The system includes specific safety requirements to mitigate potential loss, damage, or harm associated with its use, particularly in handling sensitive financial and personal data.
- Security: The system will require multi-factor authentication (MFA) to help prevent unauthorized access.

2.4 Design Constraints

The design constraints for the system focus on ensuring it works efficiently with standard computers and devices, while maintaining security for user data. It should be compatible with common hardware and across different operating systems and devices. Additionally, the system should support multiple users and provide quick access to information.

- **Hardware Constraints:** The system must be designed to operate on standard desktop computers with a minimum of 8GB of RAM and a dual-core processor to ensure efficient performance. It should also be compatible with existing printers and scanners to make document management easy. The system will need to have the proper USB or network ports to connect with these devices without any issues.
- **Security Constraints:** The system must follow data privacy rules and have security measures to protect user information. This includes secure login, encryption of personal data, and access controls to stop unauthorized users from accessing information. Additionally, regular security updates should be applied to maintain system integrity and prevent potential threats.
- **Compatibility Constraints:** The system should be compatible with the hardware and software that are mostly used by applicants, admin and ALK. It has to be compatible with different operating systems, browsers and devices in order to be accessible to everyone. The system should be optimized for both desktop and mobile devices to cater to different user needs.
- **Performance Constraints:** The system must be able to handle at least 500 simultaneous users without a noticeable decrease in performance. Data retrieval and report generation should be completed in under 5 seconds for typical queries to provide fast access to information. Additionally, the system must maintain 99.9% uptime, ensuring that it is always available for daily operations without interruptions.

3. Architectural Rationale

An architectural style refers to a set of design principles and patterns that define the structure and organization of a software system. It provides a standardized way to design system components, their relationships, and communication methods. For Koperasi KADA's automated system, Layered Architecture is used.

3.1 Architecture Style and Rationale

Choosing the right system design is important for making sure a software works well and is easy to manage. The Koperasi Kada Automated System needs a reliable and organized structure to handle membership, loan applications, and statement generation. To achieve this, the system follows a Layered Architecture, which divides it into three main parts which is View Layer, Domain Layer, and Data Access Layer.

The View Layer is the part that users interact with. It displays information and allows users to enter data, such as applying for loans or checking membership details. The Domain Layer is responsible for processing data and applying the system's rules, like approving a loan based on certain conditions. The Data Access Layer connects to the database, making sure that information is stored and retrieved correctly. Separating these layers makes the system more organized and easier to update, since changes in one layer do not affect the others.

Using a Layered Architecture makes the system easier to maintain, test, and expand. For example, if the design of the View Layer needs improvement, it can be updated without affecting the business logic or database. This also helps with security, since the Data Access Layer ensures proper handling of sensitive information. Additionally, this structure allows developers to work on different parts of the system at the same time without causing problems. By following this approach, the Koperasi Kada Automated System can efficiently replace the manual system with a faster and more reliable solution. Figure 3.1 illustrated the Layered Architecture Diagram for Koperasi KADA Automated System.

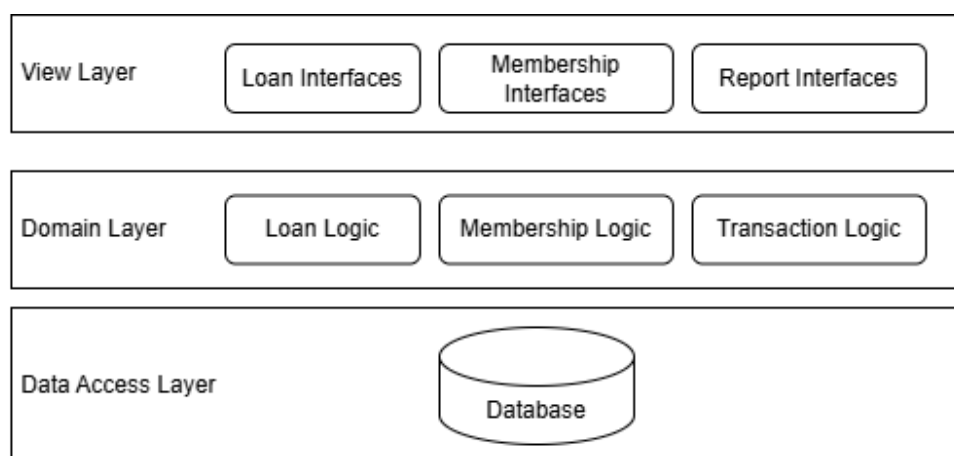


Figure 3.1: Layered Architecture Diagram for Koperasi KADA Automated System

4. Architectural Views

The 4+1 Architectural View Model is an architectural framework that organizes a system into five different views: Logical, Development, Process, Physical and Scenarios (Use Case View). The Logical View represents the system's functionality such as reviewing and approving applications and object-oriented design, defining key classes and relationships. The Development View focuses on the organization of KADA Automated System software modules, which are Membership Application Module, Terminate Membership Module, Review Module, Loan Application Module, Approve Application Module, Statement Report Production Module and Financial Tracking Module, including components and overall code structure. The Process View handles dynamic aspects of the system such as the synchronization mechanisms. The Physical View represents how software components are deployed across hardware nodes, addressing performance and scalability. Lastly, the Scenarios serve as a bridge among all the views, describing the system's functionality from the user's perspective. Figure 4.1 illustrates the 4+1 Architectural View Model.

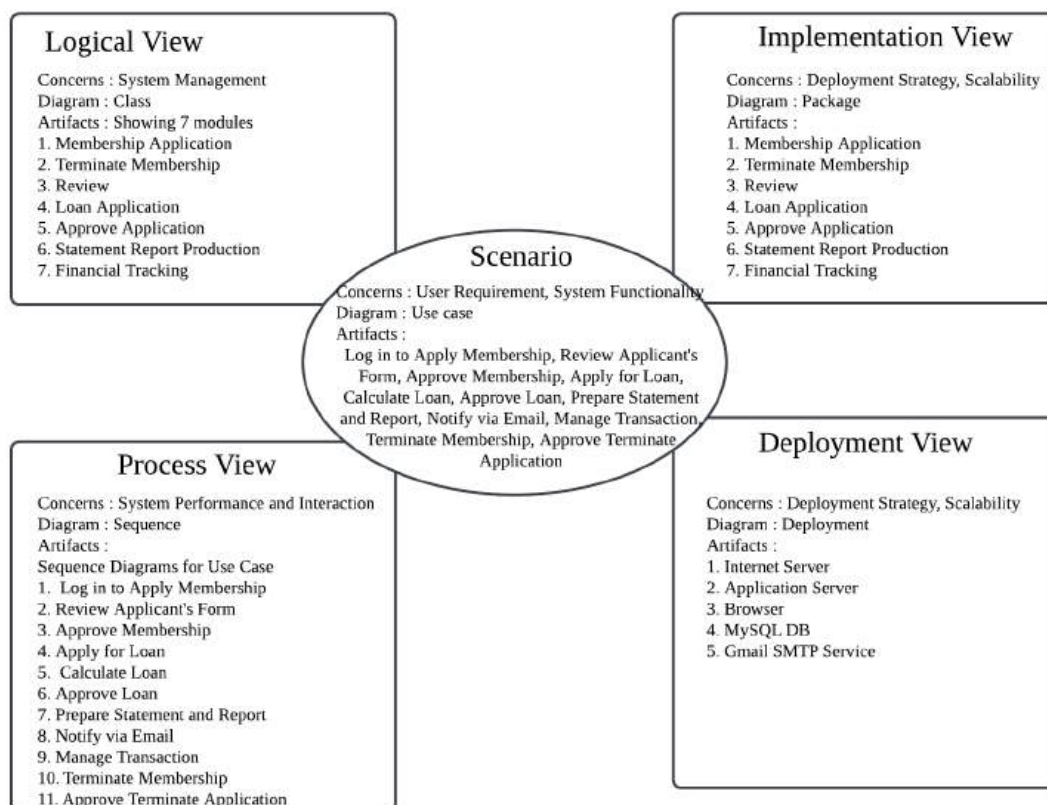


Figure 4.1: 4+1 Architectural View Model

Table 4.1 shows the diagram used for each view in KADA Automated System. Use Case is used in Scenario, Class Diagram is used in Logical View, Sequence Diagram is used in Process View, Package Diagram is used in Implementation View and Deployment Diagram is used in Deployment View. Further descriptions are described in the table below.

Table 4.1: Type of Diagram used

Type of View	Type of Diagram	Description
Scenario	Use Case	shows the interaction between Applicant, Admin, ALK and the system
Logical	Class	represent the structure of the system, focusing on business logic of membership and loan and data relationships
Process	Sequence	shows the flow of operations between different components of KADA Automated System during specific processes
Implementation	Package	represents the organization of system components and how they relate to the storage and management of data
Deployment	Deployment	shows the physical distribution of system components that includes data servers and storage systems of KADA Automated System

4.1 Scenario (Use Case View) - Use Case

This view acts as a connecting thread across the other three views, maintaining consistency and completeness. It shows the functional requirements and connects the views through real-world scenarios or use cases. These use cases serve as a link between the system's conceptual design and technical execution, connecting various components including data models, processes and user interfaces. Finally, this viewpoint contributes to a Broad view by ensuring that all system components work together effortlessly to achieve the desired results while meeting the specific requirements. Figure 4.2 illustrates a use case diagram with its module for Koperasi KADA's Automated System.

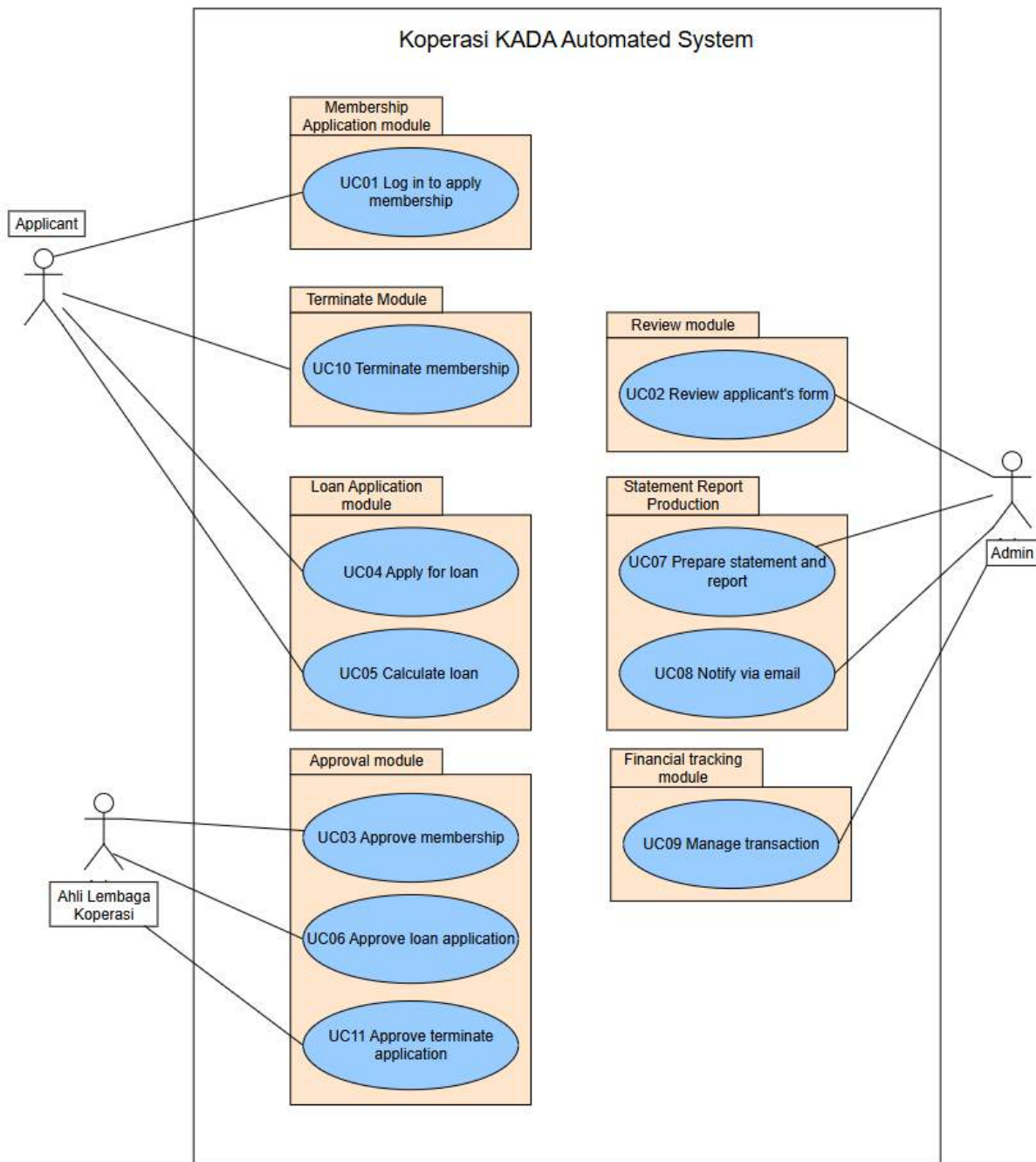


Figure 4.2: Use Case Diagram for Koperasi KADA Automated System

4.2 Implementation View - Package Diagram

Figure 4.3 is the architectural view which highlights the system's software modules that break down into high-level subsystems and their functions which arrange within its development setting. The system includes four main subsystems which are Membership Application Module, Loan Application Module, Terminate Module, Review Module, Approval Module and Statement Report Production Module.

The Membership Application Module handles user authentication and processes

membership applications. The Loan Application Module allows members to apply for loans, calculates monthly repayments loan amount and stores approved loans in a Loan List. Moreover, the Terminate Module allows users to terminate their membership of Koperasi KADA. The Review Module is in charge of going through applications from the Membership List and Loan List, making sure they are properly assessed before being accepted or rejected. The Approval Module receives the reviewed applications after which it makes final choices by accepting or rejecting them and updating the corresponding lists. Finally, data from other subsystems is used by the Statement Report Production Module to create reports and send email notifications.

These subsystems work together seamlessly to provide the intended functionality. To guarantee that only authorised members are able to apply for loans, access to the Loan Application Module and Terminate Module requires the Membership List generated by the Membership Application Module. The Review Module processes the applications for the Membership and Loan lists ensuring a thorough validation before forwarding to the Approval Module. By combining data from every subsystem, the Statement Report Production Module creates thorough reports and alerts relevant users.

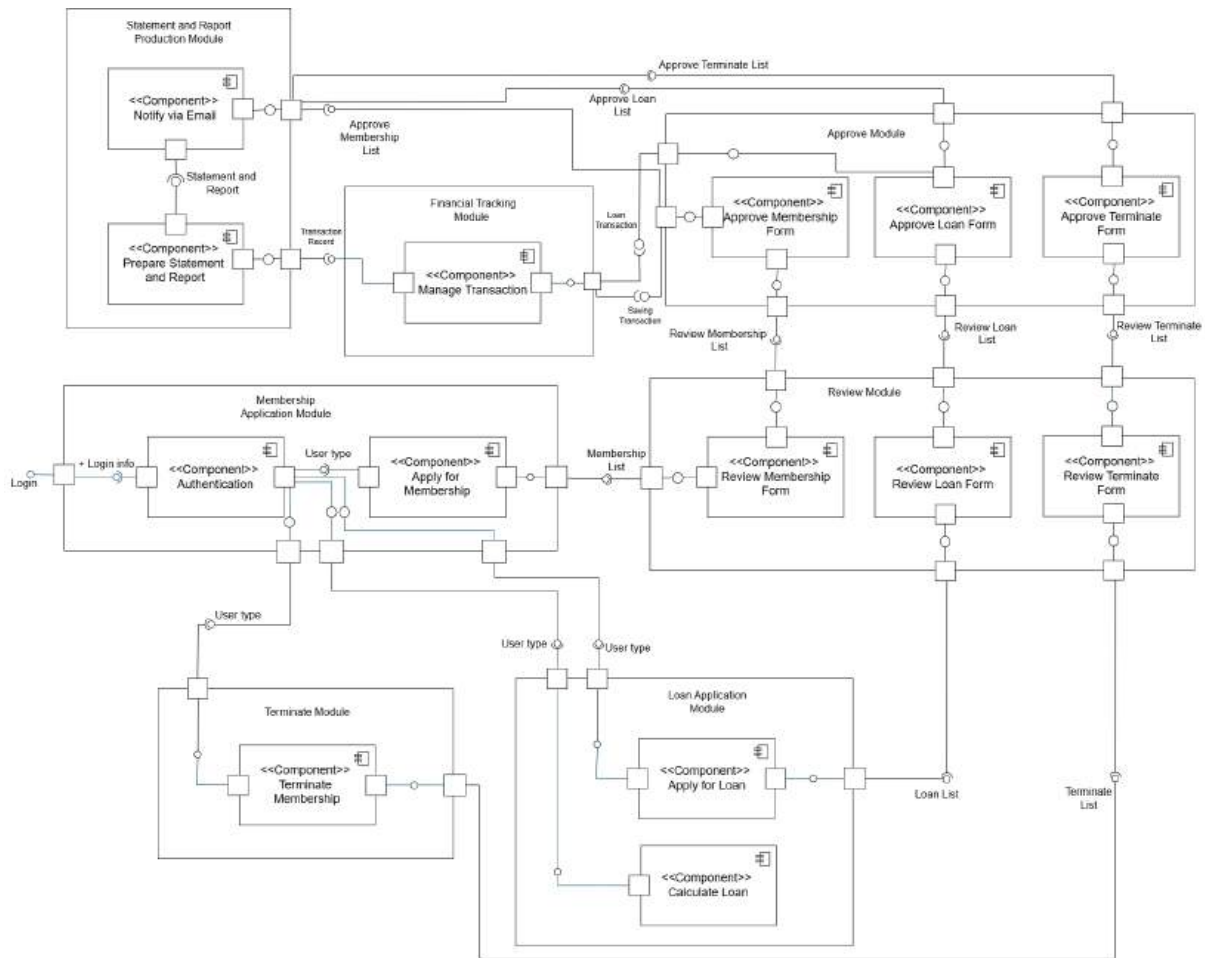


Figure 4.3: Architecture Diagram of Koperasi KADA Automated System

Figure 4.4 shows the subsystems in Koperasi KADA Automated System in Package Diagram form. The User Interface Framework handles user interaction and connects to the other subsystems like the Membership Module, Loan Application Module, Terminate Module, Review Module and Approval Module to process and manage data. The Statement Report Production Module generates reports, which are then sent back to the User Interface Framework for users to access.

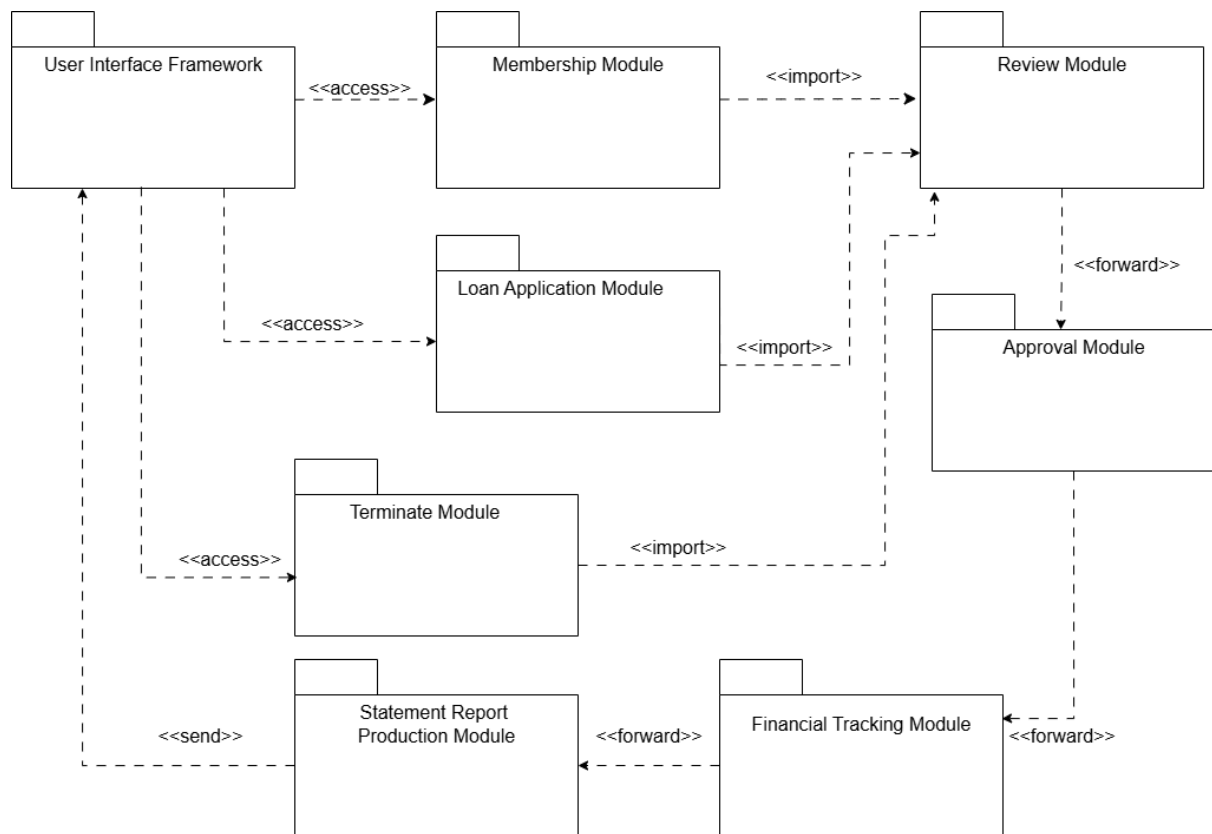


Figure 4.4: Package Diagram for Koperasi KADA Automated System

In Figure 4.5 is the Package Diagram for the Membership Application Module. The View Layer consists of user interfaces such as AuthenticationInterfaces and UserInterface and MembershipStatus Interface, which enables users to interact with the system by logging in, applying for membership and viewing the status of membership application. The Domain Layer includes components such as Applicant, Membership and FamilyInfo, which handle core business logic like processing applicant details and managing membership applications. Lastly, the Data Access Layer manages database interactions through components like MembershipData and familyInfoData, ensuring efficient retrieval and management of membership application data.

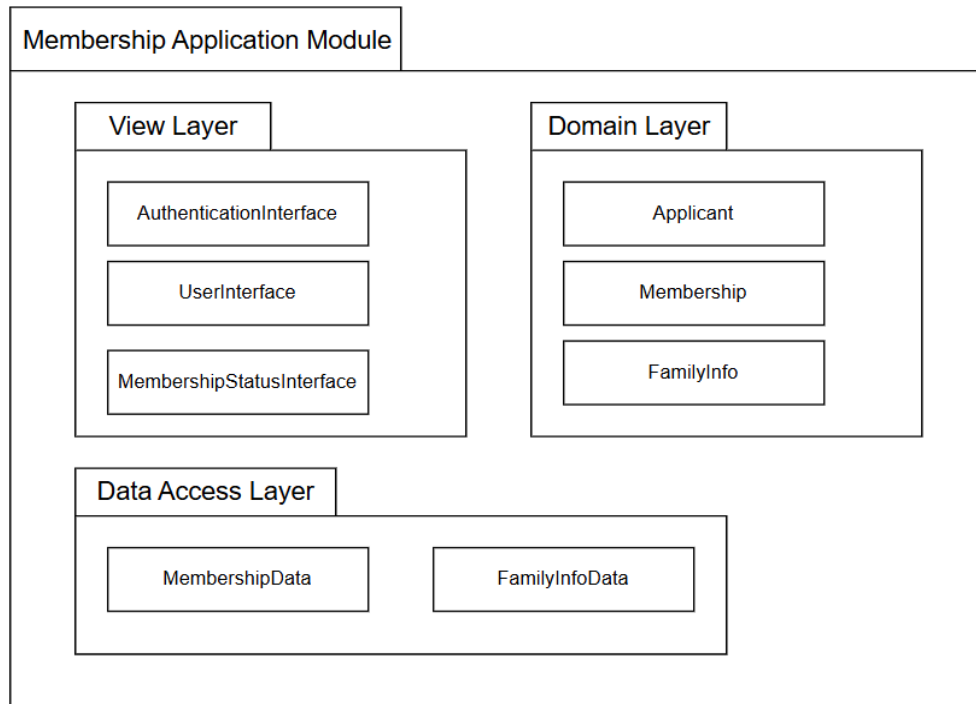


Figure 4.5: P001 Package Diagram for Membership Subsystem

In Figure 4.6 is the Package Diagram for the Review Module. The View Layer features interfaces such as `AdministratorInterface`, `MembershipStatusInterface`, `Loan StatusInterface` and `TerminateStatusInterface`, enabling administrators to interact with the system to review membership application, loan application, terminate membership application and update the status for all the applications. The Domain Layer consists of components such as `Loan`, `GuarantorInfo`, `Membership`, `FamilyInfo`, and `ApplicationManagement` which handle core business logic for loan, membership, and membership termination application reviewing process. Lastly, the Data Access Layer manages database interactions through components like `Loan Data`, `GuarantorInfoData`, `MembershipData`, `familyInfoData`, and `TerminateMembershipData` ensuring efficient retrieval and management of application data.

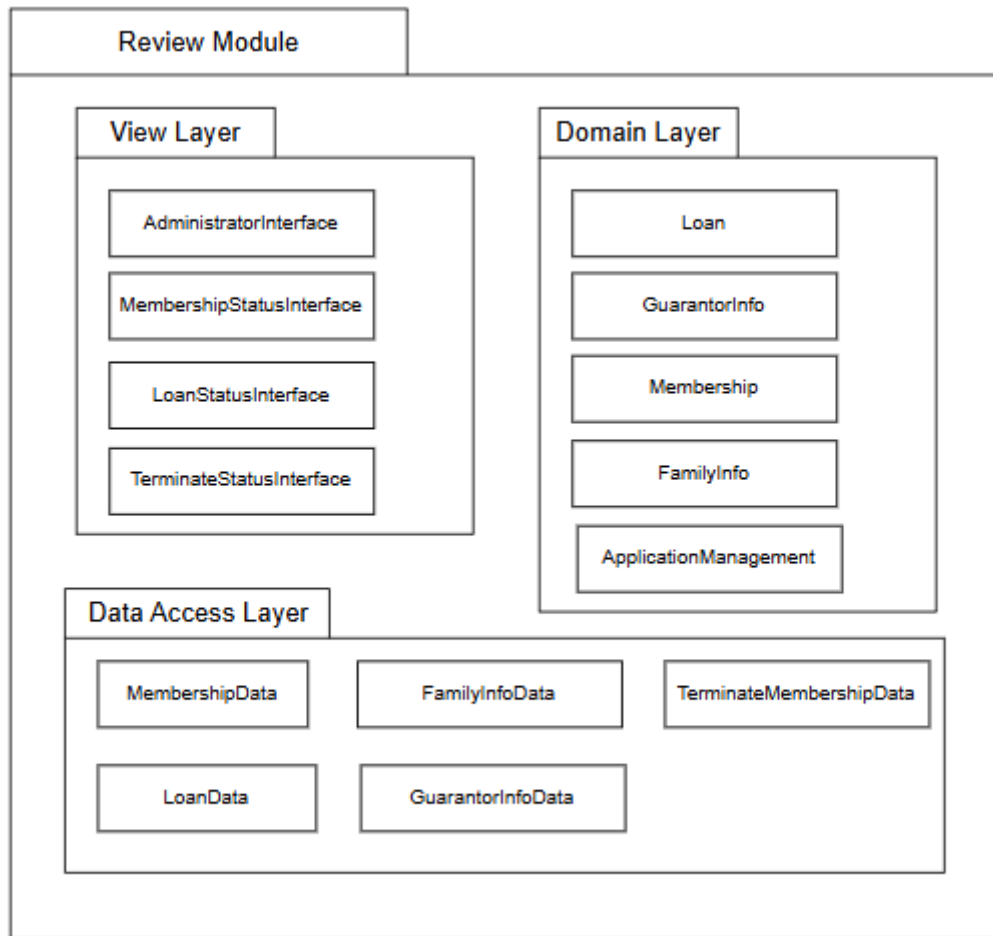


Figure 4.6: P002 Package Diagram for Review Subsystem

In Figure 4.7 is the Package Diagram for Loan Application Module. The view layer has user interfaces such as `LoanApplicationInterfaces`, `LoanStatusInterface` and `LoanCalculatorInterface`, which enables users to interact with the system by applying loans, checking loan status and calculating monthly loan repayment amount. Finally, the Data Access Layer coordinates database interactions within components such as `LoanData`, `LoanStatusData` and `ApplicantData`. This layer assures data storage and retrieval for loan and applicant information.

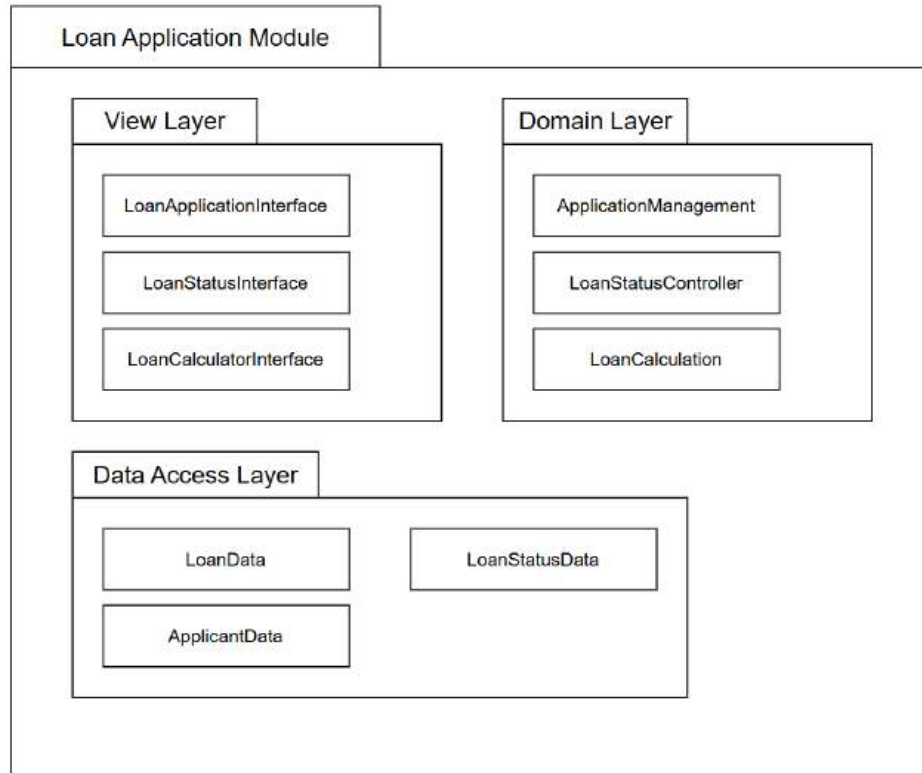


Figure 4.7: P003 Package Diagram for Loan Application Subsystem

Figure 4.8 shows the Package Diagram for Approval subsystem. The view layer represents the view component that interacts with the user. The layer includes ListMembershipUI to display the pending list of membership approval, ListLoanUI to display the pending list of loan approval, MembershipApprovalUI to interact in performing membership approval, LoanApprovalUI to interact in performing loan approval, historyMembershipUI to display past membership applications and historyLoanUI to display past loan applications. Then, the Domain Layer contains the business logic for data processing which includes MembershipApprovalController, LoanApprovalController, LoanComparisonProcessor and ModificationController. All of these are involved in approving or rejecting membership and loan applications. Lastly, the Data Access Layer is used to handle data operations. This layer includes MembershipDataRecord, LoanDataRecord and TerminateMembershipData.

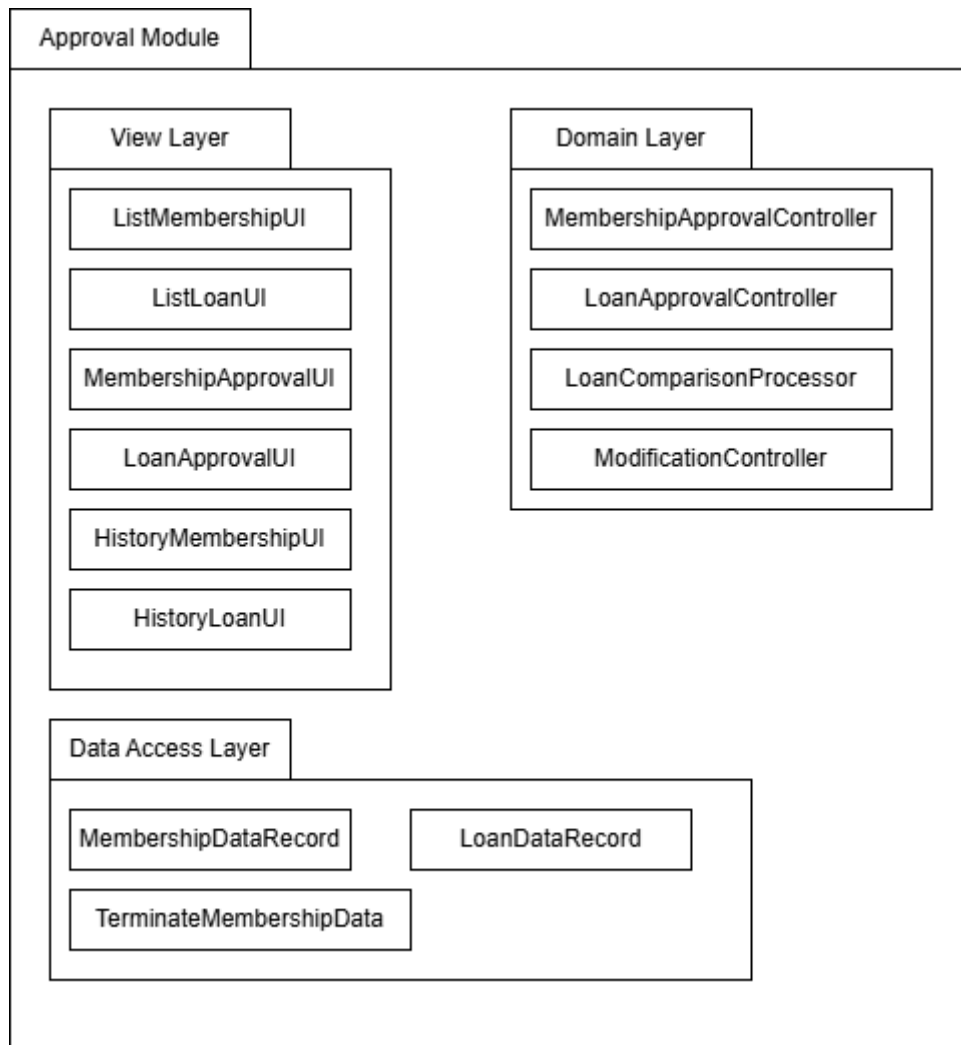


Figure 4.8: P004 Package Diagram for Approval Subsystem

Figure 4.9 illustrates the Package Diagram for the Statement Report Production subsystem. The view layer represents the components that interact with the user, including GenerateReportInterface, which provides a button for the admin to generate the statement report, NotificationInterface, which includes a button for sending email notifications, and TransactionListInterface, which displays the list of transactions related to loans and memberships. The Domain Layer contains the business logic responsible for data processing, including GenerateStatementController, which manages the process of generating the report, NotificationController, which handles the creation and sending of email notifications, and ExportController, which manages the exporting of the report into various formats like PDF or Excel. Finally, the Data Access Layer manages data operations and includes LoanTransactionData, LoanType, SavingTransactionData, and MembershipData, which handle the retrieval and storage of financial transaction information.

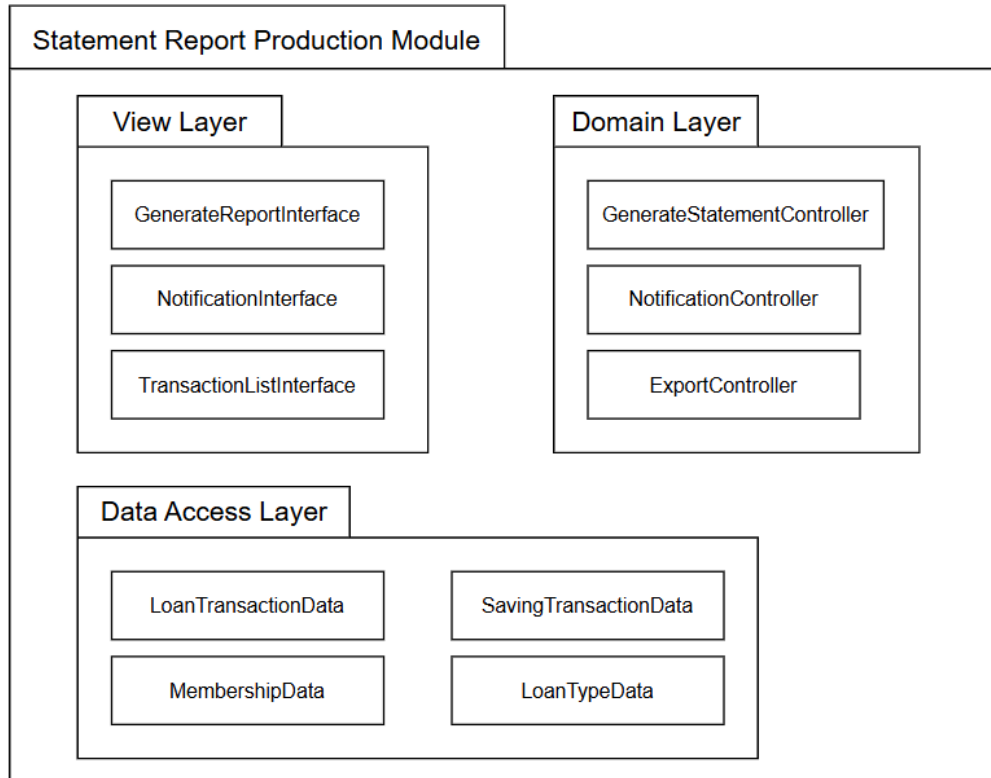


Figure 4.9: P005 Package Diagram for Statement Report Production Subsystem

Figure 4.10 is the Terminate Module packet diagram that organizes the use case into three layers. The View Layer contains interfaces for submitting termination requests and viewing their status. The Domain Layer handles the business logic including termination management and status updates. The Data Access Layer manages data related to members, termination requests and their statuses to ensure seamless interaction across the module.

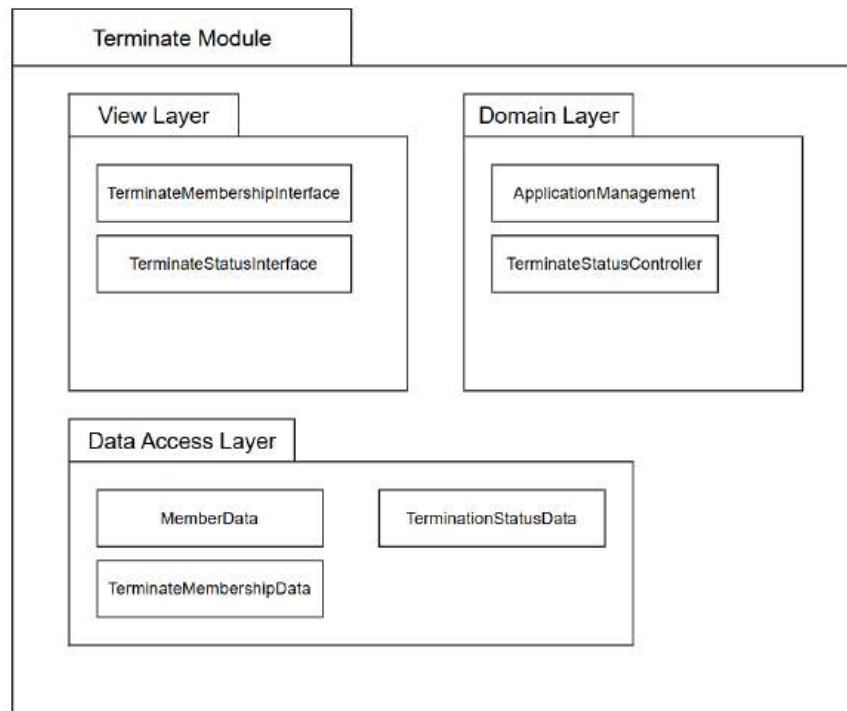


Figure 4.10: P006 Package Diagram for Terminate Module Subsystem

Figure 4.11 presents the Financial Tracking Module packet diagram, which structures the use case into three layers. The View Layer includes interfaces for monthly transactions and non-fixed transactions. The Domain Layer manages business logic, handling the processes for both types of transactions. The Data Access Layer is responsible for managing data related to savings and loan repayments to ensure accurate report statements.

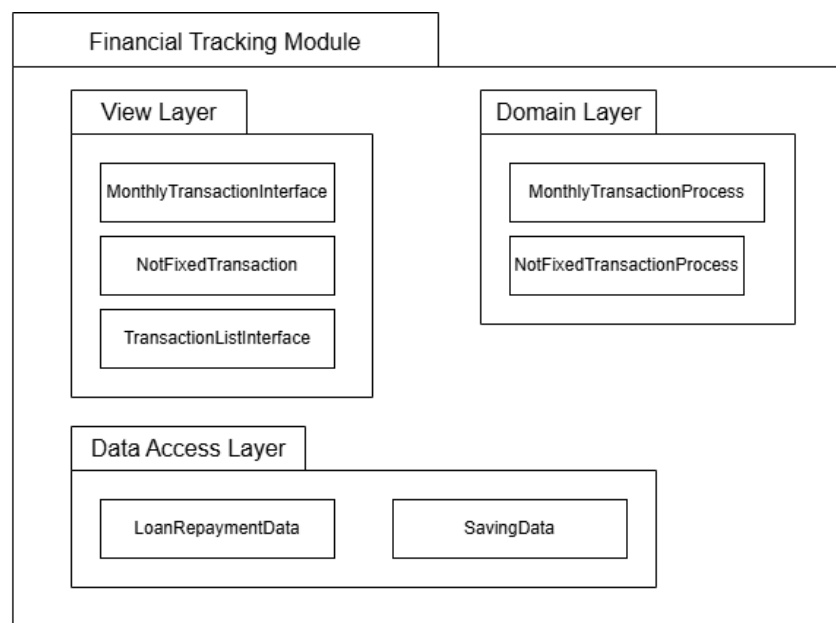


Figure 4.11: P007 Package Diagram for Financial Tracking Module Subsystem

4.3 Logical View - Class Diagram

The logical view provides a perspective on the system's structure from the developer or designer's point of view. It highlights the major abstractions in the system, which in an object-oriented design approach would normally be modeled as objects or object classes. This view is represented through class diagrams, which shows the structure of the system by showing the relationships among classes, along with their attributes and methods.

Figure 4.12 class diagram shows the class diagram for Login to Apply Membership . This Authentication class has a one-to-one relationship with the User class, ensuring each user has unique login credentials. The Applicant class has an association relationship with the User class. The Applicant class has an aggregation relationship with the Membership class, meaning membership details can exist independently of the Applicant. Then, the Membership class has a composition relationship with the familyInfo class, meaning the family information is an essential part of the membership and cannot exist separately.

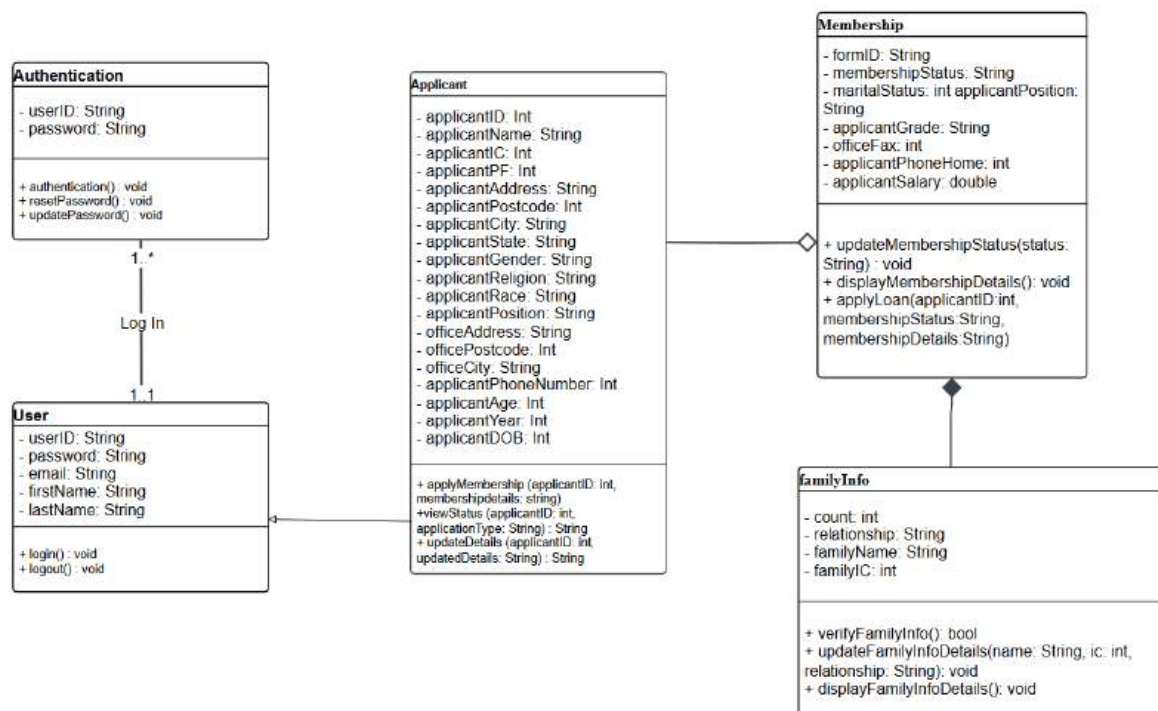


Figure 4.12: Class diagram for Login to Apply Membership Subsystem

Figure 4.13 class diagram shows the class diagram for Review Applicant's Form. The Administrator class has a one-to-many relationship with both Loan and Membership classes, meaning one administrator can review multiple loans and membership. The loan class has a composition relationship with the GuarantorInfo class, indicating that a guarantor is tightly bound to the loan and cannot exist independently. Similarly, the Membership class has a

composition relationship with the familyInfo and saving class, meaning the family information is an essential part of the membership and cannot exist separately.

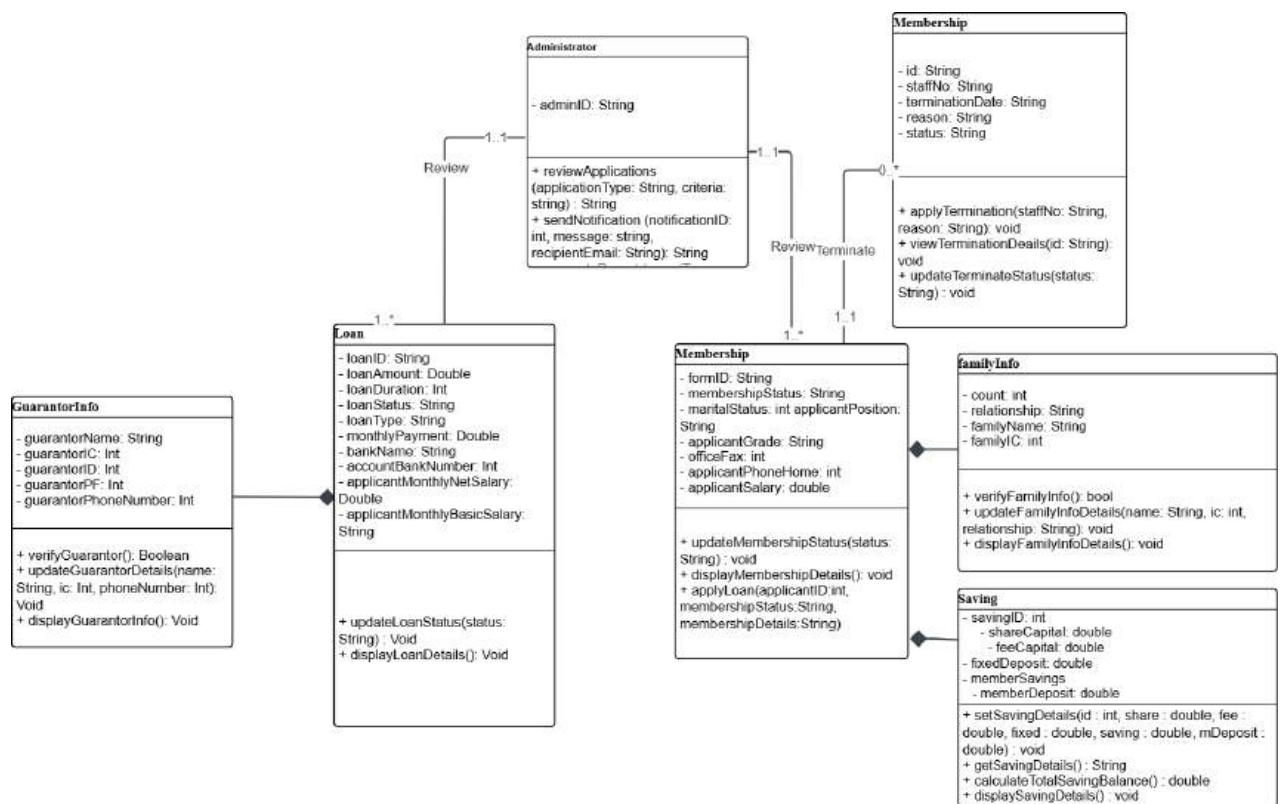


Figure 4.13: Class Diagram for Review Applicant's Form Subsystem

Figure 4.14 shows the class diagram for Approve Membership which includes the relationship between ALK class and Membership class. ALK class provides methods like approveMembership() to manage membership approvals and clarifications. The membership class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple membership applications.

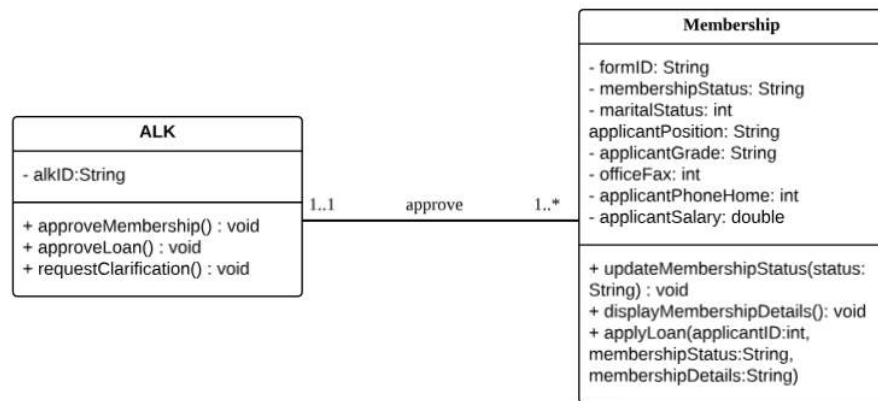


Figure 4.14: Class Diagram for Approve Membership Subsystem

Figure 4.15 class diagram shows the class diagram for Apply for Loan that shows two key relationships. First, the Membership and Loan classes have a one-to-many relationship indicating that a single membership can apply for several loans. Second, there is a composition relationship between the Loan and GuarantorInfo classes, which means that each loan application must include guarantor information because the guarantor is a critical component of the loan process.

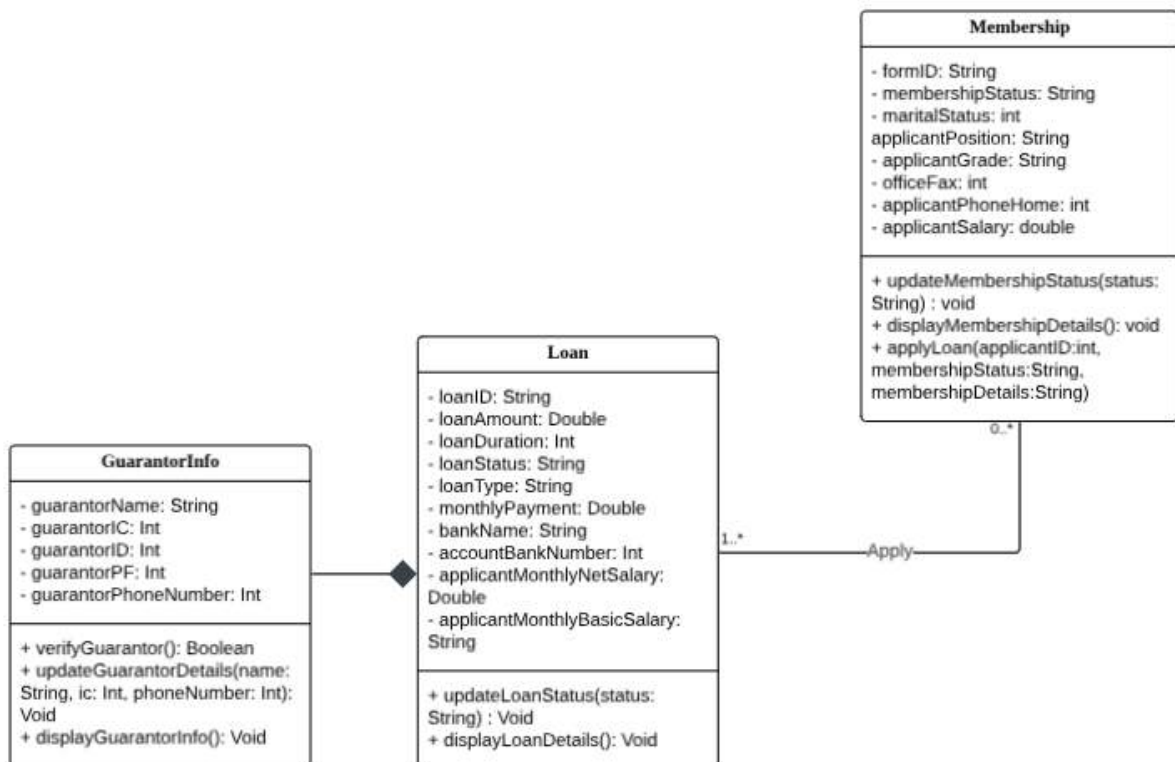


Figure 4.15: Class Diagram for Apply for Loan Subsystem

Figure 4.16 class diagram shows the class diagram for Calculate Loan that shows one key relationship. The user class has a one-to-one relationship with the loanCalculator class which allows users to calculate the estimation of monthly repayment.

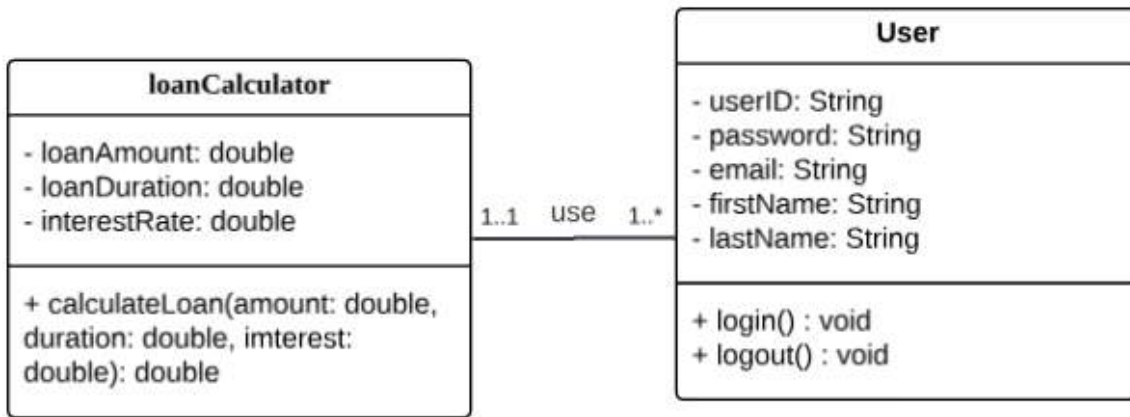


Figure 4.16: Class Diagram for Calculate Loan Subsystem

Figure 4.17 shows the class diagram for Loan Approval which includes the relationship between ALK class and Loan class. ALK class provides methods like `approveLoan()` to manage loan approvals. The loan class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple loan applications.

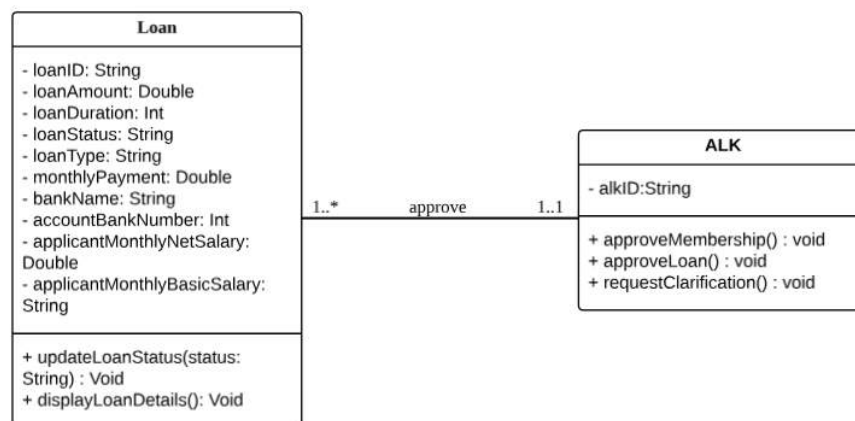


Figure 4.17: Class Diagram for Loan Approval Subsystem

Figure 4.18 illustrates the class diagram for preparing statements and reports, detailing the relationships between various entities involved in loan, savings, and membership management. The diagram includes classes such as **LoanRepayment**, **Loan**,

Applicant, Membership, Saving, and SavingType, each with specific attributes and methods. The methods provided facilitate updates and retrieval of relevant data, supporting efficient statement and report generation.

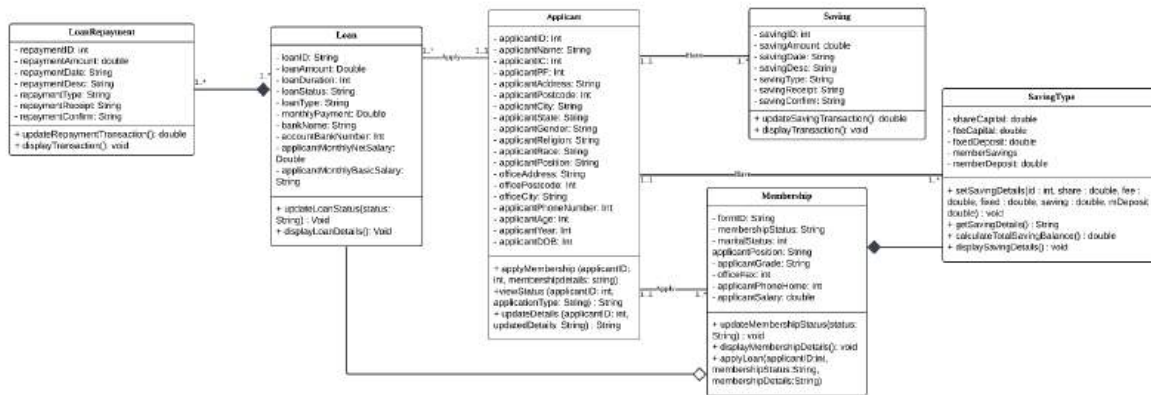


Figure 4.18: Class Diagram for Prepare Statement and Report Subsystem

Figure 4.19 presents the class diagram for the Notify via Email. In this diagram, there are two primary classes: Admin and Notification. The Admin class is responsible for sending notifications, establishing a one-to-many relationship with the Notification class. This means that one Admin can send multiple notifications to different recipients. The Notification class holds the details about each notification, such as the content and recipient information.

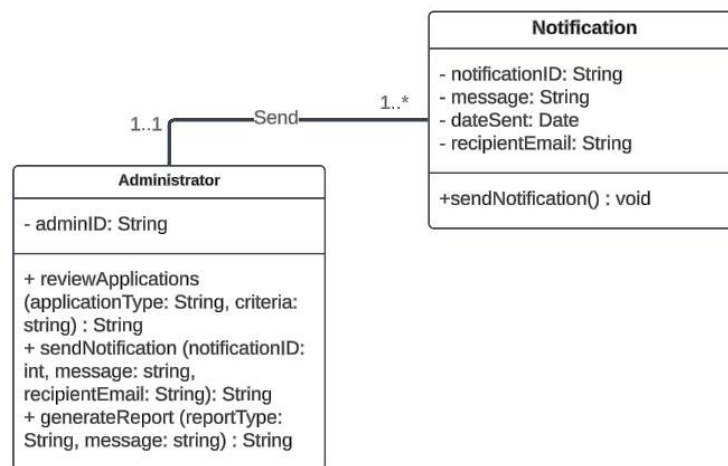


Figure 4.19: Class Diagram for Notify via Email Subsystem

Figure 4.20 is the Terminate entity representing the process of membership termination, linking it to the Membership Entity. Each membership can have one termination request while multiple termination requests can exist for different memberships. The entity

includes attributes such as id, staffNo, terminationDate, reason and status ensuring a structured termination process.

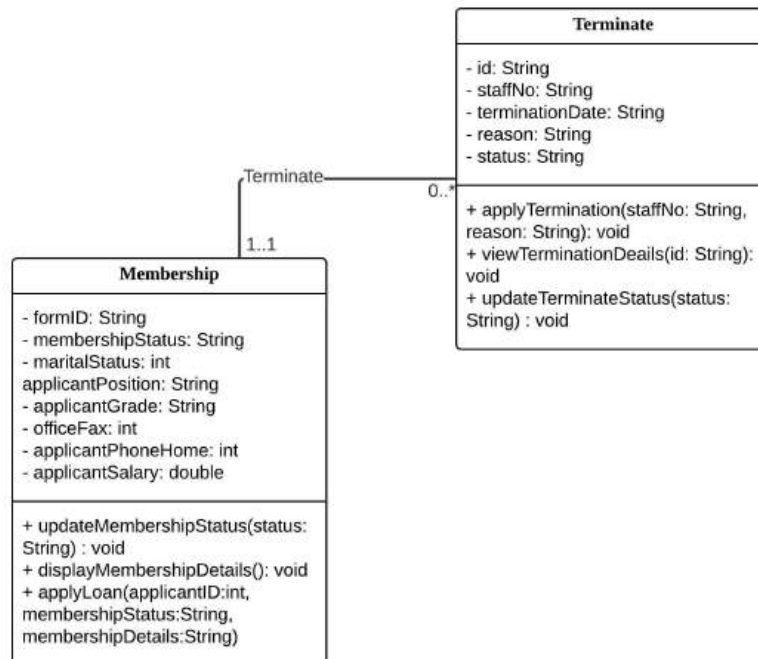


Figure 4.20: Class Diagram for Terminate Membership Subsystem

Figure 4.21 shows the class diagram for managing transactions. It includes classes like LoanRepayment, Loan, Membership, and Saving, each handling different financial activities. These classes store important details and have functions to update and display transaction records, making it easier to manage loan repayments, savings, and memberships.

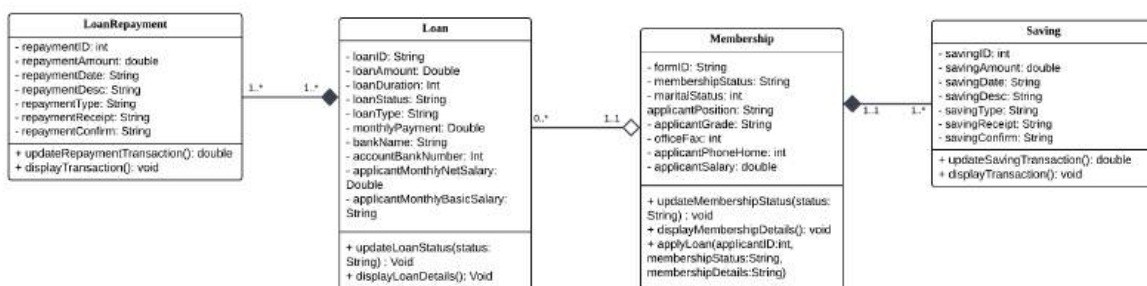


Figure 4.21: Class Diagram for Manage Transaction Subsystem

Figure 4.22 shows the class diagram for approve terminate membership application. It includes classes like ALK, Terminate and Membership. Each application will be undergoing

approval process by ALK. Each application will have updated status once ALK approved or rejected application.

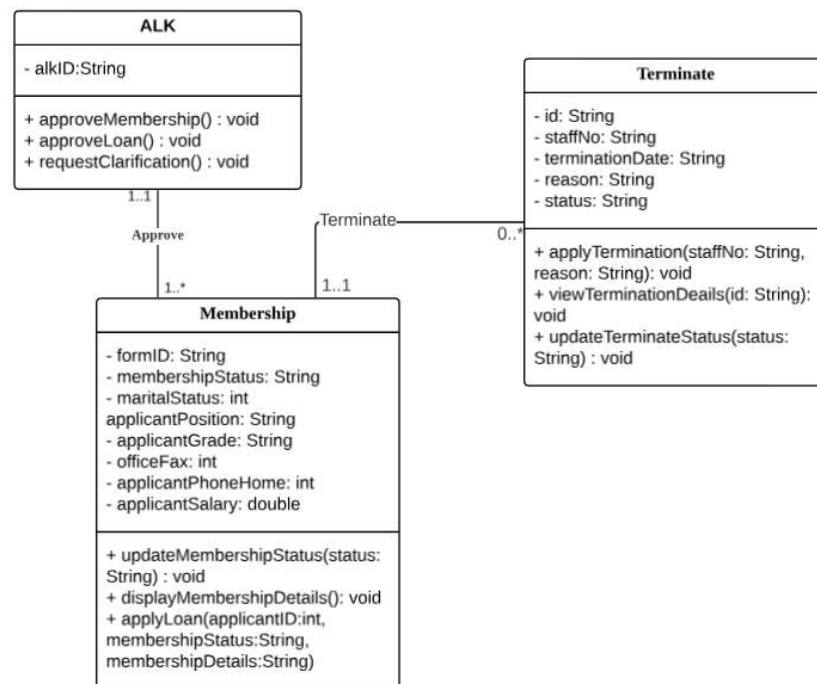


Figure 4.22: Class Diagram for Approve Terminate Application Subsystem

4.4 Process View - Sequence Diagram

This view illustrates how the system operates at runtime and focuses on how its components interact through communication and collaboration. It shows how different processes work together to achieve the system's goals, with each process responsible for specific tasks. To represent these interactions, sequence diagrams will be used for relevant use cases, showing the step-by-step flow of messages between components. Branching will capture alternate flows within the same diagram, and if the diagram becomes too cluttered, separate diagrams will be created for each scenario or alternate flow. This view provides a clear understanding of the system's dynamic behavior and the relationships between its components.

Figure 4.23 illustrates the applicant's login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

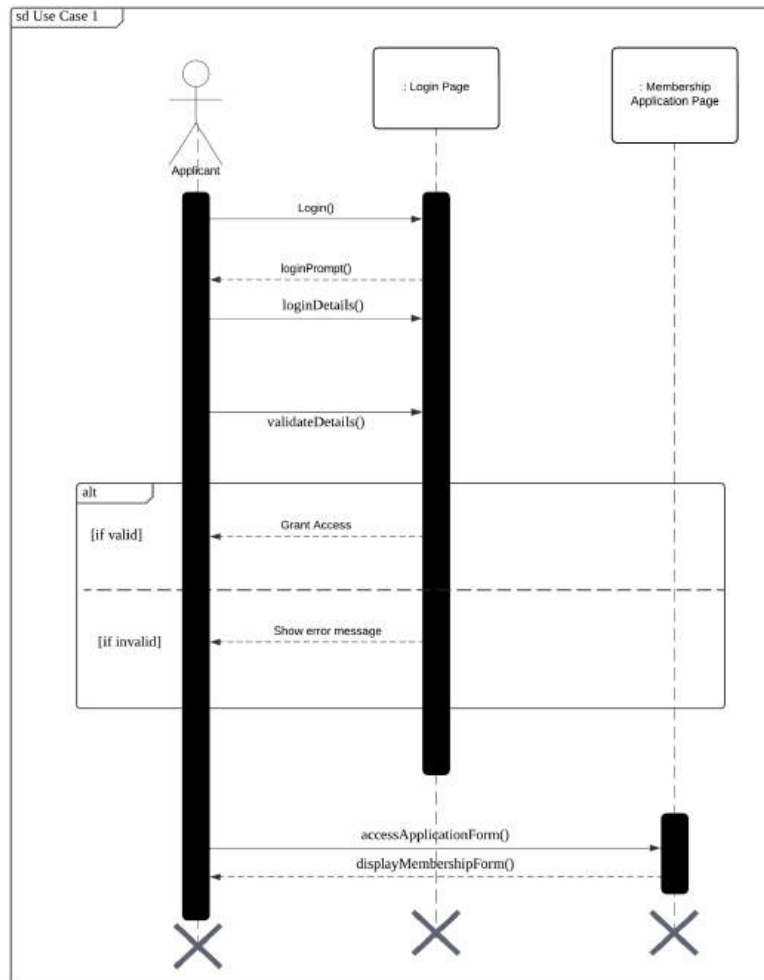


Figure 4.23: SD001 Sequence Diagram for Login for Apply Membership

Figure 4.24 shows the process of an admin reviewing membership, loan, or membership termination applications. The admin logs in, navigates to the membership list, and views batch application details. The system loops through each application in the batch, marking it as either “reviewed” or “incomplete”.

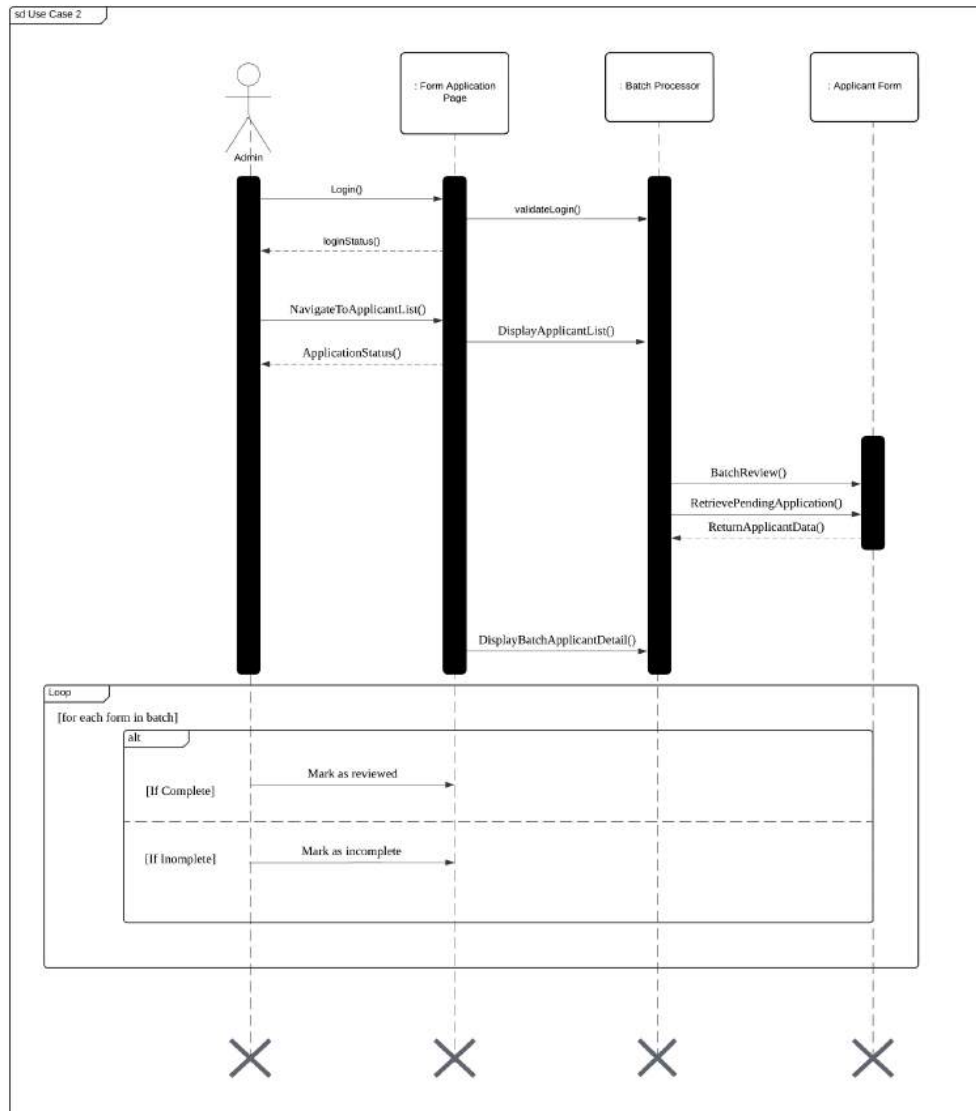


Figure 4.24: SD002 Sequence Diagram for Review Applicant's Form Subsystem

Figure 4.25 shows a sequence diagram for use case approved membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any membership application from the pending list, evaluates and decides whether to approve or not. Finally, the system updates the membership status

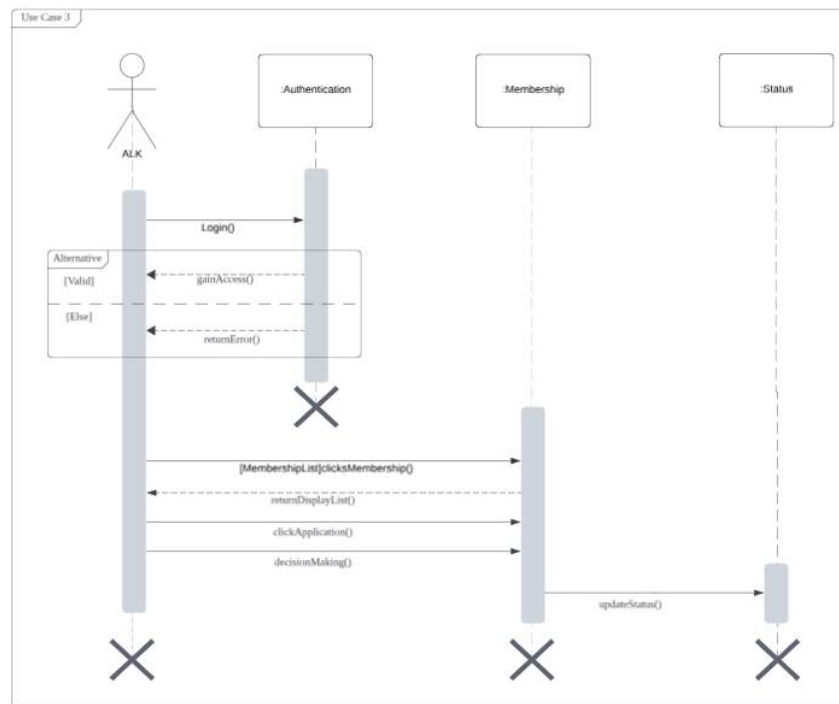


Figure 4.25: SD003 Sequence Diagram for Approve Membership Subsystem

Figure 4.26 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system's database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

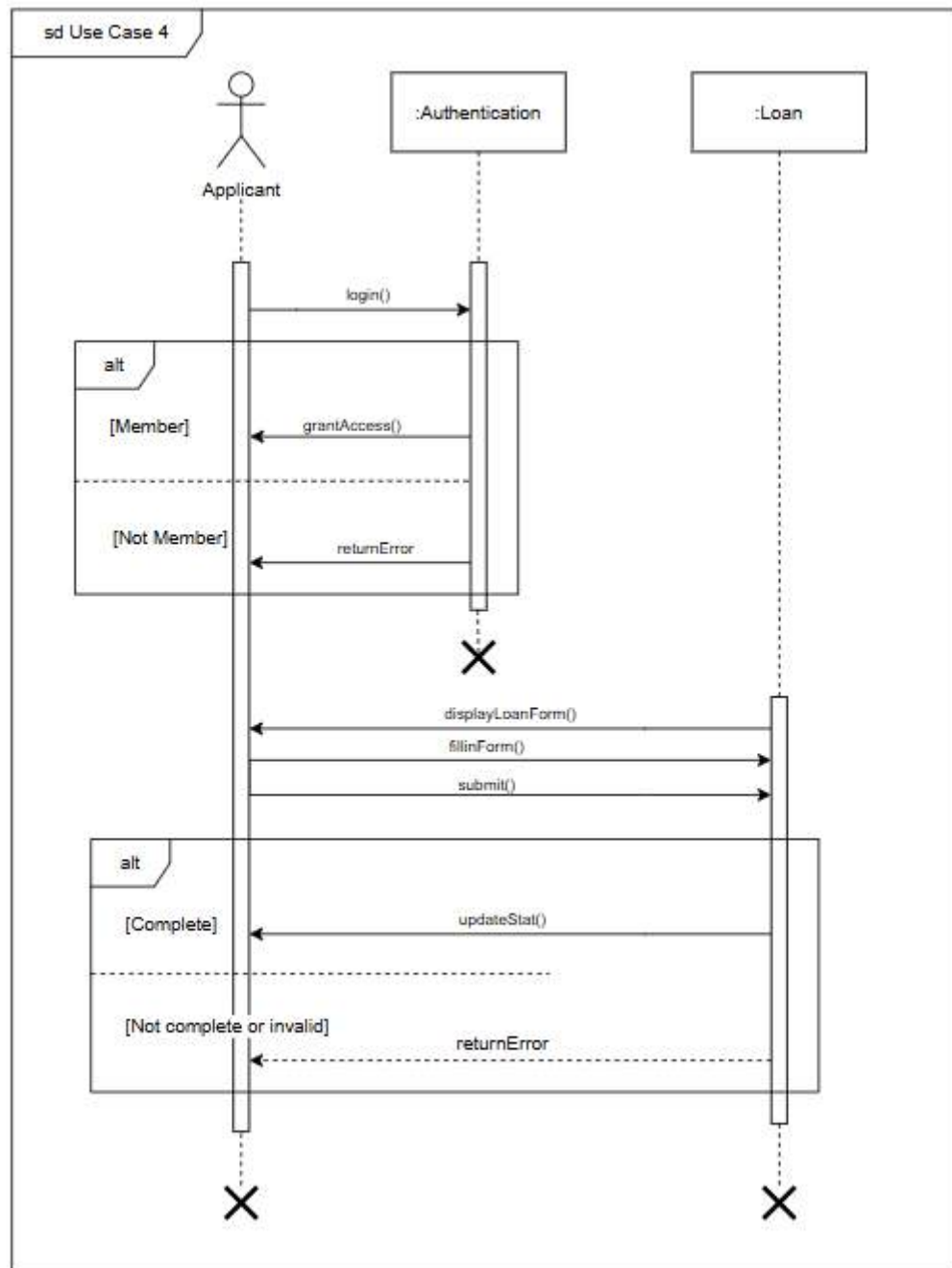


Figure 4.26: SD004 Sequence Diagram for Apply for Loan Subsystem

Figure 4.27 is a sequence diagram for calculation loan highlighting the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects the "Loan Calculator" button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that

requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

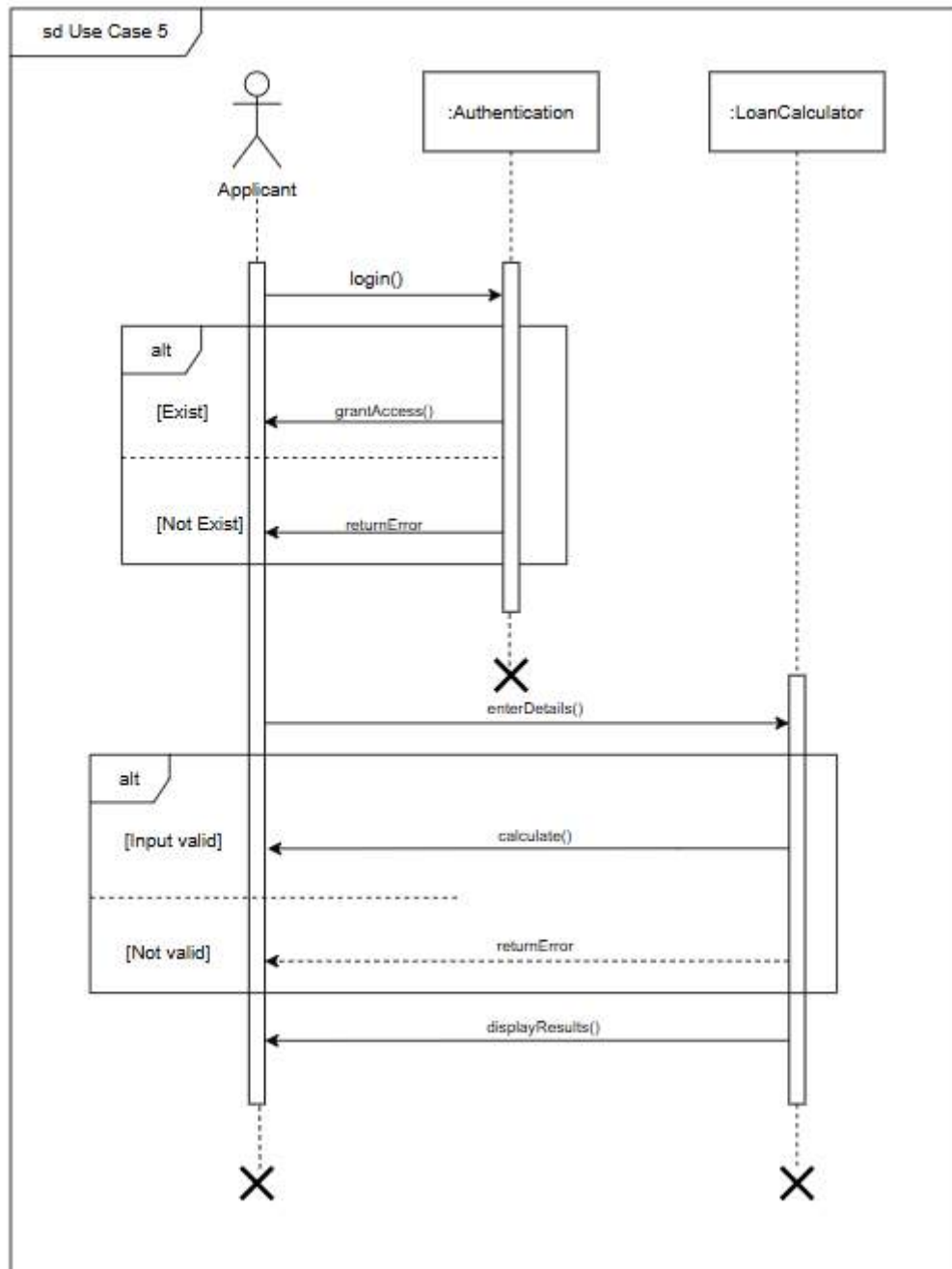


Figure 4.27: SD005 Sequence Diagram for Calculate Loan Subsystem

Figure 4.28 shows a sequence diagram for use case approved loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision. Finally, the system updates the loan status.

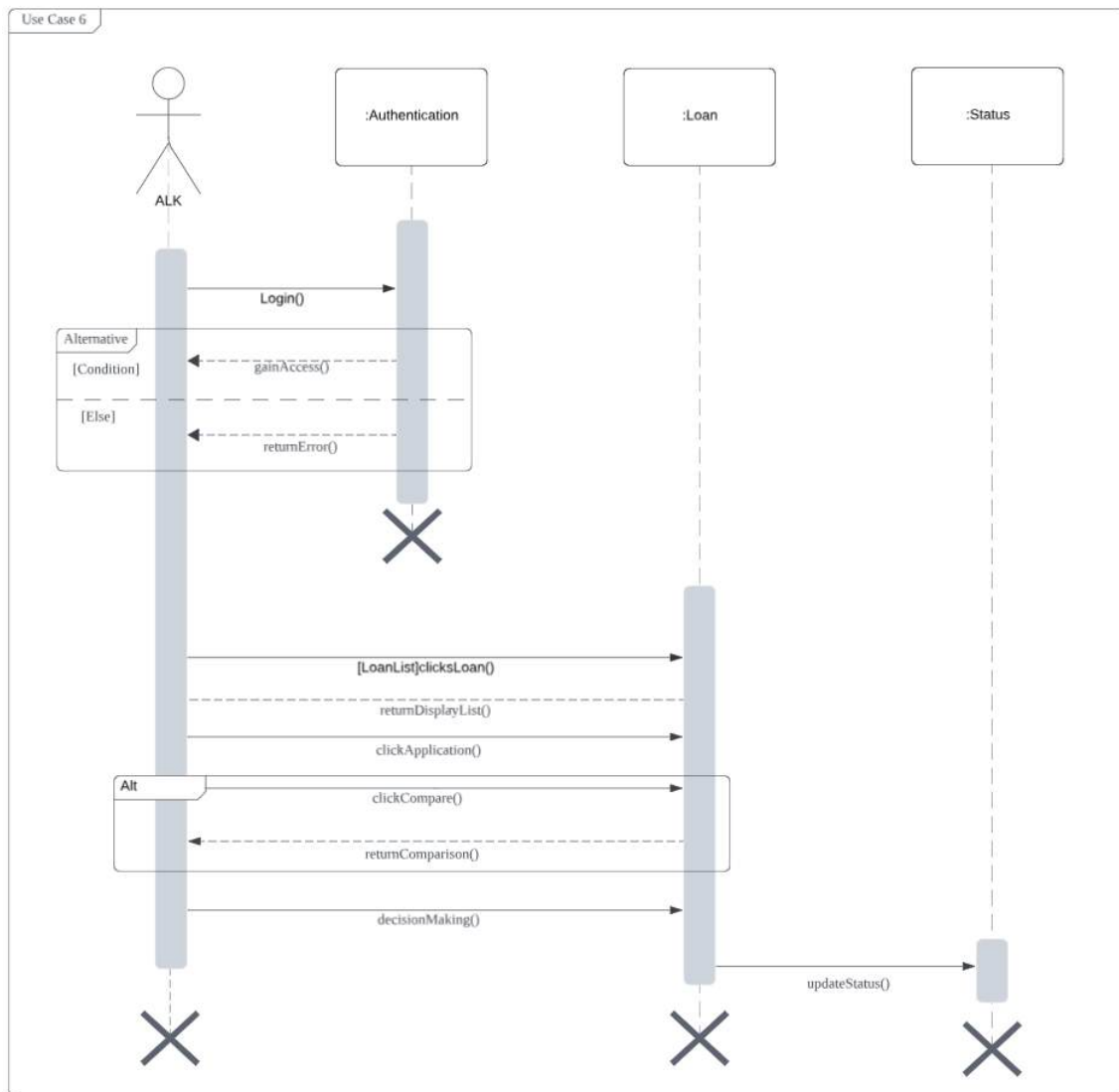


Figure 4.28: SD006 Sequence Diagram for Approve Loan Application Subsystem

Figure 4.29 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems. The admin begins by logging in, where successful authentication grants access to the admin page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

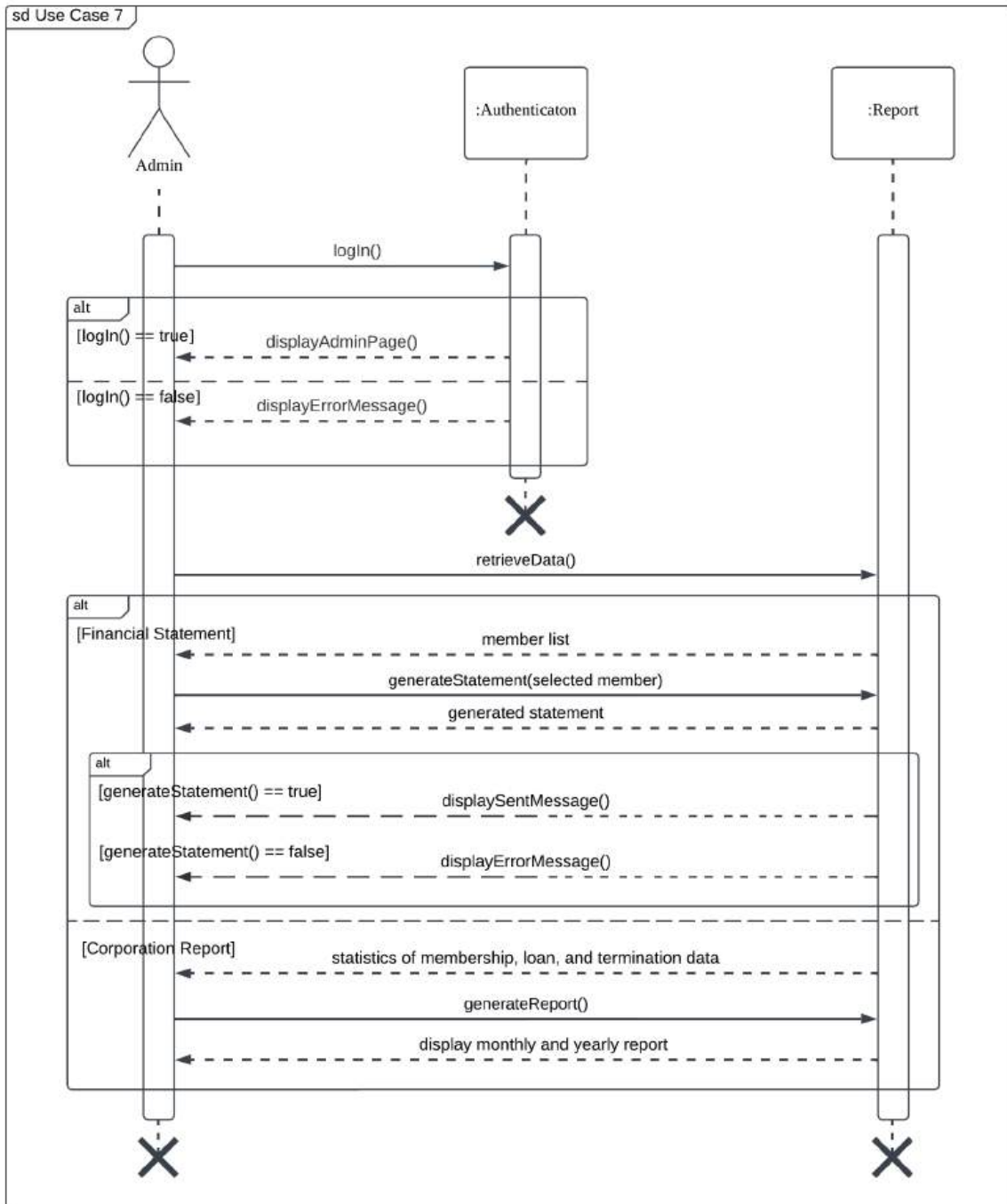


Figure 4.29: SD007 Sequence Diagram for Prepare Statement and Report Subsystem

Figure 4.30 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected

applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

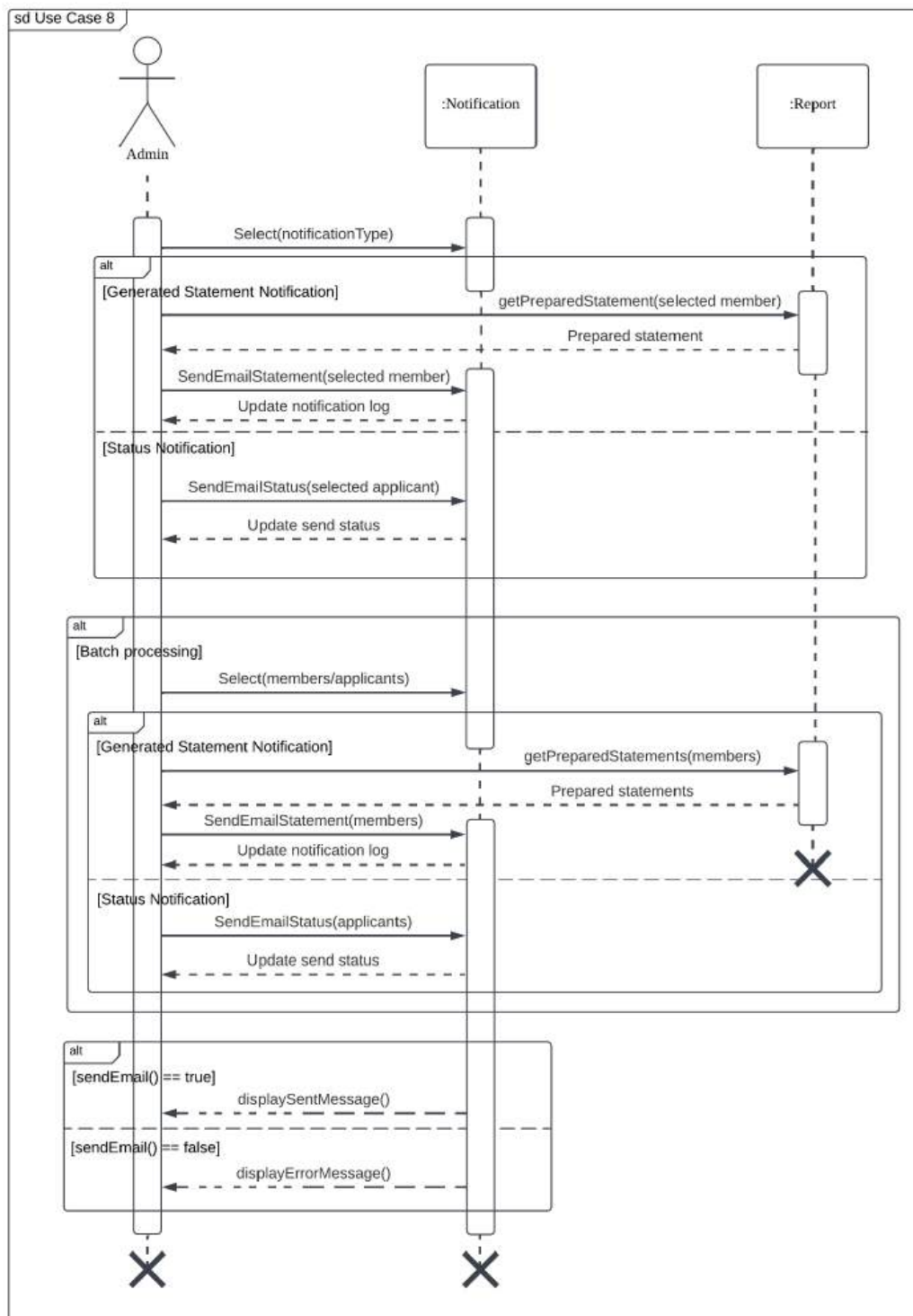


Figure 4.30 SD008 Sequence Diagram for Notify via Email Subsystem

Figure 4.31 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

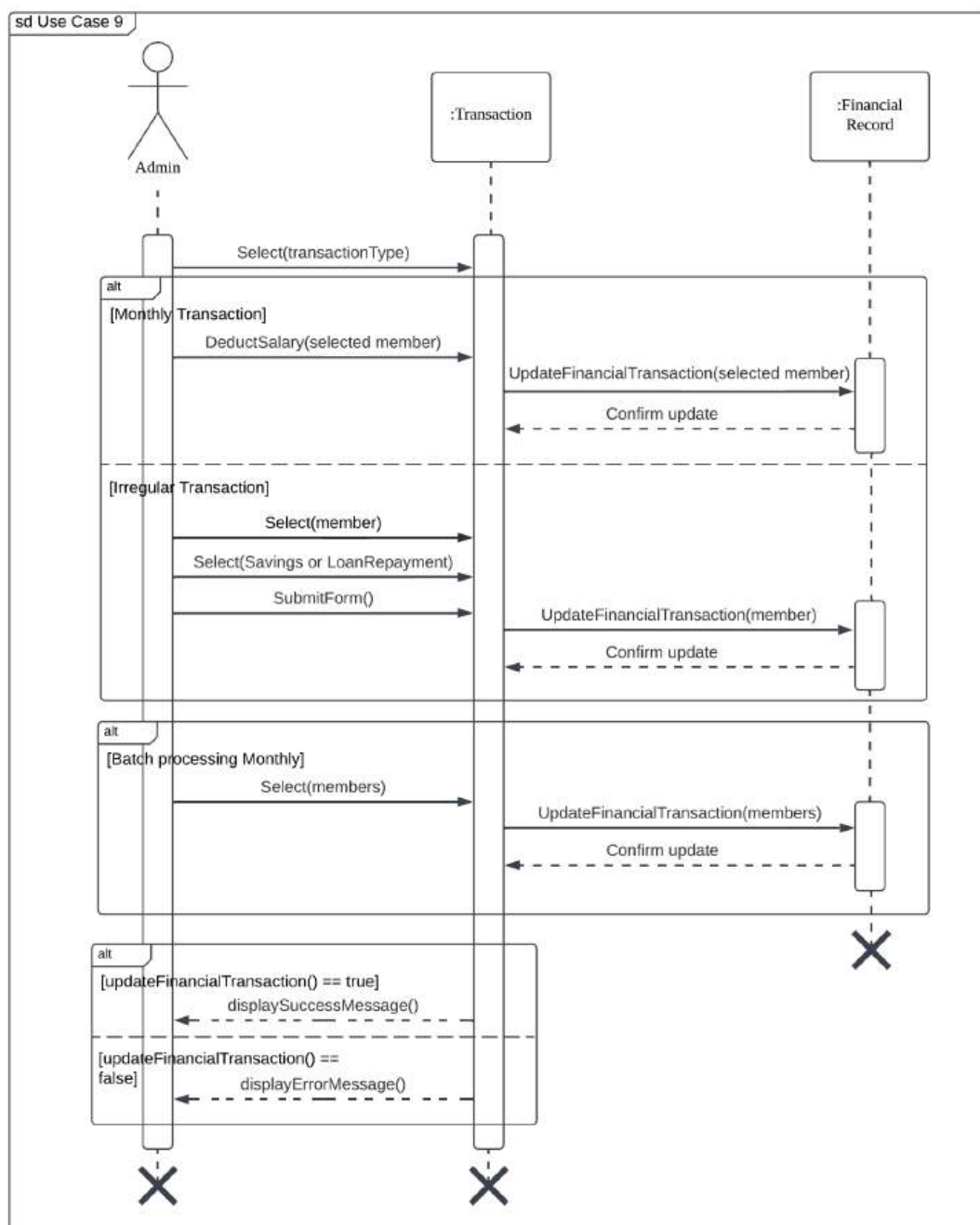


Figure 4.31: SD009 Sequence Diagram for Manage Transaction Subsystem

Figure 4.32 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

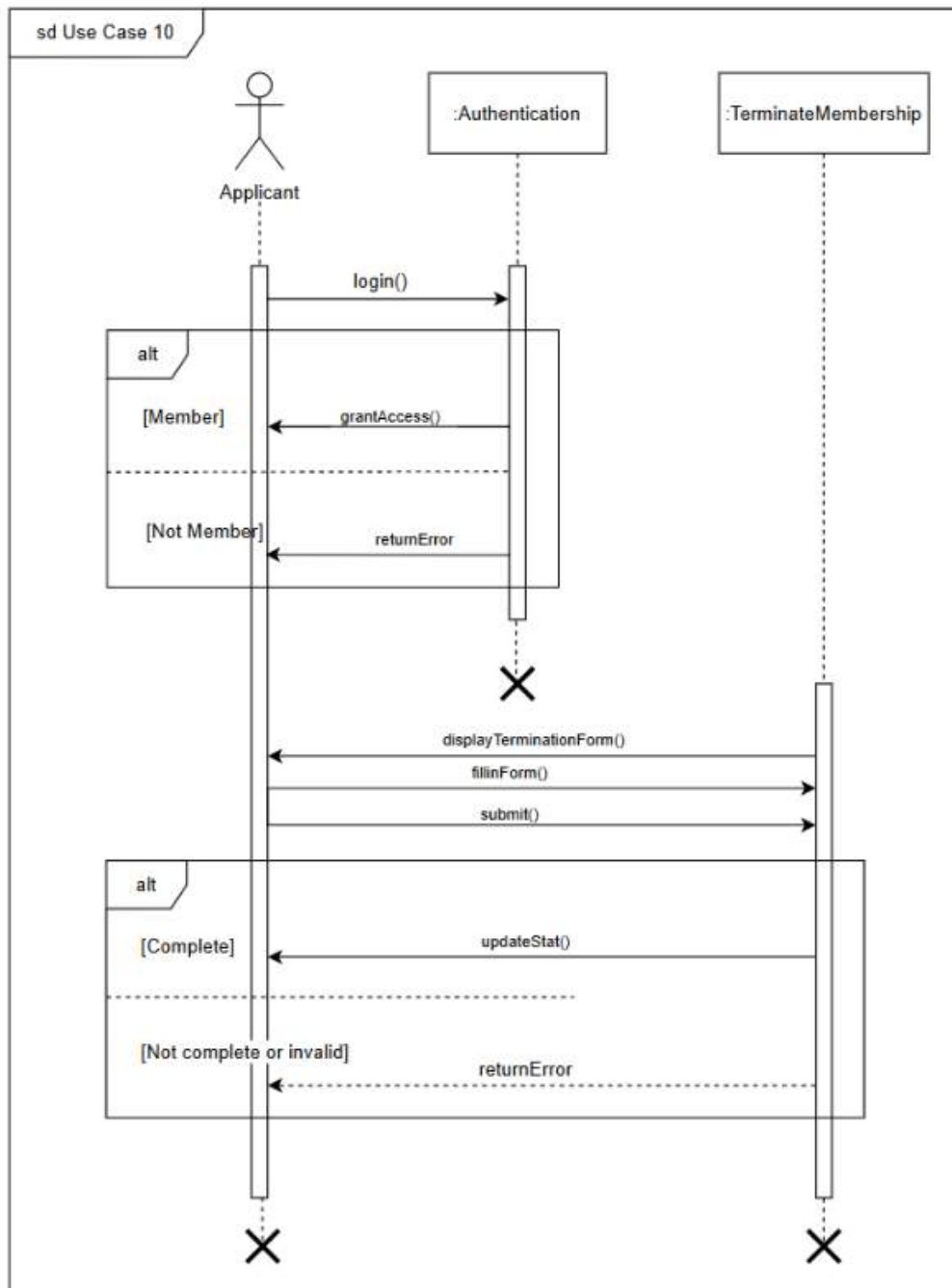


Figure 4.32: SD010 Sequence Diagram for Terminate Membership Subsystem

Figure 4.33 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin. The ALK then performs decision-making whether to approve or reject the application. If the ALK approved, then the applicant's membership will end otherwise the membership will continue as usual.

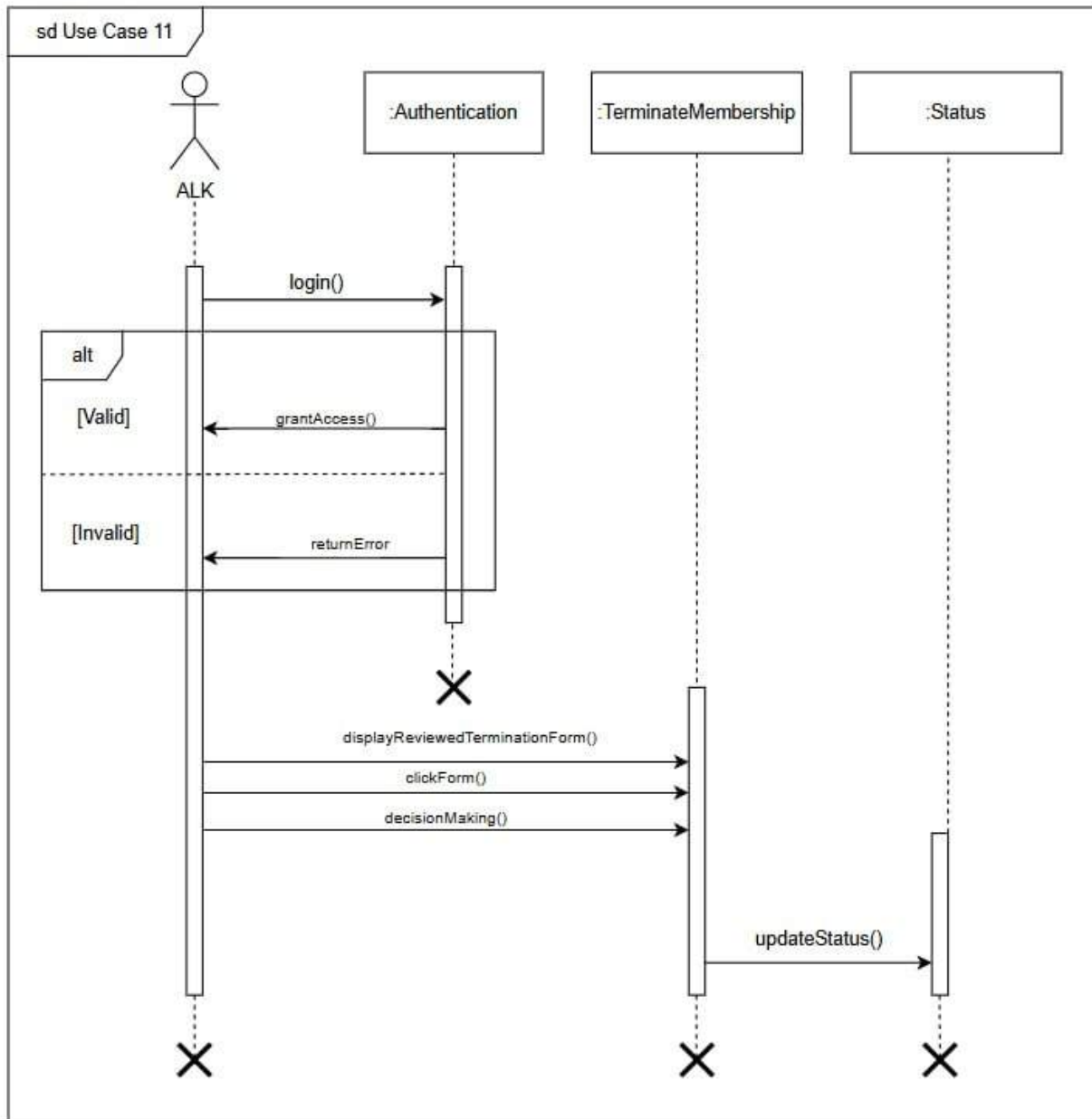


Figure 4.33: SD011 Sequence Diagram for Approve Terminate Application Subsystem

4.5 Deployment View - Deployment Diagram

This view focuses on how the system's components are distributed across hardware and software environments. It shows how the software is deployed on servers, devices, or systems, and how they interact with each other. Represented through a deployment diagram, this view illustrates the physical distribution of components across nodes and their communication paths. It helps to understand the system's layout, showing how components are allocated to machines or servers and how they interact over networks. The diagram also highlights the relationships between hardware, software, and external systems, ensuring the system runs efficiently in its environment.

Figure 4.34 illustrates the deployment diagram of the Kada Koperasi Automated System and highlights its key components along with their interactions. The system comprises four main nodes: the Client, Server, Database Server, and External Email Server. The Client node, represented by the artifact `<<artifact>> Browser`, communicates with the Server node, which includes `<<artifact>> Internet Server` and `<<artifact>> Application Server` to handle requests and execute business logic. The Application Server interacts with the Database Server, containing `<<artifact>> MySQL DB`, to manage and store cooperative data securely. Additionally, the External Email Server, represented by the artifact `<<artifact>> Gmail SMTP Service`, enables email notifications to send statement reports to members of koperasi KADA via the SMTP protocol.

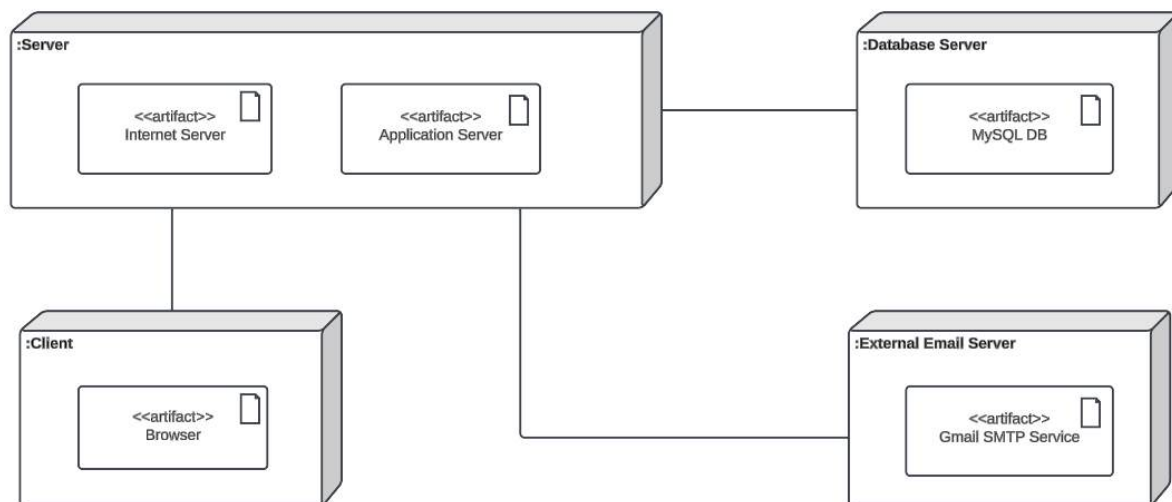


Figure 4.34: Deployment diagram for Koperasi KADA'S Automated System

5. Data Design

This section deals with the system by describing the information domain, using data entities and their relationships to represent the structured data. Various key entities have been listed, including Admin, ALK, Applicant, Loan, Membership and Notification. All of these entities are attributed according to their respective properties and behaviours. The major data or systems entities are stored into a relational database. Table 5.1 shows the description of entities in the database.

Table 5.1: Description of Entities in the Database

No.	Entity Name	Description
1	Administrator	General term describing the staff responsible for managing and reviewing membership and loan application
2	AdminApplicationFeedback	General term describing the reason of each application rejection by Admin
3	ALK	General term describing the authorized personnel responsible for approving membership and loan application
4	AlkApplicationFeedback	General term describing the reason of each application rejection by ALK
5	Applicant	General term describing all users logged in as applicants to apply membership.
6	BankInfo	General term describing the bank account information provided by applicants for processing membership applications.
7	FamilyInfo	General term for the details of the family members associated with an applicant.
8	Guarantor	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.
9	Historymodimem	General term describing the past membership applications and the modification allowed
10	historymodiloan	General term describing the past loan applications and the modification allowed

11	Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.
12	LoanRepayment	General term describing the financial activities related to loans, such as disbursements, repayments, and transaction history.
13	LoanType	General term describing the different categories or plans of loans offered
14	Managing	General term describing a notification update related to an approval process
15	Membership	General term describing to the record of applicants who are members of the Koperasi and
16	MembershipEnd	General term describing details of membership termination including reason, date of application, approval and review.
17	Saving	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.
18	SavingType	General term describing type of savings the company had
19	Status	General term represents transaction records of saving.
20	User	General term describing all the people who used the system.
21	UserType	General term describing type of user who accesses the system

5.1 Data Dictionary

This section covers a data dictionary for the Entity Relation Diagram (ERD) in figure 5.1 which is a structured document that provides detailed information about the data used in a system. It describes the system's entities, attributes and type, and serves as a reference for understanding the data structure. By ordering attribute names alphabetically, developers, designers and stakeholders can communicate with greater consistency, clarity and ease. The data dictionary describes classes and attribute names in the classes to offer a clear overview of the system's data components.

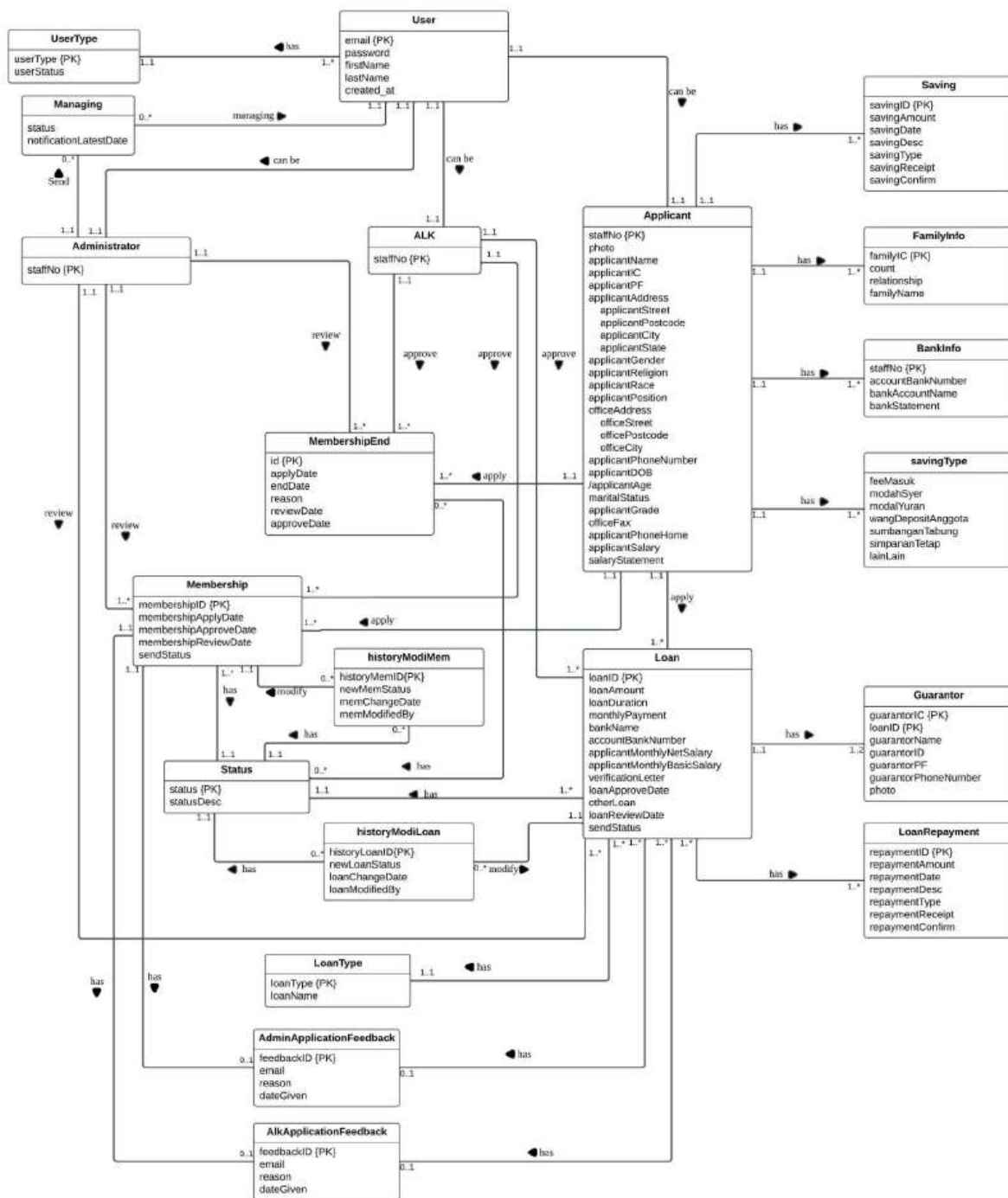


Figure 5.1: Entity Relationship Diagram (ERD)

5.1.1 Entity: Admin

Table 5.2: Data Dictionary for Admin

Attribute Name	Type	Description
email	VARCHAR(50)	Admin's email address

firstName	VARCHAR(25)	Admin's first name
lastName	VARCHAR(25)	Admin's last name
password	VARCHAR(50)	Password for admin's system login
staffNo	INT(10)	Admin's staff number
userID	INT(10)	Unique identifier for admin

5.1.2 Entity: AdminApplicationFeedback

Table 5.3: Data Dictionary for AdminApplicationFeedback

Attribute Name	Type	Description
feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

5.1.3 Entity: ALK

Table 5.4: Data Dictionary for ALK

Attribute Name	Type	Description
email	VARCHAR(50)	ALK's email address
firstName	VARCHAR(25)	ALK's first name
lastName	VARCHAR(25)	ALK's last name
password	VARCHAR(50)	Password for ALK's system login
staffNo	INT(10)	ALK's staff number
userID	INT(10)	Unique identifier for ALK

5.1.4 Entity: AlkApplicationFeedback

Table 5.5: Data Dictionary for AlkApplicationFeedback

Attribute Name	Type	Description
----------------	------	-------------

feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

5.1.5 Entity: Applicant

Table 5.6: Data Dictionary for Applicant

Attribute Name	Type	Description
applicantAddress	VARCHAR(50)	Home address of the applicant
applicantAge	INT(3)	Current Age of the applicant. This is the derivation of date of birth.
applicantCity	VARCHAR(30)	City of the applicant's address
applicantDOB	DATE	Date of birth of the applicant.
applicantGender	CHAR(1)	Gender of the applicant
applicantGrade	VARCHAR (10)	Professional grade of the applicant's job position
applicantIC	INT(12)	Identification Card number of the applicant
applicantName	VARCHAR(30)	Full name of applicant
applicantPF	VARCHAR(30)	Personal File number of the applicant
applicantPhoneHome	VARCHAR (15)	Home phone number of the applicant
applicantPhoneNumber	VARCHAR(15)	Primary contact number of the applicant
applicantPosition	VARCHAR(30)	Job position of the applicant
applicantPostcode	INT(5)	Postcode of the applicant's address

applicantRace	VARCHAR(25)	Race or ethnicity of the applicant
applicantReligion	VARCHAR(25)	Religion of the applicant
applicantSalary	DECIMAL (10,2)	Monthly Income of the applicant
applicantState	VARCHAR(30)	State of the applicant's address
maritalStatus	VARCHAR (10)	Marital status of the applicant
officeAddress	VARCHAR(50)	Address of the applicant's workplace
officeCity	VARCHAR(30)	City of the applicant's workplace
officeFax	VARCHAR (15)	Fax number of the applicant's workplace
officePostcode	INT(5)	Postcode of the applicant's workplace

5.1.6 Entity: BankInfo

Table 5.7: Data Dictionary for BankInfo

Attribute Name	Type	Description
accountID	INT(20)	Account Number for transactions
bankAccountType	VARCHAR(20)	Name of the bank

5.1.7 Entity: FamilyInfo

Table 5.8: Data Dictionary for FamilyInfo

Attribute Name	Type	Description
count	INT(20)	Number of family member
familyIC	VARCHAR(15)	Family member Identification Number
familyName	VARCHAR(25)	Name of the family member
relationship	VARCHAR(10)	Relationship of applicant with family member

5.1.8 Entity: Guarantor

Table 5.9: Data Dictionary for Guarantor

Attribute Name	Type	Description
guarantorIC	VARCHAR(20)	Guarantor identification card
guarantorID	INT(20)	Unique identifier for guarantor
guarantorName	VARCHAR(30)	Guarantor full name
guarantorPF	VARCHAR(20)	Guarantor personal file number
guarantorPhoneNumber	VARCHAR(20)	Guarantor phone number
loanID	INT (5)	Loan identification number
photo	LOB	Guarantor photo

5.1.9 Entity : HistoryModiMem

Table 5.10: Data Dictionary for historymodimem

Attribute Name	Type	Description
historyMemID	INT(5)	Unique identifier for each past applications
newMemStatus	INT(3)	New status after modification
memChangeDate	DATE	Date status modified
memModifiedBy	INT(5)	Modifier's staff number

5.1.10 Entity : historymodiloan

Table 5.11: Data Dictionary for historymodiloan

Attribute Name	Type	Description
historyLoanID	INT(5)	Unique identifier for each past applications
newLoanStatus	INT(3)	New status after modification
loanChangeDate	DATE	Date status modified
loanModifiedBy	INT(5)	Modifier's staff number

5.1.11 Entity: Loan

Table 5.12: Data Dictionary for Loan

Attribute Name	Type	Description
accountBankNumber	BIGINT(25)	Account bank number
applicantBasicSalary	DECIMAL(10,2)	Applicant basic salary
applicantNetSalary	DECIMAL(10,2)	Applicant net salary
bankName	VARCHAR(30)	Bank name
loanAmount	DECIMAL(10,2)	Loan Amount
loanApplyDate	DATE	Loan apply date
loanApproveDate	DATE	Loan approve date
loanDuration	INT(2)	Loan Duration
loanReviewDate	DATE	Loan review date
loanID	INT(5)	Loan identification number
monthlyPayment	DECIMAL(10,2)	Monthly payment of loan
otherLoan	INT(5)	Other loan type
sendStatus	INT(3)	Status for send email
verificationLetter	LONGBLOB	Applicant verified employer

5.1.12 Entity: LoanRepayment

Table 5.12: Data Dictionary for LoanRepayment

Attribute Name	Type	Description
RepaymentID	INT(5)	Uniquely identifies saving transaction
RepaymentAmount	DECIMAL(10, 2)	Repayment amount
RepaymentDate	DATE	Repayment date
RepaymentDesc	VARCHAR(200)	Repayment description
RepaymentType	VARCHAR(30)	Repayment type
RepaymentReceipt	VARCHAR(15)	Repayment receipt
RepaymentConfirm	VARCHAR(15)	Repayment confirmation

5.1.13 Entity: LoanType

Table 5.14: Data Dictionary for LoanType

Attribute Name	Type	Description
loanName	VARCHAR(25)	Loan name
loanType	INT(7)	Unique identifier for the loan type

5.1.14 Entity: Managing

Table 5.15: Data Dictionary for Managing

Attribute Name	Type	Description
notificationLatestDate	VARCHAR(25)	Date of notification being sent
status	VARCHAR(25)	Status of notification

5.1.15 Entity: Membership

Table 5.16: Data Dictionary for Membership

Attribute Name	Type	Description
membershipStatus	VARCHAR(20)	Current status of the membership request
membershipApplyDate	DATE	Date when the membership starts
membershipEndDate	DATE	Date when the membership ends
membershipReviewDate	DATE	Date when the membership form is review
membershipApproveDate	DATE	Date when the membership is approved

5.1.16 Entity: MembershipEnd

Table 5.17: Data Dictionary for MembershipEnd

Attribute Name	Type	Description
applyDate	DATE	Terminate membership apply date
approveDate	DATE	Approve date

endDate	DATE	Membership end date
id	INT (5)	Uniquely identifies membership end application form
reason	VARCHAR(255)	Reasons message to end membership
reviewDate	DATE	reviewDate

5.1.17 Entity: Saving

Table 5.18: Data Dictionary for Saving

Attribute Name	Type	Description
savingID	INT(5)	Uniquely identifies saving transaction
savingAmount	DECIMAL(10, 2)	Saving amount
savingDate	DATE	Saving date
savingDesc	VARCHAR(200)	Saving description
savingType	VARCHAR(30)	Saving payment type
savingReceipt	VARCHAR(15)	Saving receipt
savingConfirm	VARCHAR(15)	Saving confirmation

5.1.18 Entity: SavingType

Table 5.19: Data Dictionary for SavingType

Attribute Name	Type	Description
feeMasuk	DECIMAL(10, 2)	Fee capital amount
modahSyer	DECIMAL(10, 2)	Fixed deposit amount
modalYuran	DECIMAL(10, 2)	Member deposit amount
wangDepositAnggota	DECIMAL(10, 2)	Member savings amount
sumbanganTabung	DECIMAL(10, 2)	Share capital amount
simpananTetap	DECIMAL(10, 2)	Fixed saving amount

5.1.19 Entity: Status

Table 5.20: Data Dictionary for Status

Attribute Name	Type	Description
status	INT (3)	Status
statusDesc	VARCHAR (25)	Status description

5.1.20 Entity: User

Table 5.21: Data Dictionary for User

Attribute Name	Type	Description
email	VARCHAR(25)	Email for communication and receiving any update information.
firstName	VARCHAR(25)	Given name of the user
lastName	VARCHAR(20)	Family Name of the user
password	VARCHAR(15)	Password for user authentication
staffNo	INT(10)	Staff Number of the user
userID	INT AUTO_INCREMENT	Unique identifier assigned for each user

5.1.21 Entity: UserType

Table 5.22: Data Dictionary for UserType

Attribute Name	Type	Description
UserType	INT(5)	Type of user
UserStatus	VARCHAR(10)	Type of user description

6. User Interface Design

The interface includes sections designed for different types of users which are applicants, administrators, and ALK.

For applicants, the interface starts with a login page where users enter their credentials to access the system. Once logged in, they see a menu with options like applying for membership, applying for a loan, applying for a termination of membership and using a loan calculator. The membership application involves filling out a form in two steps where users provide personal details and submit their application. For loan applications, users can calculate repayment amounts, fill out forms with personal and guarantor information, sign agreements. Furthermore, the termination of membership application follows a similar process where users fill out a form with their details and state their reason. All applications can check the status of their application after submitting the application. These features ensure users can easily manage their membership and loan applications.

For administrators, the interface focuses on managing the application, including membership and loan applications, financial transactions, notifications, and reporting. Admins log in through an admin page to access tools for reviewing applications, updating membership statuses, and verifying applicants' financial and guarantor information before proceeding with approvals. They also manage transactions by recording payments, tracking outstanding balances, and updating payment statuses. Email notifications are sent to inform users about application statuses and financial statements. Additionally, administrators can generate detailed reports on transactions, loan approvals, and membership activities, which are then emailed to relevant parties for record-keeping and decision-making.

The ALK interface is designed for approving or rejecting applications. After logging in, ALK users can choose to review either membership or loan application. For membership forms, ALK users can approve or reject applications directly. For loan applications, they can check the loan details, compare them with the terms, and review supporting documents like payslips. The system clearly shows whether the application is successful or not, allowing ALK users to make quick decisions.

In summary, the interfaces are designed to help users manage applications, administrators handle reviews, and ALK staff approve or reject the applications efficiently.

6.1 Screen Images

The Koperasi KADA's Automated System's first-time user should see the login page when he/she opens the website like in Figure 6.1. If the user has not registered, he/she should be able to do that on the login page. If the user is already registered, the user can login and will be redirected to which type of user they are either user, admin or Ahli Lembaga Koperasi (ALK). After Logging in, Figure 6.2 allows users to choose from various menu options, including applying for membership, applying for a loan, and calculating loan details as well as applying for a membership termination.

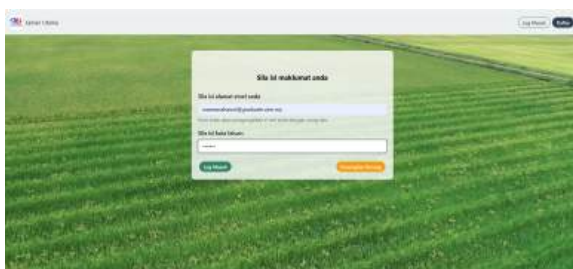


Figure 6.1: Login Page



Figure 6.2: Applicant's Menu

When users choose to apply for membership, they are directed to Figure 6.3, where they must fill out the membership form. After clicking the "Next" button, they proceed to Figure 6.4 to complete their family information, bank information, and saving information. Once all required information is filled in and they have to agree with an agreement like in Figure 6.7 before submitting the form. Lastly, the application status will be shown as shown in Figure 6.8, indicating whether the membership application is pending approval, is being reviewed, or needs further information.



Figure 6.3: Membership Form



Figure 6.4: Family Information

Figure 6.5: Bank Information

Figure 6.6: Saving Information

Figure 6.7: Applicant's Agreement

Figure 6.8: Membership Status

The prerequisite to apply for a loan is, the user must have a Koperasi KADA membership in order to apply for a loan using the website. Before filling out the loan application form as in Figure 6.10 with the loan type, amount, and duration, they can utilize the loan calculator to estimate monthly repayments like in Figure 6.9. After filling out loan information, the system will display member's information as in Figure 6.11. In Figure 6.12, the user must provide the details of a guarantor and have an agreement to the terms like in Figure 6.13 before submitting the form. Lastly, the application status will be shown as shown in Figure 6.14, indicating whether the loan is pending approval, is being reviewed, or needs further information.

Figure 6.9: Loan Calculator

Figure 6.10: Loan Application Form

Figure 6.11: Applicant's Information

Figure 6.12: Guarantor Information

Figure 6.13: Agreement

Figure 6.14: Loan Status

Furthermore, if a member wishes to terminate their membership, they can do so by filling out the application in Figure 6.15 and providing reason as in Figure 6.16. After completing the application, the applicant needs to have an agreement by clicking in the checkbox in Figure 6.17. Then, the application status will also be shown like the other two applications, as in Figure 6.18, indicating whether the loan is pending approval, is being reviewed, or needs further information.

Figure 6.15: Terminate Application

Figure 6.16: Reason of Termination

Figure 6.17: Agreement

Figure 6.18: Terminate Application Status

Admin is prompted to login first by entering Admin ID and Password as shown in Figure 6.1. After successfully logging in as an admin, it will show options to manage various aspects of the organization. These options include reviewing the membership list, reviewing the loan application list, reviewing membership termination application list, and reviewing members approved by ALK as shown in Figure 6.19.



Figure 6.19: Admin Menu

After logging in as an admin and selecting the "Membership list", "Loan Application List" or "Membership Termination Application List" option as shown in Figure 6.20, 6.21, and 6.22 admin will see a list of members and their review status (either "Completed" or "Not Completed"). To review a member's form, click the applicant's name. After completing the review of all details, return to the main review form to update the review status to "Completed" if the application is completed. Otherwise, Admin will update the review status to "Not Completed". Last but not least, Admin can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously.



Figure 6.20: Membership Applicant's Name List Figure 6.21: Loan Application Name List

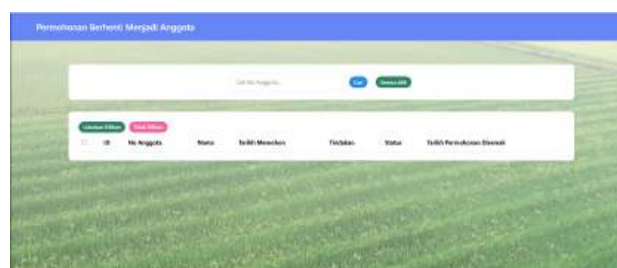


Figure 6.22: Membership Termination Application Name List

Next, Figure 6.23 illustrates the ALK's menu which shows the activity handled by ALK. To approve membership, ALK can click the Senarai Permohonan Anggota to view the

list of applications reviewed by Admin. Figure 6.24 shows the list of names that are waiting to be approved or rejected. ALK can simply click the button ‘Diterima’ or ‘Ditolak’ to perform decision making for each application.



Figure 6.23: ALK's Menu



Figure 6.24: Membership List

If the decision-maker is unsure and wants to recheck the application, ALK can click the ‘Semak Semula’ button to view the applicant’s application. Figure 6.25 shows the applicant’s information after clicking ‘Semak Semula’. Last but not least for approving membership, ALK can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously. Figure 6.26 shows the batch processing function.



Figure 6.25: Application's Information

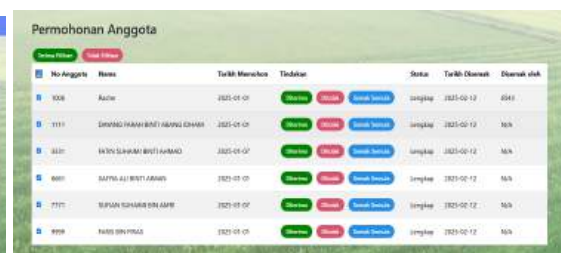


Figure 6.26 Batch Processing

Moving to approving loans, ALK can choose ‘Senarai Permohonan Pembiayaan in ALK’ menu to access the loan applications reviewed by Admin. Same goes for loans, the ALK can directly perform decision-making by clicking the ‘Diterima’ or ‘Ditolak’ button directly from the list. ALK also can view the applications by clicking the ‘Semak Semula’ button. Figure 6.27 and Figure 6.28 show the list of names with buttons and applicant’s loan application respectively. After viewing applicant’s information, ALK can use the ‘Banding’ button to perform comparison of that particular application with the company's loan policies. Figure 6.29 shows the location of the ‘Banding’ button and Figure 6.30 illustrates the comparison result.

Permohonan Pinjaman

Permohonan Anggota

No. Anggota	Nama	Tarikh Menerima	Terima	Status	Tarikh Disetujui	Disetujui oleh
1	1234	2023-01-01	Terima Ditolak	Lengkap		
2	5678	2023-01-02	Terima Ditolak	Lengkap		
3	9012	2023-01-03	Terima Ditolak	Lengkap		

Figure 6.27: Loan List

Permohonan Pinjaman

Maklumat Pemohon

Nama Pemohon: 12345678901234567890

No. ID: 12345678901234567890

Tempat Lahir: 12345678901234567890

Jenis: 12345678901234567890

Alamat: 12345678901234567890

Agama: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Figure 6.28: Applicant's Info

Banding Button

Tarikh Lahir: 12345678901234567890

Jenis: 12345678901234567890

Tempat Lahir: 12345678901234567890

Agama: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Figure 6.29: Banding Button

Keputusan Perbandingan

Syarat Pembiayaan

Agar dapat pinjaman, anda perlu membayar RM 1000 setiap bulan (RM 1000 x 12 = RM 12000)

Agar dapat pinjaman, anda perlu membayar RM 1000 setiap bulan (RM 1000 x 12 = RM 12000)

Agar dapat pinjaman, anda perlu membayar RM 1000 setiap bulan (RM 1000 x 12 = RM 12000)

Agar dapat pinjaman, anda perlu membayar RM 1000 setiap bulan (RM 1000 x 12 = RM 12000)

Agar dapat pinjaman, anda perlu membayar RM 1000 setiap bulan (RM 1000 x 12 = RM 12000)

Figure 6.30: Comparison Result

ALK is also responsible for approving membership termination applications. Figure 6.31 shows the list of applications reviewed by Admin. ALK can perform decision-making simply by clicking the 'Diterima' or 'Ditolak' button. ALK also can view the applicant's information as shown in Figure 6.32. Lastly, all membership, loan and terminate membership approval supports batch processing using checkbox to perform decision-making to more than one applications simultaneously.

Permohonan Berhenti Menjadi Anggota

Permohonan Berhenti Menjadi Anggota

No. Anggota	Nama	Tarikh Menerima	Terima	Status	Tarikh Disetujui	Disetujui oleh
1	1234	2023-01-01	Terima Ditolak	Lengkap		
2	5678	2023-01-02	Terima Ditolak	Lengkap		
3	9012	2023-01-03	Terima Ditolak	Lengkap		

Figure 6.31: Terminate List

Permohonan Berhenti Menjadi Anggota

Maklumat Pemohon

No. Anggota: 12345678901234567890

Nama: 12345678901234567890

No. ID: 12345678901234567890

Jenis: 12345678901234567890

Alamat: 12345678901234567890

Agama: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Tempat: 12345678901234567890

Figure 6.32: Application's Info

Figure 6.33 and Figure 6.34 display the interface for administrators to manage monthly salary deductions and irregular transactions. Administrators can review and update transactions by clicking the "Keluarkan" button or filling the form. Meanwhile, Figure 6.35 and Figure 6.36 show the applicant's interface for viewing savings and loan repayment records. Applicants can check their transaction details to ensure the displayed financial information is clear and easily accessible.

No. Akaun	Nama	Bilik Simpanan	Simpanan Semasa	Bayaran	Simpanan Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.33: Monthly Salary Deduction

Figure 6.34: Irregular Transaction

No. Akaun	Nama	Bilik Simpanan	Simpanan Semasa	Bayaran	Simpanan Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.35: Savings Record

No. Akaun	Nama	Bilik Pinjaman	Pinjaman Semasa	Bayaran	Pinjaman Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.36: Loan Repayment Record

Next, Figure 6.37 displays a list of membership, loan, and termination applications that have been reviewed by ALK and require applicants to be notified via email. Figure 6.38 provides an option to generate a statement for the approved loan applicant list. When the administrator clicks the "Hasilkan" button, the system automatically generates the report statement, as shown in Figure 6.39. The administrator can then send the report statement via email for applicant reference. Finally, Figure 6.40 presents the corporation report, which includes both monthly and yearly summaries. This report helps in monitoring the overall financial performance and membership activities within the cooperative.

No. Akaun	Nama	Bilik Simpanan	Simpanan Semasa	Bayaran	Simpanan Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.37: List of Applicant's Status

No. Akaun	Nama	Bilik Pinjaman	Pinjaman Semasa	Bayaran	Pinjaman Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.38: List of Member of Koperasi

No. Akaun	Nama	Bilik Simpanan	Simpanan Semasa	Bayaran	Simpanan Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.39: Generated Statement

No. Akaun	Nama	Bilik Pinjaman	Pinjaman Semasa	Bayaran	Pinjaman Baki
1234	HAZWAN ALIAH BINTI HASNUN	1234	RM 100.00	RM 100.00	RM 100.00

Figure 6.40: Corporation report

Lastly, ALK can modify the status of each application. This function was used when ALK unintentionally performed a wrong decision-making process. Figure 6.41 and Figure 6.42 illustrate the modification interface of membership and loan application respectively.

Permohonan Anggota

Cari Nama Anggota

Go! Simpan Baru

Permohonan Anggota

No Anggota	Nama	Tanggal Menerima	Status	Status	Status Baru	Tanggal Lulus Permohonan	Status dan	Tanggal
1000	Rachet	2025-01-01	Ya	Ya	Ya	2025-02-12	Ya	Ya
1111	DISKUSI TANGGAL BANYU KEMAS KEMAS	2025-01-01	Ya	Ya	Ya	2025-02-13	Ya	2025-02-0
2001	PERMISI KEMAS BANYU KEMAS	2025-01-01	Ya	Ya	Ya	2025-02-13	Ya	Ya
2001	PERMISI KEMAS BANYU KEMAS	2025-01-01	Ya	Ya	Ya	2025-02-13	Ya	2025-02-0

Figure 6.41: Membership Modification

Permohonan Pembiayaan

Cari Nama Anggota

Go! Simpan Baru

Permohonan Pembiayaan

ID	No Anggota	Nama	Tanggal Menerima	Status	Status Baru	Tanggal Lulus Pembiayaan	Status dan	Tanggal
1	1000	DISKUSI TANGGAL BANYU KEMAS KEMAS	2025-01-01	Ya	Ya	2025-02-12	Ya	2025-02-0
2	1111	PERMISI KEMAS BANYU KEMAS	2025-01-01	Ya	Ya	2025-02-13	Ya	2025-02-0
3	2001	PERMISI KEMAS BANYU KEMAS	2025-01-01	Ya	Ya	2025-02-13	Ya	2025-02-0
4	2001	PERMISI KEMAS BANYU KEMAS	2025-01-01	Ya	Ya	2025-02-13	Ya	2025-02-0

Figure 6.42: Loan Modification

7. Traceability

This section generally ensures that every requirement, feature and functionality of a system is connected to its corresponding design, implementation and testing processes. Traceability is often maintained through tools like the Traceability Matrix, which maps the relationships between requirements and their corresponding artifacts.

Table 7.1 and Table 7.2 shows the requirement traceability matrix, identifies functional requirements, which are mapped to specific system components, such as the use cases, sequence diagram and related entities. For example, the P001 is associated with the entities User, Applicant, Membership, and FamilyInfo since a membership application should address the authentication of a user, details of an applicant and information about the family. P002 will review the membership form. Similarly, P004 includes Admin, ALK, Membership, Loan and Approval to handle the approval process of memberships and loans. Further, P003 includes entities such as Loan, LoanType, LoanCalculator and GuarantorInfo for loan applications, tracking of loan types and transactions and verification guarantors. Lastly, P005 is associated with Admin, Applicant, Saving, LoanTransaction, Notification for preparing statement reports and sending email notifications.

Table 7.1: Requirement Matrix

Package, UC, Sequence Diagram /Entity Class	User	User Type	Applicant	Administrator	ALK	Membership	FamilyInfo	BankInfo	SavingType	MembershipEnd	Managing
P001, UC001, SD001	X	X	X			X	X	X	X		
P002, UC002, SD002	X	X		X		X	X	X	X		
P004, UC003, SD003	X	X			X	X	X	X	X		
P003, UC004,	X	X	X			X					

SD004											
P003, UC005, SD005	X	X	X								
P004, UC006, SD006	X	X			X						
P005, UC007, SD007	X	X		X		X			X		
P005, UC008, SD008	X	X	X	X		X					X
P007, UC009, SD009	X	X	X	X		X			X		
P006, UC010, SD010	X	X	X	X						X	
P004, UC011, SD011	X	X			X	X				X	

Table 7.2: Requirement Matrix (Continuation)

Packag e, UC, Seque nce Diagra m /Entity Class	Savin g	Loa n	Guar antor	Loan Type	LoanRe paymen t	history ModiM em	histor yModi Loan	AdminA pplicati onFeed back	AlkAppli cationFe edback	status
P001, UC001, SD001										X
P002, UC002, SD002										X
P004, UC003,						X			X	X

SD003										
P003, UC004, SD004		X	X	X						X
P003, UC005, SD005		X		X		X				
P004, UC006, SD006		X	X	X			X		X	X
P005, UC007, SD007	X	X		X	X					
P005, UC008, SD008		X							X	X
P007, UC009, SD009	X	X		X	X					
P006, UC010, SD010						X	X	X	X	X
P004, UC011, SD011									X	X